

Can language models forecast inductively?

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Abstract

Judgmental forecasting requires integrating sparse evidence with knowledge about the actors, institutions, and processes that shape how events unfold. The same observation can imply different futures depending on these underlying relationships. Here, we investigate whether language models (LMs) recover those relationships or rely on surface regularities. Across a battery of experiments spanning real-world prediction markets and controlled synthetic environments, we test whether models update their beliefs coherently, recover latent predictive relationships, and transfer what they learn to new settings. Our results provide evidence for each of these capabilities. Models selectively update their forecasts in response to causally relevant evidence, use contextual information to infer predictive relationships when observations are sparse, and revise these inferences as direct evidence accumulates. Training on families of predictive relationships further improves forecasting on unseen structures, although transfer is incomplete in more challenging settings. Together, these experiments provide behavioral evidence that LMs can forecast inductively: their predictions respond to evidence and context in ways that reflect underlying predictive structure, and some of this behavior transfers beyond the settings in which it was learned.¹

1 Introduction

Judgmental forecasting is a demanding form of reasoning central to decisions about political, social, and economic futures. Unlike settings with abundant historical data for statistical extrapolation, judgmental forecasting requires integrating sparse, noisy evidence with contextual knowledge to form calibrated beliefs about future outcomes [Tetlock and Gardner, 2016, Mellers et al., 2014, Lawrence et al., 2006]. This makes judgmental forecasting fundamentally inductive: forecasters must infer from limited observations the relationships that govern how evidence impacts outcomes, and use those relationships to anticipate how events may unfold [Tenenbaum et al., 2006, Griffiths et al., 2010, Tenenbaum et al., 2011, Lake et al., 2017]. Figure 1 illustrates this identification problem in a low-stakes setting. A researcher observes a carnival coin-flipping machine and, after seeing two heads, must estimate the probability that the next flip will be heads. A judgmental forecast may incorporate contextual knowledge (e.g., whether the game is rigged) to infer the process generating the observed outcomes.

FORECASTING WITH A WORLD STRUCTURE

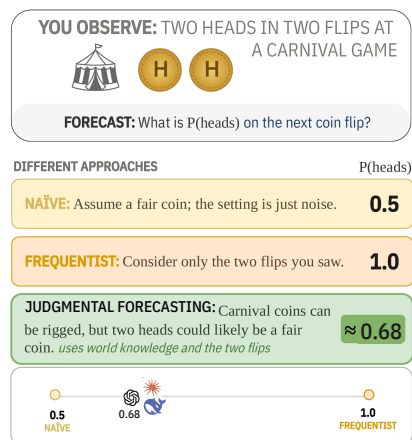


Figure 1: **Motivation.** Judgmental forecasts combine observations with contextual knowledge about the process that generated them. Frontier models, asked to forecast this question, answer with $P(H) = 0.68$ (see App. A).

¹ 📄 [Study dataset and artifacts](#) 🗄️ [Code repository](#) 🔗 [Links anonymized for review.](#)

Language Models (LMs) now approach crowd accuracy in forecasting tasks [Halawi et al., 2024, Schoenegger et al., 2024], raising a deeper question about what underlies their predictive accuracy. Prior work has shown that in some settings, models recover latent structure that supports their prediction [Li et al., 2023], while in others, they predict accurately despite learning relationships that fail to generalize beyond their observed setting [Vafa et al., 2025]. This distinction is particularly consequential for judgmental forecasting, where decision-makers often care not only about what will happen, but how the forecast should change under alternative evidence or events. Such counterfactual questions require models to capture predictive relationships that generalize across changes in context, rather than merely reproduce accurate probabilities in the observed setting.

Present work. We study whether LMs exhibit inductive reasoning in forecasting through progressively stronger behavioral experiments spanning real-world prediction markets and controlled synthetic environments. First, we test whether LMs react selectively to evidence by studying whether they distinguish outcome-relevant evidence from topically related but orthogonal information (Exp. 1). Second, we test whether LMs recover predictive relationships from context by varying contextual information while holding numerical evidence fixed in a controlled synthetic environment (Exp. 2). Finally, we test whether LMs transfer what they learn to new settings (Exp. 3 and Exp. 4) by fine-tuning models on forecasting tasks and evaluating transfer to unseen synthetic structures and held-out historical prediction markets.

Results. Across four experiments, LMs show behavioral signatures of inductive forecasting. All nine systems update $1.9\text{--}7.5\times$ more to outcome-relevant than topical evidence (Exp. 1). In Coin City, our controlled task, forecasts track context-selected relationships ($r = .52\text{--}.70$) and revise them as direct evidence arrives; larger models recover episode-local conventions that affect forecasts under intervention (Exp. 2). Episode-matched training improves joint domain-and-mechanism transfer beyond an answer-matched shuffled control at all three tested Qwen3 sizes (8B MAE 4.46 vs. 4.71; Exp. 3). On 318 held-out markets, proper-score training improves every tested base; several trained checkpoints reach crowd performance (Exp. 4). Together, LMs select, revise, and transfer predictive relationships, and training improves forecasts on unseen events.

2 Related work: why accuracy is not enough

Judgmental forecasting. Strong human forecasters combine base rates, decomposition, diverse evidence, and repeated revision [Tetlock, 2005, Mellers et al., 2014, Tetlock and Gardner, 2016]. LM benchmarks increasingly use time-indexed or leakage-controlled questions, and retrieval or ensembling can bring systems near crowd accuracy [Zou et al., 2022, Karger et al., 2025, Halawi et al., 2024, Schoenegger et al., 2024]. These results establish what probabilities models can produce. They do not tell us what would make those probabilities change.

Induction and relational reasoning. Transformers can infer functions from examples, and work on analogy and world models asks whether learned representations capture relations or dynamics [Garg et al., 2022, Chan et al., 2022, Webb et al., 2023, Yasunaga et al., 2024, Ha and Schmidhuber, 2018, LeCun, 2022, Li et al., 2023]. At the same time, shortcut research shows that surface success can fail under structural change [Geirhos et al., 2020, Mitchell, 2023, Vafa et al., 2025]. A forecast can therefore be right for the wrong reason: two models may report the same number while relying on different populations, evidence, or pathways.

Social agents and conditional validity. Generative agents can reproduce plausible individual and collective behavior [Park et al., 2023, Aher et al., 2023, Argyle et al., 2023], yet matching a population average is not the same as preserving the relationships within that population [Bisbee et al., 2024]. For systems intended to inform decisions, we care about both the *probability reported* and the *relationship that makes the evidence predictive*. Our experiments therefore move through increasingly stringent tests: relevance and direction, contextual relationship selection, revision from local evidence, induction of a novel convention, and transfer across worlds. Each step rules out a weaker account of the same forecast without assuming a general internal world model.

3 Do forecasts change for the right reason?

Forecasting is not only about assigning a probability; it is also about knowing which evidence should change it. A model can still move for the wrong reason: a headline may share a market’s topic

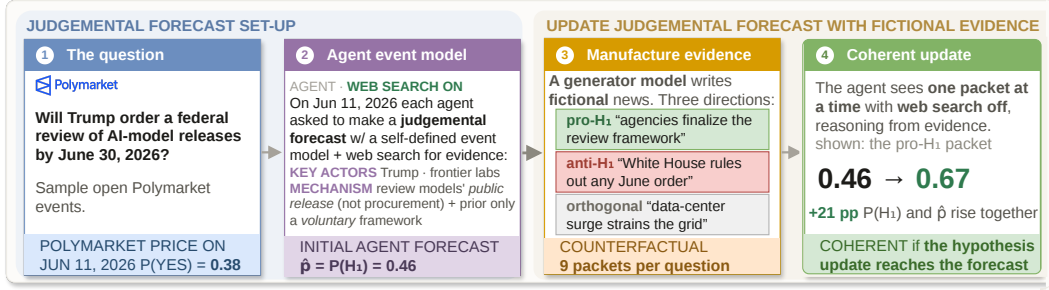


Figure 2: **When should a forecast change?** An agent forecasts an unresolved question, then receives one fictional update with search off. Some updates support the outcome, some oppose it, and some concern the same topic but do not bear on the outcome. The labels remain hidden; we ask whether the model moves when the news matters and whether it moves the right way.

without changing the event’s likelihood. We therefore begin by asking whether models distinguish evidence that bears on an outcome from topical resemblance and move in the direction it warrants. Exp. 1 tests this on unresolved markets. Exp. 2 raises the bar: when populations respond differently, can context select the relationship that governs a new case, and can local evidence revise it?

3.1 Exp. 1: can the model separate evidence from topical chatter?

Design. We test this before events resolve using 100 unresolved binary Polymarket questions. Each model first forecasts with web search, then sees one fictional update with search turned off. Some updates support YES, some support NO, and others mention the same topic without bearing on the outcome. The model does not see these labels. Its task is to decide whether to move and, if so, in which direction. A word-only classifier evaluated on separate markets shows what surface cues can accomplish; the full sampling design is in App. B. Figure 2 shows the sequence. Across nine systems, this yields 30,094 valid forecast updates.

Finding: models first decide whether an update is warranted. Figure 3 shows that every system leaves its prior unchanged more often for topical controls. When models update, they move $1.9\text{--}7.5\times$ farther for directional evidence across all nine systems. They also move in the implied direction: directional consistency spans .887–.990, compared with .688 for the word-only classifier. Human validation and uncertainty estimates are in App. B.3.4.

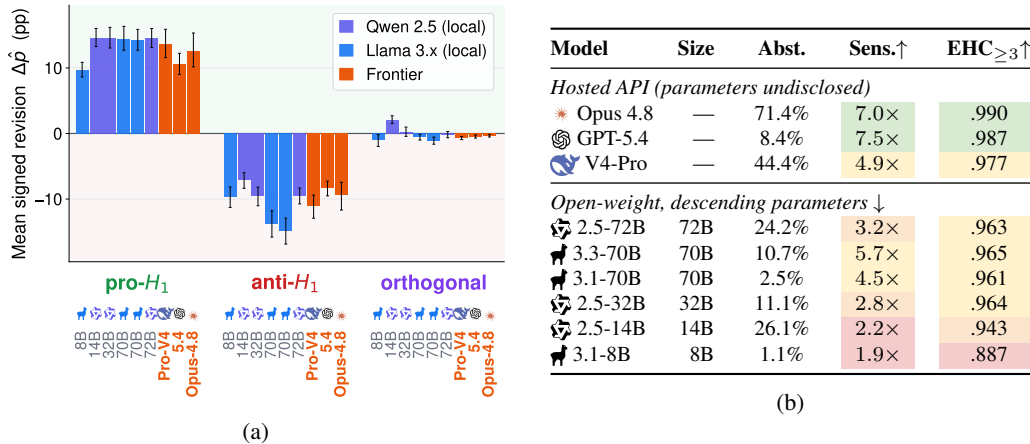


Figure 3: **Models move much more for evidence than for topical chatter, and usually move in the implied direction.** (a) Average revision after supporting, contrary, and topical updates. (b) The table summarizes how often models leave topical updates alone, how much farther they move for directional evidence, and how often a substantial move has the right sign. Intervals and robustness checks are in App. B.3.4.

Shared prompt in every arm

Each row is a separate polling case rather than a time series. Positive net news favors the candidate and negative net news harms the candidate. Poll change is the ending poll minus the starting poll. Within a city, typical responsiveness to net news is stable across cases, although individual end polls remain noisy. Rows with the same Pair label have equal-sized news in opposite directions.

Query: A new City C case begins with a poll of 53.9 and has net news +5. What poll do you predict at the end of this new case? Think carefully about the expected poll, but do not provide step-by-step working. Respond with one JSON object only: {"rationale": "one short sentence", "predicted_poll": <number from 0 to 100>}

Illustrated arm: References + target context

CITY A		CITY B		CITY C · TARGET CONTEXT	
Background: Residents generally encounter campaign developments through national news coverage , where campaign stories receive sustained attention and tend to produce larger immediate poll movements.		Background: Residents generally encounter campaign developments through local news coverage , where campaign stories compete with local issues and tend to produce smaller immediate poll movements.		Background: City C residents generally encounter campaign developments through national news coverage .	
Pair	Start	News	Delta	End	
1	45.4	-8	-6.5	38.9	
1	45.4	+8	+12.2	57.6	
2	55.1	-6	+0.9	56.0	
2	55.1	+6	+11.4	66.5	

Pair	Start	News	Delta	End	
1	35.0	-10	-5.6	29.4	
1	35.0	+10	+9.3	44.3	
2	51.8	-10	-5.5	46.3	
2	51.8	+10	+7.2	59.0	

Pair	Start	News	Delta	End	
1	57.8	+8	+9.4	67.2	
1	57.8	-8	-1.8	56.0	
2	47.4	+10	+8.1	55.5	
2	47.4	-10	-10.6	36.8	

Figure 4: **Coin City turns contextual reasoning into a paired comparison.** Every arm shares the query and numerical cases. The context arm adds a description connecting City C to one displayed media environment; the opposite-context arm changes only that sentence. Target cases then accumulate, allowing local evidence to confirm or correct the contextual choice.

109 **What Exp. 1 establishes.** Across the tested systems, forecasts distinguish outcome-relevant evidence
110 from topical chatter and move in the direction implied by the evidence. This does not yet show
111 whether the model knows whose response pattern makes that evidence predictive. Coin City exposes
112 that hidden choice.

113 3.2 Exp. 2: let context select a relationship, then challenge it

114 **Coin City makes the process ambiguity observable.** Figure 1 began with a familiar ambiguity:
115 the same two heads can imply different forecasts depending on what process the forecaster believes
116 generated them. Coin City turns that intuition into a controlled test. The same campaign story moves
117 polls strongly in City A but weakly in City B; context identifies the better analogue for target City C.
118 If the model uses that link, changing only the context should change its forecast, while direct City
119 C cases should eventually override a misleading cue. Each episode supplies noisy A/B examples, a
120 one-sentence link, and zero to four target cases. Each city’s poll response follows

$$\Delta p_{ij} = g_j x_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, 5^2),$$

121 where x_{ij} is signed campaign news and g_j is that city’s stable sensitivity. The prompt never states
122 g_j ; the model has to infer it from examples. Figure 4 shows the shared query, reference cities, and
123 target-city context in one illustrated arm.

124 We compare five versions of the same problem: target cases alone; references without a link; the
125 correct link; the opposite link; and episode-varying KIV/ZOR labels whose meaning must be learned
126 from the references. Forecast error tells us whether context helps. The model’s implied response to
127 campaign news tells us which city it is treating as the analogue. App. C.5 gives the full model roster
128 and sampling design.

129 **Finding: the forecast follows the selected analogue.** Figure 5 shows that correct context lowers
130 error before any City C cases for every system; changing only that sentence reverses the selected
131 regime. The implied response correlates .51–.86 with the named reference after controlling for the
132 true regime, while the other reference is near zero. As City C cases accumulate, both the contextual
133 advantage and inversion harm shrink: forecasts shift from context toward diagnostic local behavior.

134 3.2.1 Infer a convention with no stable semantics

135 The context result still admits a weaker explanation: familiar descriptions such as “national news”
136 might cue a response without the model inferring the relationship. KIV and ZOR remove that
137 route. Their meanings change in every episode, so the model can use them only by inferring from
138 the reference cases which label names which response pattern. Twelve of fifteen systems recover

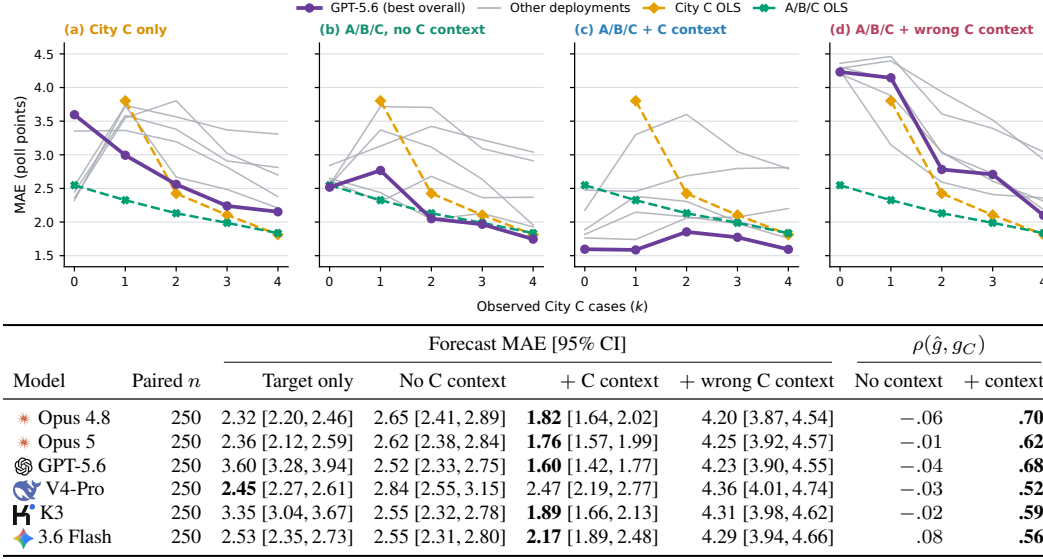


Figure 5: **Context improves forecasts before target cases and helps assign the response pattern across six systems.** Giving the correct analogue lowers error before City C has a history; naming the wrong analogue raises it. As direct City C cases accumulate, forecasts rely less on either contextual cue. Full results by model are in App. C.5.

the convention before any target cases. When the two populations are made harder to distinguish, performance falls but the ordering of models remains similar, as an induction account predicts (App. C.8).

This behavior shows that models can use a newly induced convention, but not whether they represent that convention internally or whether it helps produce the forecast.

Figure 6 asks the first question. Across seven open checkpoints, the four larger models carry the episode’s inferred KIV/ZOR meaning in their internal state, while the three smaller models struggle. That state follows the contextual relationship before City C has a history, then yields as direct City C evidence identifies the local response. The representation therefore tracks which source of information should govern the forecast, rather than a fixed meaning of either label.

Decoding is still correlational. Figure 7 asks the causal question by swapping the label-state representation between otherwise matched Qwen3-14B prompts. Before City C provides its own history, the intervention changes both the analogue the model names and its numerical forecast. After four City C cases, the same swap has no reliable effect. Across episodes, the shift is $+.333$ before target cases and $-.035$ after four, the latter inside the random-swap range. This timing is the point: the induced convention helps produce the forecast when context is needed, then gives way when direct evidence reveals the relationship (App. C.7.7).

What Exp. 2 establishes. Models use context to select which population’s response governs a forecast and revise that choice when target evidence arrives. Twelve of fifteen systems infer a convention with no stable semantics; larger open models encode that episode-local mapping, and changing it causally changes Qwen3-14B’s forecast when local evidence is absent.

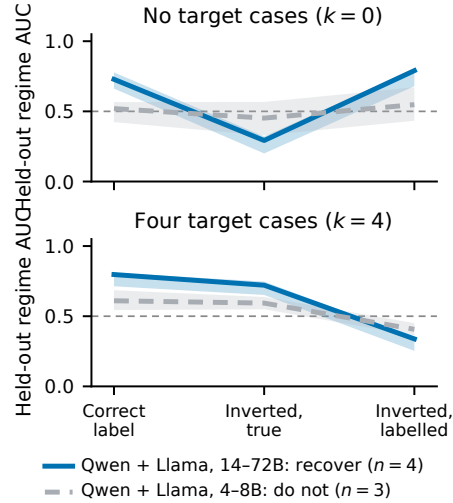


Figure 6: **Larger models infer an episode-local convention, then yield to direct evidence.** On 88 new episodes, four larger Qwen and Llama checkpoints (14–72B) recover the changing KIV/ZOR mapping; three smaller checkpoints (4–8B) struggle.

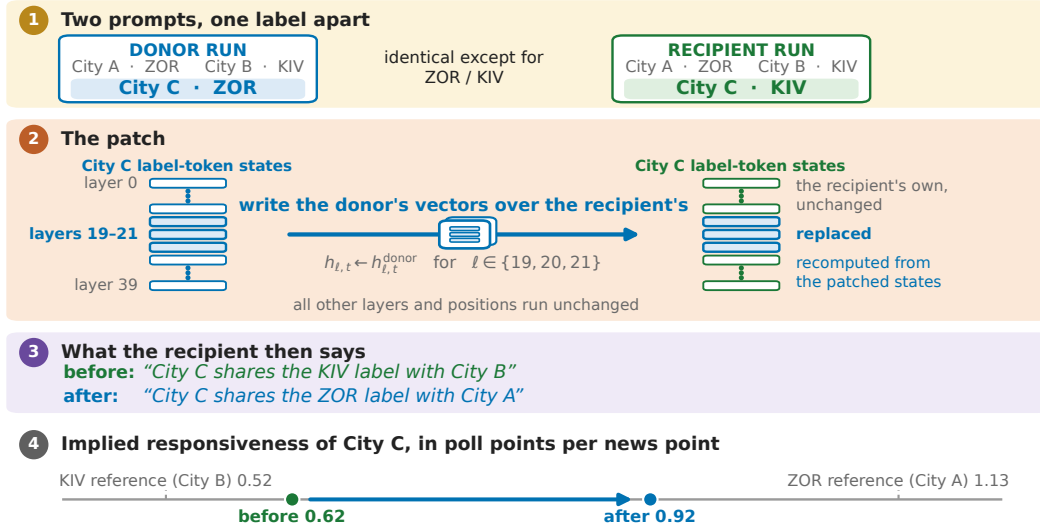


Figure 7: **Changing the induced convention changes the forecast before target evidence arrives, but not after.** In the illustrated pair, swapping the internal representation moves the forecast from 51.3 to 54.0 and changes which reference city the model names; reversing the swap reverses the movement. Once four target cases reveal City C’s own behavior, arbitrary swaps are no more influential than the intervention. Full controls and across-episode results are in Table 34.

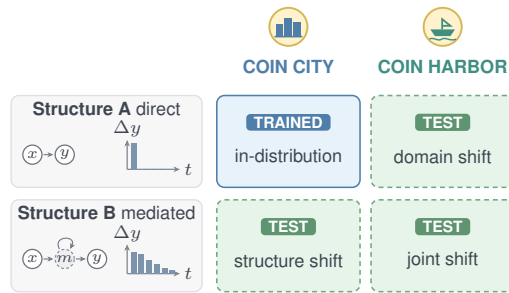
170 Section 4 asks whether training carries this form of relationship use into a new world.

171 4 Can relationship use survive a change of world?

172 An improvement after finetuning is ambiguous: the model may have learned how evidence produced
 173 the outcome, or only the usual format and answers. Section 3 tested relationship use within a prompt.
 174 Section 4 asks whether training carries that relationship into a new world, then whether it helps on
 175 unseen real events.

176 4.1 Exp. 3: transfer under domain and mechanism shift

177 **Why matching matters.** A training gain could re-
 178 flect the prompt format or answer distribution rather
 179 than learning which evidence produces which out-
 180 come. The two arms therefore use byte-identical
 181 prompts, the same target-answer multiset, and the
 182 same training budget; only the case-answer pairing
 183 differs. Shuffled-pair training captures generic task
 184 learning. Any matched-over-shuffled gain comes
 185 from preserving the evidence-outcome link; sur-
 186 vival under a new domain and response mechanism
 187 supports reuse of that link beyond Coin City.



188 We test that survival by changing both the setting
 189 and the way effects unfold. Coin City campaign
 190 news and candidate support become Coin Harbor
 191 shipping bulletins and a cargo-flow index over tide
 192 cycles, with different populations, noise, units, and time scale. A persistent two-step response replaces
 193 the direct one-step response. The 2×2 design in Figure 8 tests each change separately and jointly;
 194 full details are in App. F.

195 **Finding: episode matching adds a transfer gain.** Figure 9 shows the same qualitative result across
 196 the transfer grid and tested Qwen sizes: preserving the case-answer pairing adds value beyond
 197 exposure to the same prompts and answer distribution. The gain narrows with scale, and reversing

Figure 8: **Transfer grid.** Training uses shading; all four are evaluated. Columns vary domain; rows vary response mechanism.

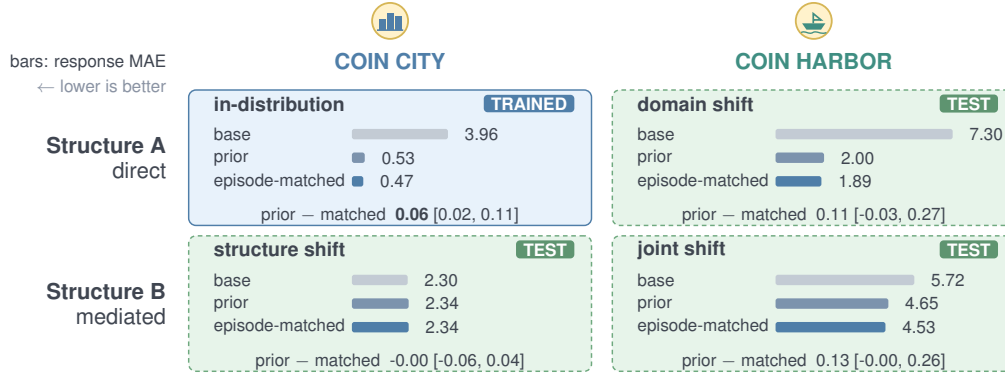


Figure 9: **Qwen3-8B Coin City transfer at round 300 under five-draw stochastic decoding, eight training seeds.** Bars show task-averaged response MAE over four evidence depths; trained arms average eight seeds. Each cell prints the paired seed-first prior-minus-matched contrast, bold when its 95% interval excludes zero. At three seeds three of the four contrasts excluded zero; at eight only the in-distribution cell does. The structure-shift row is a control rather than a transfer test: training contains no structural variation, so the trained policy emits no delayed response anywhere (Section 4.1).

the pathway description hurts, so the evidence supports reusable relationship use within the tested Qwen family rather than a format-learning effect or scale invariance. Full cell estimates, intervals, seed-level patterns, the cross-size synthesis, and the Llama comparisons are in App. D.1.5.

What Exp. 3 establishes. For the tested Qwen models, episode-matched training adds a transfer gain beyond learning the task’s format and answer distribution, including when both the domain and response mechanism change.

4.2 Exp. 4: external grounding in unseen historical markets

Experiment 3 makes transfer legible by deliberately changing a known domain and a known response mechanism. Historical markets do not come with those factors labeled, and we cannot enumerate which mechanisms differ between training and evaluation. Experiment 4 asks whether the same principle survives that loss of control: train on information paired with its realized outcome, then forecast family-disjoint events whose governing relationships are unknown. We train on resolved markets using only information available before each outcome, then evaluate 318 unseen questions from different event families. Error is the squared distance between the forecast probability and what happened, so lower Brier scores are better (App. E).

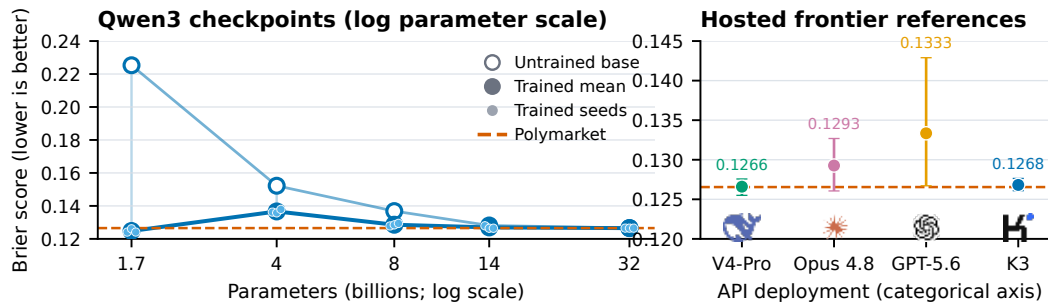


Figure 10: **Training improves the tested bases, but scale alone does not guarantee a better trained forecaster.** Across Qwen3 checkpoints from 1.7B to 32B, several trained models reach crowd-level performance, while the gain narrows at the strongest base. Local model size is shown on a log parameter axis; hosted systems at right are reference points. The Llama replication is in App. Figure 16.

Figure 10 first compares each trained model with its base, then tests whether the gain persists as the base becomes stronger. Training lowers error at every tested size, but Qwen performance does not improve steadily and the gap shrinks from 1.7B to 32B. At 32B, trained mean Brier is .1265 versus .1266 for the base, the smallest observed gain; the trained-minus-base interval $[-.00045, .00029]$ includes zero. A Llama 3B/8B replication shows the same qualitative convergence pattern (App. Figure 16). Several trained checkpoints reach contemporaneous Polymarket performance. Hosted systems provide context, not additional points on the local scaling curve.

***What Exp. 4 establishes.** The central idea from the synthetic study extends to naturally heterogeneous forecasting: pairing evidence with its outcome lowers mean error on held-out event families even when domains and causal mechanisms are neither labeled nor deliberately swapped. This makes the Experiment 3 result a candidate general training principle rather than a Coin City-specific effect, although its incremental benefit narrows as the Qwen base strengthens and is uncertain at 32B.*

Experiments 3 and 4 answer complementary parts of the same claim. The synthetic study identifies why matching evidence to outcomes matters under controlled shifts; the historical study asks whether that principle has predictive value when the world supplies the shifts and their structure is unknown. Together they connect relationship use to external accuracy.

5 Discussion: why inductive reasoning matters

Forecasting is our testbed for inductive reasoning, not the boundary of the claim. It makes otherwise hidden choices observable: which evidence matters, which population offers the relevant comparison, when local behavior should override context, and whether a relationship survives a new world. Each choice implies a different numerical forecast. The same ability matters whenever a social agent must generalize from sparse evidence rather than repeat a familiar pattern.

In practice, a forecast is rarely the last question. Users need to know which evidence should change a decision, whether the conclusion survives in a different population, and what observation should trigger revision. A model that reaches the right number through a topical cue or a familiar base rate may still be unable to support those judgments. Average accuracy can hide that weakness precisely because most evaluation cases resemble the past. It becomes costly when a decision concerns a new group, an unusual institution, or a policy that changes the usual pathway.

We designed the experiments around alternatives left open by the preceding study. The market study tests whether forecasts ignore topical chatter and follow evidence up or down, but leaves open a verbal shortcut. Coin City makes the hidden comparison visible; conflicting local cases test revision, arbitrary labels remove familiar semantics, and an intervention connects the induced representation to behavior. Transfer to a new domain and response pathway then tests whether the result is specific to one prompt, while historical markets test accuracy outside training. No single step is sufficient; together, these successively stronger contrasts narrow the explanations for success.

These distinctions extend beyond forecasting. A persuasion model should respond to both a message and who delivers it. A coordination model should distinguish shared payoffs from what participants know about one another. A policy model should distinguish similar language from the institutional pathway through which an effect occurs. In synthetic populations, these paired changes can reveal whether conditional relationships survive even when aggregate averages look human-like [Park et al., 2023, Aher et al., 2023, Argyle et al., 2023, Bisbee et al., 2024]. For social agents in particular, the relationship is often the object of interest: who knows what, which group treats a message as credible, or which institution transmits an action into an outcome. Matching a population average while missing those dependencies can produce a convincing static simulation and a misleading response to intervention.

For evaluation, this implies measuring both outcomes and reasons for change. Hold surface content fixed while changing the relevant analogue; provide local evidence that contradicts context; replace meaningful cues with a new convention; and test the relationship again after the setting changes. Headline accuracy and parameter count cannot substitute for these checks because the relevant capability depends on the population, pathway, and decision at hand.

263 6 Limitations: what these studies do and do not establish

264 The evidence supports a specific claim: models use context-dependent predictive relationships in
265 these tasks. It does not establish human-like understanding or a general internal world model. The
266 market packets are controlled pieces of text, Coin City offers two simple response patterns, and the
267 transfer study spans two constructed worlds. Real settings contain strategic interaction, unreliable
268 cues, changing variables, and feedback over longer horizons.

269 The internal result is narrower still. Decoding identifies a representation associated with the selected
270 convention, and one intervention shows that this representation affects one checkpoint’s forecast.
271 That is stronger than a correlation alone, but not a claim about every model or every form of reasoning.
272 Likewise, the historical study covers one market cohort and training recipe. Additional architectures,
273 later cohorts, interactive settings, and deliberately misleading evidence would test how far the pattern
274 extends.

275 Performance is heterogeneous. Smaller checkpoints often struggle when the KIV/ZOR meaning must
276 be induced anew, and historical-market gains narrow at the strongest Qwen base. Because the scale
277 comparison uses one training recipe across model sizes, its nonmonotonic shape should not be read
278 as a law of capacity: larger models may need different learning rates, schedules, or amounts of data.
279 The sequence is useful partly because it shows where performance changes rather than collapsing
280 inductive reasoning into one yes-or-no score.

281 7 Ethics, artifacts, and conclusion

282 The studies use public market data, model outputs, synthetic worlds, and an unpaid adult review of
283 synthetic materials. No identifying or sensitive information was collected. Counterfactual packets
284 were evaluated offline, never posted to markets, and never used for trading. We report aggregate
285 behavior rather than recommend a deployment or trading strategy.

286 An anonymized artifact contains prompts, fixed datasets, saved model responses, analysis code, and
287 figure builders. The appendix records the implementation and statistical details omitted from the
288 main narrative.

289 Forecast accuracy is a starting point, not a sufficient test of inductive reasoning. Stronger tests ask
290 whether a model notices relevant evidence, moves in the implied direction, chooses and revises
291 comparisons, infers new conventions, and preserves relationships under change. The larger tested
292 models pass these contrasts, subject to the limits above. For social agents, these experiments test both
293 real-world accuracy and relationship-sensitive updating without presuming a general theory of the
294 world.

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401 Appendices

402 The supplement is organized by experiment. A descriptive coin-flip elicitation precedes the Exper-
 403 iment 1 evidence-response details. Experiment 2 documents contextual relationship selection and
 404 revision, including a linear-probe representation analysis; Experiments 3–4 document controlled and
 405 historical transfer, replication, structure-selection diagnostics, supervised learnability, and mechanism
 406 evidence use. Shared training details and the artifact map close the appendix.

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A Descriptive coin-flip forecast elicitation

This appendix reports the forecasts used to position the model markers in Figure 1; it is not one of the four experiments. We queried GPT-5.4, Claude Opus 4.8, and DeepSeek V4-Pro eight times each (24 calls total) at temperature 0.7 and a 600-token output ceiling. All responses parsed. The figure places each model’s logo at its eight-sample mean; the green $\approx .68$ annotation rounds the pooled mean of .679.

The exact user prompt was:

```
You are at a carnival game. A coin you have never seen before is flipped
twice in front of you, and it comes up heads both times. These two flips
are everything you know about this coin.

What is the probability that the next flip comes up heads?

Think about it intuitively, the way you actually would in the
moment -- do NOT use any formula, rule, or calculation, and do not
work out any numbers in your reasoning. Just reason in plain words
about what you'd expect and why. Then END your reply with your gut
answer as a JSON object on its own line:
{"rationale": "one short sentence, no math",
 "p_heads": <number between 0 and 1>}
```

Table 1: Forecasts underlying the three model markers in Figure 1. Each row summarizes eight independent temperature-0.7 calls.

Model	Mean	Median	Range
GPT-5.4	.644	.670	[.60, .67]
Claude Opus 4.8	.681	.675	[.65, .75]
DeepSeek V4-Pro	.712	.750	[.60, .75]
Pooled ($n = 24$)	.679	.670	[.60, .75]

What this probe does and does not separate. It has one condition. The 0.5 and 1.0 endpoints in Figure 1 are reference policies rather than measurements: no fair-coin arm and no no-carnival arm were collected, so the probe cannot attribute the departure from 0.5 to the carnival framing. It also does not separate a prior about carnival coins from ordinary sequential updating on an unknown bias. Laplace’s rule of succession gives $3/4$ after two heads from a coin with an unknown bias, using no world knowledge at all, and the observed .679 sits below that rather than above it. Finally, the instruction not to use any formula or calculation discourages explicitly deriving either the 0.5 or 0.75 reference policy, changing the elicitation mode and complicating comparison to both. We report the probe as a descriptive illustration of the distinction the paper draws, not as evidence for it; the four experiments carry the argument. `eval/run_coin_probe.py` implements three further arms—a stipulated fair coin, an unfamiliar coin with no carnival, and the carnival prompt without the no-calculation instruction—which would supply the missing contrast.

The prompt, temperature, output limit, parsed probabilities, rationales, and raw completions are stored in `exp2_simulated_worlds/biased_news/data/coin_probe/coin_probe_2026-06-30.jsonl`; the collection and parsing code is `exp2_simulated_worlds/biased_news/eval/run_coin_probe.py`. The prompt names neither a fair nor a rigged coin. It requests qualitative reasoning while retaining a numeric forecast for the plotted quantity.

B Experiment 1: prospective forecast updating

Experiment 1 tests whether forecast revisions follow evidence direction and whether directional packets produce larger changes than orthogonal packets. All reported counts refer to parseable records in the frozen consistency report unless stated otherwise.

The main comparison reports the nine deployments with scale-valid archived revisions. This appendix retains Qwen 2.5-7B for directional-compliance diagnostics and documents why its magnitude-based statistics are excluded.

B.1 Study design, sample, and agent roster

Protocol summary. Experiment 1 crosses ten forecasting agents with a frozen set of unresolved binary Polymarket questions. The study has four stages: (1) freeze the market sample; (2) obtain one or more initial forecasts from each agent; (3) generate nine synthetic evidence packets for each of the 100 core markets (three pro- H_1 , three anti- H_1 , and three outcome-orthogonal); and (4) ask the same agent to revise each parseable initial forecast after receiving one packet at a time. The unit of analysis is an *agent* $\times$ *market* $\times$ *initial run* $\times$ *counterfactual packet* update. Requested repeat counts were five initial runs for hosted agents and three for local agents; realized valid counts are uneven because of provider failures, parsing failures, and targeted top-ups. Section B.3 specifies how those records are normalized and aggregated.

B.1.1 Sampling frame and market selection

The primary sampling frame is a Polymarket API snapshot taken on June 9, 2026. Selection was deterministic given that snapshot. We retained markets that were binary, active and unresolved, scheduled to resolve 14–180 days after sampling, priced strictly between 0.10 and 0.90 on the YES contract, and supported by at least \$1,000 in USD volume. A keyword-based relevance filter retained politics, government, elections, economics, technology, and geopolitics while excluding sports, cryptocurrency, entertainment, and other out-of-scope topics. When several contracts belonged to the same event, we retained the highest-volume contract.

The recorded filter funnel makes the sampling loss explicit. The snapshot contained 60,989 binary records; 37,386 were active, 8,639 passed the topic filter, 3,570 passed the resolution-window filter, 1,403 passed the price filter, 839 passed the volume filter, and event-level deduplication left 436 candidates. A diversity pass selected the 100-market core using a cosine-similarity threshold of 0.25 and a maximum of 16 markets per topical bucket. The frozen core contains 16 global-election, 16 other, 15 Iran-nuclear, 12 U.S.-federal-policy, 10 AI/technology, 9 U.S. state-election, 7 Russia–Ukraine, 5 Brazil, 5 U.S.-midterm, 3 macroeconomic, and 2 China–Taiwan markets.

Counterfactual generation and Stage 3 updating target this 100-market core for every model, but per-model coverage of it is not uniform. Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast, and Claude Opus 4.8 has usable updates on all three packet arms in 73 markets; Section B.5.3 reports the coverage table and the resulting robustness checks.

Table 2: Representative markets from the frozen 100-market core. Snapshot mid is the stored YES midpoint on June 9, 2026 rather than an eventual outcome.

Question	Topic	Snapshot mid
Will Renan Santos win the 2026 Brazilian presidential election?	Brazil	0.17
US–Iran nuclear deal by June 30?	Middle East	0.20
US $\times$ Iran permanent peace deal by July 31, 2026?	Iran	0.32
Will a new Gemini flagship be released by June 30, 2026?	AI/technology	0.89

B.1.2 Agent roster and execution modes

The roster comprises three hosted API systems and seven open-weight local systems (Table 3). The hosted models ran through Azure AI Foundry and used its native search-supported Responses workflow [Microsoft, 2026]. Local models were served by Ollama [Ollama, 2026] on cluster GPUs; their research turn was supplied by Tavily search. All agents received the same substantive three-step contract: build an event model, gather current evidence, and emit a structured forecast. Provider-specific search orchestration and model runtimes were not identical.

Table 3: Experiment 1 agent roster and requested initial-forecast repeats. Requested k is the collection target rather than an analysis denominator; exact realized coverage appears in Table 6.

Model	Runtime	Research search	Nominal params	Requested k
GPT-5.4	Azure Foundry	Foundry native	—	5
Claude Opus 4.8	Azure Foundry	Foundry native	—	5
DeepSeek V4-Pro	Azure Foundry	Foundry native	—	5
Qwen 2.5-7B	Ollama	Tavily	7B	3
Qwen 2.5-14B	Ollama	Tavily	14B	3
Qwen 2.5-32B	Ollama	Tavily	32B	3
Qwen 2.5-72B	Ollama	Tavily	72B	3
Llama 3.1-8B	Ollama	Tavily	8B	3
Llama 3.1-70B	Ollama	Tavily	70B	3
Llama 3.3-70B	Ollama	Tavily	70B	3

For local models, reasoning calls used the frozen configuration temperature of 0.1 and the final JSON-formatting call used temperature 0.0. Search queries, tool responses, token counts, parse errors, and provider errors were retained with each run. Some smaller models returned probabilities on a 0–100 rather than 0–1 scale. The initial-forecast parser converted those values, but the frozen Stage 3 data retain a mixed-scale Qwen 2.5-7B subset documented in Section B.3.1.

B.2 Forecast, intervention, and validation protocol

B.2.1 Initial forecast elicitation

Design rationale. The elicitation asks each agent to restate the resolution criterion, construct explicit YES and NO hypotheses, identify a status-quo trajectory, research current evidence, and map its final YES probability to the posterior assigned to H_1 . Question restatement and status-quo anchoring follow prior forecasting-prompt evidence [Halawi et al., 2024, Wilson, 2025].

Executed three-turn sequence. Every frozen initial forecast followed the same substantive sequence, with provider-specific tool plumbing:

1. **Event model.** The agent received the question, full resolution description, days to resolution, and category. It restated what would count as YES versus NO; identified actors, mechanisms, and latent variables; formed H_1 (YES) and H_2 (NO); and described the status-quo outcome.
2. **Current-evidence research.** The agent gathered evidence for and against each hypothesis, checked whether the required chain of events was plausible within the remaining time, and revised its assessment. Hosted systems used Azure Foundry’s native search workflow; local systems generated queries that were executed through Tavily and received the returned evidence in context.
3. **Structured forecast.** A final call converted the accumulated analysis to the common JSON contract below. A run entered downstream processing only when this object was parseable.

The research turn included this calibration instruction:

Given the days remaining, what chain of events is required for H1, and is that pace of change plausible? Give extra weight to the status quo; depart substantially only when concrete directional evidence warrants it.

The common output contract was:

```
{
  "event_model": {
    "key_actors": [str],
    "key_mechanisms": [str],
    "latent_variables": [str]
  },
  "hypotheses": [
    {"id": "H1", "description": str,
     "prior_probability": float, "posterior_probability": float,
```

```

556     "supporting_evidence": [str], "contradicting_evidence": [str]},
557     {"id": "H2", ...}
558 ],
559 "hypothesis_to_forecast_mapping": str,
560 "yes_prob": float,
561 "confidence": "low"|"medium"|"high",
562 "rationale": str,
563 "evidence_sources": [str]
564 }

```

565 The contract makes the intended invariant explicit: $\hat{p} = P(H_1)$. The ICS metric tests whether the
566 returned fields obey that invariant after an intervention. The substantive prompt contract was common,
567 but prompts were not byte-identical across runtimes because the hosted and local search interfaces
568 require different orchestration.

569 B.2.2 Counterfactual packet construction and validation

570 For each of the 100 core markets, GPT-5.4 generated nine synthetic evidence packets: three pro- H_1 ,
571 three anti- H_1 , and three orthogonal. Each generation call received the market question, resolution
572 description, and a canonical parseable initial event model and forecast. It produced:

- 573 • **evidence_headline**: one sentence that states the new development;
- 574 • **evidence_text**: a 2–4 sentence synthetic wire-style item;
- 575 • **mechanism_targeted**: the causal pathway by which the item should affect, or remain irrelevant
576 to, resolution;
- 577 • **direction**: pro- H_1 , anti- H_1 , or orthogonal;
- 578 • **slot_type**: one of the six roles in Table 4; and
- 579 • **plausibility_rating**: the generator’s 1–5 self-rating.

580 Within each direction, sequential exclusion required a new packet to target a different actor and
581 mechanism from earlier packets. Search was disabled so the packets remained controlled synthetic
582 premises rather than paraphrases of live reporting.

Table 4: Counterfactual slot taxonomy. Each market receives one packet of every direction–slot combination shown, for nine packets total.

Slot type	Direction	Role in the intervention
DATA	pro, anti	A measurement, price signal, statistic, or poll that shifts the likelihood of H_1 .
DECISION	pro, anti	An actor decision, official statement, policy announcement, commitment, or refusal bearing on H_1 .
583 STRUCTURAL	pro, anti	A technical, logistical, legal, or contextual change that makes H_1 more or less likely.
PROCEDURAL	orthogonal	A same-domain administrative or process update without a direct causal path to H_1 versus H_2 .
PERSONNEL	orthogonal	A staffing or leadership change involving a relevant actor but an issue unrelated to H_1 versus H_2 .
BACKDROP	orthogonal	Contextual economic, logistical, or social information that sets the scene without shifting the outcome odds.

584 **Packet-level integrity checks.** The frozen packet file contains 900/900 parseable records: 100
585 markets multiplied by nine packets, with exactly 300 pro- H_1 , 300 anti- H_1 , and 300 orthogonal
586 records. Every packet has a nonempty headline, and the logs contain zero search calls. All 900
587 generator-assigned plausibility ratings equal 4/5. That field documents the generator’s compliance
588 claim but provides neither variation nor independent validation; the separate human materials review
589 is reported below. These checks verify structural completeness; they do not establish that directional
590 strength or difficulty is perfectly matched across packets.

591 As an illustration, for the Hormuz market the pro- H_1 headline *Oman says Iran agrees daily protected*
592 *windows for commercial Hormuz transits* elicited the largest single update reported in the viewer:

Claude Opus 4.8 revised from 10% to 45% ($\Delta\hat{p} = +35$ pp). The orthogonal IMO incident-reporting item produced negligible movement.

B.2.3 Stage 3 update and pairing

Each packet was presented separately with the agent’s own parseable initial structured forecast. The agent was told to reason only from that forecast and the packet and to return the same structured forecast with revised H_1 , H_2 , and YES probabilities. Search was disabled.

One deployment received an extra instruction. The Stage 3 template was not common across the roster. GPT-5.4 alone was run with the runner’s novelty option, which appends a rule asking for a `novelty_assessment`: *a single sentence stating whether this evidence was likely already reflected in your initial forecast as background knowledge (suggesting a small update) or is genuinely new and surprising (warranting a larger revision)*. The frozen records show the asymmetry exactly: 3,838 of 3,921 GPT-5.4 updates carry the field and no other deployment carries it in any record. The instruction tells a model to modulate revision size by perceived novelty, which is close to the construct Section B.5.3 estimates, and GPT-5.4 is the deployment that behaves differently on exactly that dimension: it returns the prior unchanged on 8.4% of orthogonal packets against 71.4% and 44.4% for the other two hosted systems, writing a small revision where they abstain. We cannot separate that difference from the prompt difference, so GPT-5.4’s abstention rate is not comparable to the rest of the roster and we do not read the hosted ordering on abstention as a model difference. Its conditional magnitude, directional correctness, and AUC use the same scoring as every other deployment.

For an agent–market cell with k parseable initial runs, the intended crossed design contains $9k$ update records. Each update is conditionally independent in prompt context: it contains one initial run and one packet, never another packet or another agent’s answer. The novelty field is not a reported endpoint and should not be viewed as a validated correction.

Provider interruptions, parse failures, and subsequent top-ups make realized cells uneven. We therefore report exact valid record and market denominators by model in Table 6 rather than treating requested k or the theoretical $9k$ as observed sample sizes.

B.3 Estimands, record processing, and aggregation

B.3.1 Record construction, deduplication, and missingness

The frozen analysis artifact is `exp1_prospective/data/results/consistency_report_2026-06-17.json`; tables in this paper use its stored denominators rather than a fresh scan of the evolving collection directories. During report construction, date suffixes are stripped from task identifiers so daily collection files join to the same market. Counterfactuals are deduplicated by `cf_id`; updates are deduplicated by `update_id`, with a parseable record preferred over an error record when both exist. Files are read newest first for the initial-forecast and market-price lookup. Malformed JSON lines are skipped. Missing values are handled per estimand: a record may therefore contribute to ICS but not EHC, or vice versa, depending on which fields parse.

Stage 3 update fields were evaluated on their stored scale in the frozen report. For Qwen 2.5-7B, 207 of 8,636 deduplicated records store `updated_yes_prob` above 1.5; no other model does. Its directional sign metrics remain usable, but its absolute revision magnitudes and sensitivity ratio are scale-contaminated and are omitted from the paper tables.

B.3.2 Consistency estimands

For update record i , let $d_i \in \{+1, -1\}$ denote the intended direction (+1 for pro- H_1 , −1 for anti- H_1). Orthogonal packets do not have an intended sign and are excluded from EHC and HFC. The 3 pp movement threshold excludes cases in which a model effectively leaves the forecast unchanged.

Evidence–hypothesis consistency (EHC). Let $\Delta p(H_1)_i = p(H_1)_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with $|\Delta p(H_1)_i| \geq 0.03$,

$$\text{EHC}_i = \mathbf{1}[\Delta p(H_1)_i d_i > 0]. \quad (1)$$

EHC asks whether the updated hypothesis field moves in the direction implied by the packet; it does not score the size of that movement.

642 **Hypothesis–forecast consistency (HFC).** Let $\Delta\hat{p}_i = \hat{p}_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with
643 $|\Delta\hat{p}_i| \geq 0.03$,

$$\text{HFC}_i = \mathbf{1}[\Delta\hat{p}_i d_i > 0]. \quad (2)$$

644 HFC asks the same directional question of the headline YES probability. Because the output contract
645 requires this probability to equal $P(H_1)$, we retain HFC in the appendix as a compliance diagnostic
646 rather than as a separate main-text result.

647 **Internal coherence score (ICS).** ICS is defined whenever both updated fields parse and is uncondi-
648 tional on movement:

$$\text{ICS}_i = \mathbf{1}[|\hat{p}_i^{\text{post}} - p(H_1)_i^{\text{post}}| < 0.02]. \quad (3)$$

649 The inequality is strict: a discrepancy of exactly 2 pp does not pass.

650 **Directional sensitivity.** Packet selectivity is summarized by

$$\text{Sensitivity} = \frac{|\Delta\hat{p}|_{\text{pro+anti}}}{|\Delta\hat{p}|_{\text{orthogonal}}}. \quad (4)$$

651 A value near one indicates comparable movement for directional and orthogonal packets; larger
652 values indicate preferential response to directional packets. This ratio is descriptive. It cancels a
653 uniform multiplicative scale within a model, but does not repair mixed scales across records, hence
654 the Qwen 2.5-7B caveat above.

655 B.3.3 Aggregation and uncertainty

656 For EHC, HFC, and ICS, the evaluator first pools eligible records within each model–market cell to
657 obtain a market-level rate. The reported model score is the unweighted mean of those market rates, so
658 a market with more successful repeat runs does not receive greater weight. Reported standard errors
659 are the sample standard deviation of market-level rates divided by the square root of the number of
660 markets.

661 Revision magnitudes and sensitivity retain the record-pooled point estimands. Uncertainty uses
662 20,000 percentile cluster-bootstrap draws over represented markets. All nine packets and every
663 repeated initial-run update from a selected market remain together, and sensitivity is recomputed
664 from pooled directional and orthogonal movement within each draw. This accounts for dependence
665 among records from the same market while preserving the published estimand.

666 B.3.4 Output compliance and threshold robustness

667 **Output-field coupling.** The final-output schema explicitly instructs the agent that $\hat{p} = P(H_1)$, and
668 the Stage 3 prompt requests the same structure after updating. EHC and HFC are therefore not
669 independent tests of an inferred hypothesis-to-forecast link. ICS is a schema-adherence diagnostic,
670 while close EHC–HFC agreement reflects a mixture of directional updating and compliance with
671 the prompted equality. Among the 26,010 relevant records with both updated fields present, 25,984
672 (99.90%) satisfy the equality exactly after scale normalisation and 25,987 (99.91%) satisfy it within
673 the 2-pp ICS tolerance. We therefore do not treat EHC–HFC parity as independent evidence for a
674 latent representation or for spontaneous propagation between separately elicited quantities.

675 **Unconditional estimand.** For threshold t , Table 5 retains a valid directional update when $|\Delta| \geq t$
676 and reports the fraction for which $d_i \Delta_i > 0$. At $t = 0$, every valid pro- and anti- H_1 update enters
677 the denominator and a zero change is scored incorrect. We first compute correctness within market
678 and then average markets equally, matching the primary analysis. The *Below 3 pp* column is the
679 record-weighted fraction and count excluded by the published 3-pp conditional metric. The table’s
680 n is the complete pre-threshold denominator: records require a parseable initial probability and the
681 applicable updated field, but are not selected on the direction or magnitude of movement.

Table 5: Unconditional directional correctness and robustness to 0-, 1-, 3-, and 5-percentage-point movement thresholds. The 0-pp column is unconditional over all valid directional updates and scores zero movement as incorrect; positive-threshold columns condition on the stated magnitude. *Below 3 pp* gives the percentage and count omitted from the corresponding published 3-pp EHC or HFC denominator. Rates are macro-averaged across markets.

Model	Metric	n valid	Directional correctness				Below 3 pp
			0 pp	1 pp	3 pp	5 pp	
Claude Opus 4.8	EHC	2,119	98.2%	98.6%	99.0%	99.5%	18.1% (383)
	HFC	2,119	98.3%	98.6%	99.0%	99.5%	18.1% (383)
GPT-5.4	EHC	2,678	98.4%	98.4%	98.7%	98.8%	6.9% (185)
	HFC	2,684	98.4%	98.4%	98.7%	98.8%	6.9% (185)
DeepSeek V4-Pro	EHC	2,929	97.3%	97.6%	97.7%	98.0%	12.7% (371)
	HFC	2,929	97.3%	97.6%	97.7%	98.0%	12.6% (370)
Qwen 2.5-72B	EHC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
	HFC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
Llama 3.3-70B	EHC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
	HFC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
Llama 3.1-70B	EHC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
	HFC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
Qwen 2.5-32B	EHC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
	HFC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
Qwen 2.5-14B	EHC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
	HFC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
Llama 3.1-8B	EHC	3,601	88.1%	88.7%	88.7%	88.9%	1.7% (62)
	HFC	3,604	88.4%	88.7%	88.7%	88.9%	1.4% (52)
Qwen 2.5-7B	EHC	5,759	90.2%	90.2%	90.6%	90.8%	2.2% (129)
	HFC	5,759	89.9%	90.2%	90.5%	90.8%	2.4% (140)

Threshold-sweep results. The threshold sweep preserves the qualitative conclusion but narrows its wording. Hosted-model unconditional correctness is 97.3–98.4%; at the published 3-pp threshold it is 97.7–99.0%. The 3-pp rule excludes 6.9% of GPT-5.4, 12.6–12.7% of DeepSeek V4-Pro, and 18.1% of Claude Opus 4.8 directional updates. Thus, the near-1.0 EHC/HFC values do not result from discarding a majority of cases, and the conclusion is stable across 0, 1, 3, and 5 pp. The precise statement is *99% directionally correct among updates of at least 3 pp*. It is not *99% coherent updating*. Small local models have fewer sub-3-pp updates (0.6–2.9%) but lower unconditional correctness (88.1–96.4%), so their errors are not hidden primarily by the movement filter.

Normalisation and reproducibility. Before recomputing deltas, the analysis independently normalises the initial headline probability, updated headline probability, and updated H_1 posterior from the documented $[0, 1]$ or $[0, 100]$ formats. This matters for 166 relevant Llama 3.1-8B records in which H_1 was emitted as a percentage while `yes_prob` was emitted as a proportion. The complete eligible record and market counts at every threshold, invalid-field counts, market-level standard errors, input-file SHA-256 hashes, and full-precision estimates are in `exp1_prospective/data/results/threshold_robustness.json`. The table is regenerated by `exp1_prospective/agent/evaluate_threshold_robustness.py`; neither script nor table filters items using realized outcomes.

B.4 Extended results

B.4.1 Analysis coverage

Table 6 gives the exact Stage 3 record and market counts. The market column counts markets with at least one valid update. Coverage of all three packet arms is not uniform across models: Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast, and Claude Opus 4.8

705 has all three arms in 73 of 100. Section B.5.3 gives the per-arm coverage and assesses whether
706 missingness is arm-selective.

Table 6: Valid Stage 3 update records and represented core markets by agent. Hosted systems first; open-weight rows in descending parameter count.

Model	Type	Records	Markets
Claude Opus 4.8	Hosted	3,125	100
GPT-5.4	Hosted	3,913	100
DeepSeek V4-Pro	Hosted	4,395	100
Qwen 2.5-72B	Local 72B	2,692	100
Llama 3.3-70B	Local 70B	2,682	100
Llama 3.1-70B	Local 70B	2,490	99
Qwen 2.5-32B	Local 32B	2,693	100
Qwen 2.5-14B	Local 14B	2,700	100
Llama 3.1-8B	Local 8B	5,404	100
Qwen 2.5-7B	Local 7B	8,636	100
Total		38,730	

708 B.4.2 Prospective settlement audit

709 The frozen-cohort evaluation includes a dated settlement audit that leaves the directional-updating
710 estimand unchanged. On August 24, 2026, we queried the public Polymarket market endpoint for the
711 exact 100 frozen IDs. We admitted a label only when the market was closed and the YES/NO token
712 prices were exactly $[1, 0]$ or $[0, 1]$. This strict rule yielded 60 scoreable markets (21 YES and 39 NO).
713 One additional closed market had an invalid 0.5/0.5 settlement and 39 remained unresolved at the
714 audit cutoff; all 40 were excluded.

715 For each agent–market cell, the forecast is the median of its parseable repeated initial YES probabili-
716 ties, matching the maintained Experiment 1 viewer. The crowd benchmark is the YES price stored in
717 the frozen June 9 selection record. We report threshold accuracy (YES iff $p > 0.5$), Brier score, and
718 natural-log loss. Proper-score contrasts are paired agent-minus-market differences; negative values
719 favor the agent. Percentile intervals use 20,000 market-level bootstrap resamples.

Table 7: Descriptive accuracy on the first 60 strictly settled markets. A/M denotes agent/June 9 market on the same rows; Correct is the number of thresholded calls matching the outcome. Lower Brier and log loss are better, and each Δ is agent minus market. Llama 3.1-70B lacks a parseable initial forecast for one settled market, so its paired denominator is 59.

Model	n	Correct A/M	Brier A/M	Δ Brier [95% CI]	Log A/M	Δ log [95% CI]
Claude Opus 4.8	60	41/39	.242/.246	-.004[-.063, .054]	.738/.700	.038[-.129, .210]
GPT-5.4	60	40/39	.225/.246	-.021[-.074, .030]	.663/.700	-.037[-.178, .103]
DeepSeek V4-Pro	60	40/39	.257/.246	.011[-.050, .069]	.793/.700	.093[-.093, .282]
Qwen 2.5-72B	60	32/39	.265/.246	.019[-.031, .066]	.728/.700	.028[-.097, .144]
Llama 3.3-70B	60	30/39	.266/.246	.020[-.038, .078]	.732/.700	.032[-.113, .177]
Llama 3.1-70B	59	33/38	.258/.249	.008[-.058, .075]	.712/.707	.004[-.155, .164]
Qwen 2.5-32B	60	32/39	.264/.246	.018[-.044, .078]	.728/.700	.028[-.126, .178]
Qwen 2.5-14B	60	37/39	.277/.246	.031[-.031, .092]	.827/.700	.127[-.049, .308]
Llama 3.1-8B	60	31/39	.266/.246	.020[-.047, .085]	.741/.700	.041[-.127, .202]
Qwen 2.5-7B	60	38/39	.230/.246	-.016[-.075, .041]	.651/.700	-.049[-.195, .090]

721 The raw correct-call counts are close for the hosted agents and crowd: Claude Opus 4.8 gets
722 41/60 correct, GPT-5.4 and DeepSeek V4-Pro each get 40/60, and the June 9 market side gets
723 39/60. GPT-5.4 has the lowest Brier score (.225 versus .246 for the market), a paired difference of
724 $-.021 [-.074, .030]$; its log-loss difference is $-.037 [-.178, .103]$. Qwen 2.5-7B has the lowest log
725 loss (.651 versus .700), but gets 38/60 thresholded calls correct. No agent’s paired Brier or log-loss
726 interval excludes zero.

727 This audit describes the outcome-ready slice of the frozen cohort. Settlement by the audit date
728 censors the cohort toward shorter-horizon and early-resolving questions, so the 60 are not a random
729 subset of the frozen 100. Moreover, the benchmark is the June 9 snapshot whereas agent forecasts
730 were collected from June 10 through June 17; later-running agents may have had additional public

information. The result therefore does not establish superiority to the crowd. It documents the prospective check; the directional-update estimates do not, by themselves, imply a detectable accuracy advantage.

B.4.3 Revision magnitudes and packet selectivity

Table 8 reports mean absolute forecast revision $|\Delta\hat{p}|$ by packet direction. For the nine models whose updates are consistently stored on 0–1, hosted-system sensitivity is 8.2–24.0 $\times$, compared with 1.9–6.4 $\times$ for local systems. These are the unconditional ratios; Table 9 separates the abstention and magnitude factors behind them, and the main text reports those two rather than the product.

Qwen 2.5-7B has mixed 0–1 and 0–100 stored updates (Section B.3.1); we therefore omit its absolute magnitudes and sensitivity ratio rather than present them as comparable estimates.

Table 8: Mean absolute forecast revision by counterfactual direction. Hosted systems first; open-weight rows in descending parameter count. Sensitivity is mean directional divided by mean orthogonal movement; n is the valid Stage 3 record count. Values are percentage points for models with consistently stored 0–1 updates.

Model	Type	pro- H_1	anti- H_1	orthogonal	Sensitivity	n
Claude Opus 4.8	Hosted	12.7 pp	9.6 pp	0.5 pp	24.0 $\times$	3,125
DeepSeek V4-Pro	Hosted	13.9 pp	11.4 pp	1.4 pp	8.9 $\times$	4,395
GPT-5.4	Hosted	10.6 pp	8.5 pp	1.2 pp	8.2 $\times$	3,913
Qwen 2.5-72B	Local 72B	15.0 pp	10.2 pp	3.0 pp	4.2 $\times$	2,692
Llama 3.3-70B	Local 70B	14.8 pp	14.3 pp	2.3 pp	6.4 $\times$	2,682
Llama 3.1-70B	Local 70B	14.8 pp	15.5 pp	3.3 pp	4.6 $\times$	2,490
Qwen 2.5-32B	Local 32B	15.0 pp	10.3 pp	4.0 pp	3.2 $\times$	2,693
Qwen 2.5-14B	Local 14B	14.8 pp	8.6 pp	4.1 pp	2.9 $\times$	2,700
Llama 3.1-8B	Local 8B	11.1 pp	12.0 pp	5.9 pp	1.9 $\times$	5,404
Qwen 2.5-7B	Local 7B	—	—	—	—	8,636

Qwen 2.5-7B magnitude cells are omitted because 207 deduplicated updates are stored above 1.5, mixing probability scales within the row. Table 8 documents the magnitude contrast; Figure 11 then places directional consistency and packet selectivity on a common visual scale.

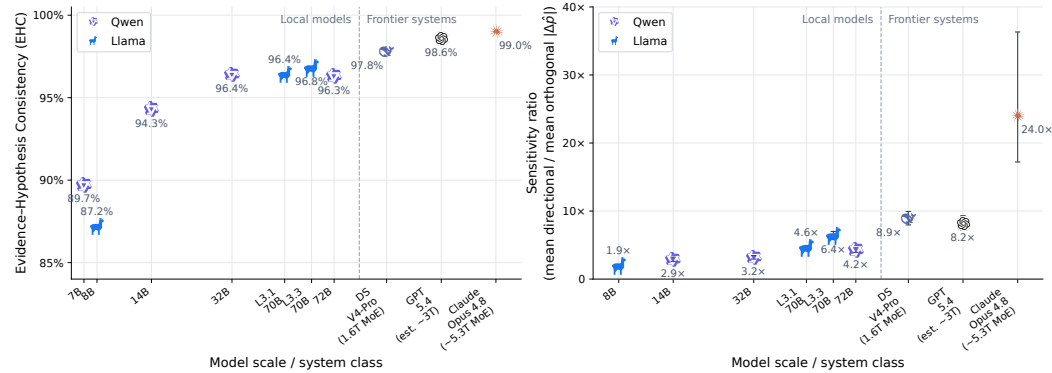


Figure 11: **Directional consistency and packet selectivity by deployment.** *Left:* EHC is 87–90% for 7–8B local models and 98–99% for hosted systems. *Right:* sensitivity is 1.9–6.4 $\times$ for local models and 8.2–24.0 $\times$ for hosted systems; Qwen 2.5-7B is omitted from this panel because its archived absolute revisions mix 0–1 and 0–100 scales and therefore do not yield a comparable ratio. The panel plots the unconditional ratio, which Table 9 decomposes.

B.5 Study limitations

B.5.1 Sampling and outcome scope

The sample is a filtered slice of liquid, medium-horizon Polymarket questions rather than a probability sample of forecasting tasks. Topic exclusions, the 0.10–0.90 price window, the 14–180 day horizon,

750 event deduplication, and the diversity cap shape its difficulty distribution. The primary Experiment 1
751 estimand evaluates conditional updating on questions that were unresolved at collection. The
752 settlement audit in Section B.4.2 is naturally censored to the first 60 strict settlements and is descriptive.
753 Experiment 4 separately evaluates accuracy on a locked sample of resolved historical markets with
754 synchronized forecast cutoffs and market-price baselines.

755 B.5.2 What the update prompt withholds

756 The Stage 3 template fills in one packet field, `evidence_text`. The packet record also holds
757 the intended direction, `target_hypothesis`, `expected_hypothesis_shift`, `slot_type`,
758 `evidence_headline`, `mechanism_targeted`, and a free-text rationale. None of these reaches
759 the agent; all are read only when the update is scored. The agent sees its own prior forecast, the
760 snippet, and the fixed instruction.

761 We checked this two ways. Over the 12,728 frozen records that kept their prompt string, spanning all
762 three hosted systems, the evidence block is identical to the packet’s `evidence_text` and contains
763 no private packet field. Prompt persistence was a per-runner logging choice, so this covers a subset
764 of records by logging coverage rather than by outcome; the runners that stored nothing use the
765 same one-field template. Turning to the snippets, none of the 900 names H_1 , H_2 , the market, or its
766 resolution. Twenty-two (2.4%, 10 pro- H_1 and 12 anti- H_1) use any likelihood wording—*odds*, *chance*
767 *of*, *moreless likely*, *probability*, *likelihood*—and each time it describes something in the world, a
768 simulation percentage or a betting line, not the market’s resolution. The sign is not stated; it must
769 be inferred from the relation between the reported event and the market outcome. Reproduced by
770 `exp1_prospective/agent/audit_packet_polarity.py`.

771 B.5.3 Alternative explanations for the selectivity gap

772 Here we state the competing accounts of the movement gap and what we find for
773 each. All checks come from `exp1_prospective/agent/audit_selectivity_robustness.py`,
774 `audit_orthogonal_abstention.py`, `audit_lexical_sign_baseline.py`, and `audit_group_contrast.py`.

776 **Where the numbers come from.** The checks read the frozen June 17 report, which enumerates every
777 scored run, so they use exactly the data behind the published numbers. The enumeration matches the
778 report’s own per-arm counts for seven of the nine models; Claude Opus 4.8 and GPT-5.4 carry 14 and
779 4 extra runs in markets absent from the per-market join, 18 of 38,730. Reproduced sensitivity ratios
780 agree with the published ones to within 0.2. Three checks need packet text and so join runs to the
781 June 10 packet set; 346 runs (1.1%) do not join because they come from a June 7–8 pilot against a
782 superseded 63-packet generation, and they are dropped from those three checks only.

783 **Attrition is much higher for the hosted systems.** Claude Opus 4.8 made 9,313 calls and yielded
784 3,232 usable updates, losing 3,656 to broken pipes, 914 to HTTP 401s after a key expired mid-run,
785 and 1,481 to unparseable responses. Of those, 1,461 come from one malfunctioning June 15 retry pass
786 that failed at 91%; the main June 10 collection parsed at 99.3%, so the published compliance figures
787 are not survivorship. DeepSeek V4-Pro made 6,338 calls for 4,598 usable updates, losing 1,629 to
788 HTTP 401s; GPT-5.4 made 4,269 for 4,057. The six local models lose almost nothing—four record
789 no failed calls, the other two lose 8 and 15 responses to parsing. Content-based exclusion is confined
790 to 28 GPT-5.4 calls across 2 markets. Recorded failures are predominantly transport or parsing
791 failures; the coverage check below assesses arm and market imbalance, not content-specific selection.
792 Separately, two update streams were corrupted by concurrent writes and repaired in August 2026,
793 after the freeze; the original bytes and unrecoverable fragments sit in `exp1_prospective/data/_recovery_audit/`.
794 Nothing in this section reads those files.

795 **The generator saw one agent’s world model.** All packets were built from a single canonical initial
796 forecast, and it is DeepSeek V4-Pro’s: its June 10 YES probability matches the stored generation
797 prior on 100 of 100 markets, where no other agent matches on more than 28. For the other eight
798 models the evidence came from a different system’s event model. The seed model gains nothing by
799 it—among hosted systems it is second on sensitivity ($8.9\times$ against Claude’s $24.0\times$), third on AUC
800 ($.967$), and last on EHC ($.977$)—but its estimates are the least independent in the roster.

801 **Coverage is uneven across arms.** Table 10 gives per-arm market coverage. Seven models cover
802 99 or 100 markets on every arm. Claude Opus 4.8 covers 82 on the orthogonal arm and 77 on the

directional arms, 73 in common, so its pooled ratio divides means taken over slightly different market sets. Restricting to its 73 common markets gives $23.7\times$ against $24.2\times$ pooled. As a proxy check, every other model has similar sensitivity on the markets Claude covers and those it misses (8.4 vs. 7.6 for GPT-5.4, 9.7 vs. 7.2 for DeepSeek V4-Pro, 4.3 vs. 4.2 for Qwen 2.5-72B, 1.9 vs. 2.1 for Llama 3.1-8B).

The ratio has an unstable denominator. Claude’s orthogonal mean is 0.5 pp, so its $24\times$ comes from a small divisor rather than unusually large directional movement, and its market-clustered interval is correspondingly wide, [17.3, 36.9]. Table 11 reports three statistics that avoid that division. The difference of means is 5.6–12.3 pp and does not order the models the way the ratio does. The within-market probability that a directional packet moves the forecast more than an orthogonal one runs from .791 for Llama 3.1-8B to .994 for GPT-5.4, putting GPT-5.4 above Claude Opus 4.8 and reversing the ratio ranking. Every model moves further on directional packets in every market it covers.

The denominator is mostly zeros. That instability has a specific cause, and naming it replaces the ratio with two statistics that are easier to read. A revision of exactly zero—the model reproduces the probability shown to it—enters the mean like any other record, so the ratio silently mixes how often a deployment declines to move with how far it moves when it does. Writing A for the share of runs with $\Delta\hat{p} = 0$ and M for the mean absolute revision among the rest, $|\Delta\hat{p}| = (1 - A)M$, and the ratio factors exactly:

$$\text{Sensitivity} = \underbrace{\frac{M_{\text{dir}}}{M_{\text{orth}}}}_{\text{magnitude}} \times \underbrace{\frac{1 - A_{\text{dir}}}{1 - A_{\text{orth}}}}_{\text{abstention}}. \quad (5)$$

Table 9 gives both factors. Directional abstention is 0.0–1.4% for every deployment, so the second factor is governed entirely by the orthogonal arm, where abstention runs from 1.1% for Llama 3.1-8B to 71.4% [63.8, 78.5] for Claude Opus 4.8. Claude’s $24\times$ is therefore $7.0\times$ [5.5, 9.1] in magnitude multiplied by 3.48 in abstention; DeepSeek V4-Pro’s $8.9\times$ is $4.9\times$ times 1.79; GPT-5.4’s $8.2\times$ is $7.5\times$ times 1.09, because GPT-5.4 writes a small revision where the other two hosted systems return the prior unchanged. Across the roster the published 1.9–24.2 $\times$ becomes 1.9–7.5 $\times$ once the zeros are separated out, and the ordering changes: GPT-5.4 rather than Claude Opus 4.8 is highest, and Llama 3.3-70B at $5.7\times$ sits above DeepSeek V4-Pro at $4.9\times$.

We report both factors rather than the product. Declining to revise at all is an appropriate response to an outcome-irrelevant premise, so the abstention rate is a selectivity measure in its own right and a better-behaved one: it is a bounded proportion, it is paired within model across arms, and it needs no division by a small number. It is also the more fragile of the two. An exact copy of a displayed prior is a discrete act adjacent to formatting, and the update prompt does instruct the agent to leave unaffected fields unchanged, so we do not read a high rate as evidence about internal computation. The no-information condition described in Section B.5 is what would separate declining to revise from copying the anchor, and it was not run. Reproduced by `exp1_prospective/agent/audit_orthogonal_abstention.py`.

Table 9: Decomposition of the sensitivity ratio into magnitude and abstention factors (Equation 5). *Orthogonal abstention* is the share of orthogonal runs with $\Delta\hat{p}$ exactly zero; the directional-arm rate is 0.0–1.4% throughout. The two *given a move* columns condition on a nonzero revision, and their ratio is the magnitude factor. Brackets are 95% market-clustered percentile intervals from 20,000 draws, the scheme used for the published movement intervals. Qwen 2.5-7B is omitted for mixed probability scales.

Model	Orthogonal	Mean $ \Delta\hat{p} $ given a move		Sens.	Sens.
	abstention	orthogonal	directional	given a move	published
Claude Opus 4.8	71.4% [63.8, 78.5]	1.6 pp	11.2 pp	$7.0 \times [5.5, 9.1]$	$24.2 \times [17.3, 36.9]$
GPT-5.4	8.4% [5.2, 11.9]	1.3 pp	9.6 pp	$7.5 \times [6.6, 8.5]$	$8.2 \times [7.2, 9.3]$
DeepSeek V4-Pro	44.4% [38.9, 50.0]	2.6 pp	12.7 pp	$4.9 \times [4.4, 5.5]$	$8.9 \times [8.0, 9.9]$
Qwen 2.5-72B	24.2% [19.6, 29.0]	3.9 pp	12.6 pp	$3.2 \times [3.0, 3.4]$	$4.2 \times [3.9, 4.6]$
Llama 3.3-70B	10.7% [7.0, 14.8]	2.6 pp	14.6 pp	$5.7 \times [5.2, 6.2]$	$6.4 \times [5.8, 7.0]$
Llama 3.1-70B	2.5% [0.8, 5.0]	3.4 pp	15.2 pp	$4.5 \times [4.2, 4.8]$	$4.6 \times [4.3, 4.9]$
Qwen 2.5-32B	11.1% [8.0, 14.6]	4.5 pp	12.6 pp	$2.8 \times [2.6, 3.1]$	$3.2 \times [2.9, 3.5]$
Qwen 2.5-14B	26.1% [20.7, 31.9]	5.5 pp	11.9 pp	$2.2 \times [2.0, 2.3]$	$2.9 \times [2.6, 3.2]$
Llama 3.1-8B	1.1% [0.5, 1.9]	6.0 pp	11.6 pp	$1.9 \times [1.8, 2.0]$	$1.9 \times [1.8, 2.1]$

The deployment contrast depends on the statistic. Comparing the three hosted systems against the six scale-valid open-weight models by market-clustered bootstrap (20,000 replicates, macro-averaged within group so a model with more runs does not dominate), the statistics disagree. On the ratio the difference is $+9.8 [7.2, 14.2]$, group means $13.7 \times$ and $3.9 \times$, and the per-model ranges do not overlap. On AUC it is $+0.070 [0.062, 0.079]$, which excludes zero, but the ranges do overlap: Llama 3.3-70B at .989 and Llama 3.1-70B at .978 both beat DeepSeek V4-Pro at .967. On raw movement there is no difference at all, $+0.8 [-0.0, 1.7]$ pp around means of 10.1 and 9.3.

The decomposition explains why the ratio is the one statistic that separates the groups completely. Both of its factors favour the hosted systems on average and neither separates them on its own: orthogonal abstention differs by $+0.286 [0.244, 0.329]$ around group means of .413 and .126, with four open-weight deployments at or above GPT-5.4’s 8.4%; the magnitude factor differs by $+3.10 [2.31, 4.08]$ around $6.5 \times$ and $3.4 \times$, with Llama 3.3-70B above DeepSeek V4-Pro. Multiplying two partially separating factors yields a product whose ranges happen not to overlap, so we do not report that non-overlap as a finding. The two classes move about equally far on directional evidence; the hosted systems more often decline to move at all on controls, and move somewhat less far when they do. We state the deployment result that way rather than as a general selectivity ordering. Group contrasts come from `expl_prospective/agent/audit_group_contrast.py`.

The arms differ in wording. By construction directional packets take the DATA, DECISION, and STRUCTURAL slots and orthogonal packets take PROCEDURAL, PERSONNEL, and BACKDROP, and the text differs accordingly: directional snippets average 96.8 words and 7.1 digits, orthogonal ones 85.7 and 3.7. Movement could track density rather than content. Two checks argue otherwise. Within the directional arm, where length cannot indicate sign, Spearman correlation between word count and $|\Delta\hat{p}|$ is $-.06$ to $.00$ and between digit count and $|\Delta\hat{p}|$ is $.01$ to $.10$. And the arms’ word counts almost fully overlap (orthogonal 66–114, directional 66–121); restricting to that band keeps 889 of 900 packets and moves every sensitivity estimate by at most 0.1 (Table 11, last column). The arms are not matched on surface form. These checks show that word and digit counts do not explain the effect; other surface differences remain. Directional correctness is not explained by length, since length cannot say which way to move.

A shallow classifier might be enough. We tested it. Multinomial naive Bayes and L2-regularised logistic regression were fit on the 900 packet texts with five-fold cross-validation grouped by market, so no market appears on both sides of a fold. Telling directional from orthogonal packets is nearly free: naive Bayes reaches .989 against a .667 majority rate. That follows from the slot-type plan rather than revealing anything—the most distinctive orthogonal tokens are *appointed*, *appointment*, *oversee*, and *accreditation*, which name the roles the generator was told to write. So a model could route on that marking without weighing content, and the sensitivity ratio alone cannot rule it out. Matching the control arm on evidence type would need a different packet design; the checks above bound only the length part of the problem.

877 Recovering the *sign* is another matter. On pro- H_1 versus anti- H_1 the same classifiers reach .688 and
 878 .675 against .500 chance, and prepending the market question does not help (.665 and .680): these
 879 bag-of-words models do not combine question polarity with event valence. The tokens it keys on
 880 carry no consistent sign—*vessel*, *venue*, *signature* for pro- H_1 , *table*, *materials*, *findings* for anti- H_1 .
 881 Against this .69 lexical baseline, the human panel reaches .762 on its smaller categorical task and the
 882 hosted agents .973–.984 on probability revision. These are not matched performance comparisons.
 883 The agent–classifier gap is evidence against this bag-of-words account, not proof that no surface
 884 heuristic can recover sign; unlike the arm contrast, it stays within pro- versus anti- H_1 packets.

885 **Does the stated reasoning track the packet?** If movement were routed on surface marking, a model’s
 886 written rationale would not need to bear any particular relation to what it does with the forecast. We
 887 tested two versions of this (`exp1_prospective/agent/audit_mechanism_tracking.py`).

888 The first asks whether a rationale identifies the specific mechanism its packet was built to target,
 889 ranking the true `mechanism_targeted` against the other eight from the same market so that question,
 890 actors, and domain vocabulary are held fixed. Rationales rank it first on .278–.430 of records against
 891 1/9 chance, but the snippet alone scores .516, so most of that is echo of the text the model just
 892 read. Removing every token the snippet supplied leaves .128–.195 under matched query length, still
 893 above chance—but the generator’s own rationale for the packet, scored the same way, lands at .119,
 894 essentially chance, because it paraphrases the mechanism rather than restating it. With the reference
 895 writer at chance the metric has no usable dynamic range, and we draw no conclusion from it.

896 Models often declare orthogonal packets irrelevant: explicit non-relevance wording appears in 4.9–
 897 61.9% of orthogonal rationales against 0.1–2.8% of directional ones. What matters is whether saying
 898 it predicts doing it. Within the orthogonal arm, records whose rationale declares irrelevance move
 899 less than those that do not, and the market-clustered gap excludes zero for eight of the nine scale-valid
 900 deployments: 0.88 pp [0.56, 1.22] for Claude Opus 4.8, 1.70 [1.45, 1.96] for DeepSeek V4-Pro, 2.73
 901 [2.28, 3.19] for Qwen 2.5-72B, down to 0.36 [0.21, 0.54] for GPT-5.4, whose unflagged orthogonal
 902 movement is already 1.3 pp. The exception is Llama 3.1-8B at 0.26 [−0.64, 1.06], where no coupling
 903 is detectable; it is also the weakest deployment on sensitivity and directional correctness, though we
 904 do not test whether the two are related. Qwen 2.5-7B is omitted for mixed probability scales.

905 For eight of nine deployments, then, the revision a model makes is not independent of the relevance it
 906 states, and the one deployment where that link is undetectable is the weakest on every other measure.
 907 Two limits: the flag is a hand-built lexicon, so it captures what a rationale declares rather than what
 908 the model understood; and consistent self-description is compatible with surface routing, so this does
 909 not show relevance was derived from the mechanism.

910 **One deployment was prompted differently.** GPT-5.4, and only GPT-5.4, received the
 911 `novelty_assessment` rule at Stage 3 (Section B.2.3). Because that rule speaks directly to re-
 912 vision size, its abstention rate is not comparable to the other deployments’. The within-GPT contrast
 913 uses one template, but the extra instruction could interact with packet arm, so its contrast should not
 914 be attributed to model differences alone.

915 **Two remaining caveats.** A single generator wrote all packets and scored its own work at a uniform
 916 4/5 for plausibility, so automated validation establishes structural completeness but not equal difficulty
 917 or independent content validity. And the local models ran through Ollama at the default tag for each
 918 name, with the served weight precision not recorded per run; rows ordered by parameter count also
 919 vary in quantization and inference stack, so we do not attribute the open-weight ordering to parameter
 920 count alone.

Table 10: Per-arm market coverage in the frozen June 17 report. Markets counts entries in the model’s per-market block, which for three Qwen rows includes 20 carrying no scored update; the arm columns count markets with at least one.

Model	Markets	pro- H_1 /anti- H_1	orthogonal	all three
Claude Opus 4.8	100	77	82	73
GPT-5.4	100	99	99	99
DeepSeek V4-Pro	100	100	100	100
Qwen 2.5-72B	120	100	100	100
Llama 3.3-70B	100	100	100	100
Llama 3.1-70B	100	99	99	99
Qwen 2.5-32B	120	100	100	100
Qwen 2.5-14B	120	100	100	100
Llama 3.1-8B	100	100	100	100

Table 11: Selectivity under statistics that do not divide by the orthogonal mean. *Sens.* is the frozen ratio; *Diff.* is mean directional minus mean orthogonal absolute revision; *AUC* is the within-market probability that a directional packet moves the forecast more than an orthogonal one, averaged over markets; *Markets* counts markets where mean directional movement exceeds mean orthogonal movement; *Length-m.* recomputes the ratio on the 66–114-word band the arms share. Qwen 2.5-7B is omitted for mixed probability scales.

Model	Sens.	Diff.	AUC	Markets	Length-m.
Claude Opus 4.8	24.0×	10.7 pp	0.984	77/77	24.3×
GPT-5.4	8.2×	8.4 pp	0.994	99/99	8.1×
DeepSeek V4-Pro	8.9×	11.2 pp	0.967	100/100	8.8×
Qwen 2.5-72B	4.2×	9.6 pp	0.950	100/100	4.2×
Llama 3.3-70B	6.4×	12.3 pp	0.989	100/100	6.4×
Llama 3.1-70B	4.6×	11.9 pp	0.978	99/99	4.6×
Qwen 2.5-32B	3.2×	8.7 pp	0.906	100/100	3.2×
Qwen 2.5-14B	2.9×	7.7 pp	0.852	100/100	2.9×
Llama 3.1-8B	1.9×	5.6 pp	0.791	100/100	1.9×

B.5.4 Human materials review

Nine adult reviewers submitted the 18-item materials protocol. It samples one report from each of 18 distinct markets, balanced across six pro-YES, six anti-YES, and six orthogonal packets. One reviewer met the disclosed response-pattern exclusion below, leaving eight quality-eligible reviewers. We treat the study as descriptive materials evidence. Before the synthetic report, every item shows the market question, resolution criteria, and matching summary from the frozen June 10 initial forecast. The consent screen and every item instruct reviewers to treat June 10, 2026 as the current date and use no external information or assistance. Intended directions, source IDs, generator metadata, and model responses remain hidden.

Reviewers record the report’s conditional direction, confidence, clarity, real-world plausibility, and usability as an explicit hypothetical premise. We report exact agreement with the frozen direction and Cohen’s κ between the panel-majority labels and the frozen key. Reviewer-level Cohen’s κ , the component ratings, and every item-level judgment appear below. Ambiguous judgments remain in every denominator.

Practice responses are stored separately and excluded from the analysis.

The analyzed sample contains eight quality-eligible reviewers (R1 and R3–R9), exceeding the six-reviewer target for this descriptive materials check. Table 12 includes every eligible reviewer.

Exact-direction agreement is 108/144 (75.0%). The panel-majority label matches the frozen direction on 17 of 18 items (94.4%; Cohen’s $\kappa = .919$): all six anti- H_1 items, all six orthogonal items, and five of six pro- H_1 items. Across individual judgments, exact agreement is 35/48 for pro- H_1 , 37/48 for anti- H_1 , and 36/48 for orthogonal packets. Reviewers marked the premise usable on 124/144 judgments (86.1%). The corresponding all-reviewer sensitivity is reported below. These descriptive results indicate that readers usually recovered the intended conditional direction and could use the

946 reports as stipulated premises. They also bound how legible that direction is: 75.0% exact agreement
 947 and mean pairwise Cohen’s $\kappa = .47$ indicate substantial but imperfect recovery across the four
 948 response labels. The panel is not a matched baseline for the agents—reviewers emit a categorical
 949 label on 18 items, agents emit a revised probability against their own prior on all 900—so we read it
 950 as evidence that sign had to be inferred, not as a human–model comparison.

Table 12: **Stage 3 human materials judgments for eight quality-eligible reviewers.** Direction is exact agreement with the frozen conditional direction and κ is Cohen’s κ against that key over the four response options. Confidence, clarity, and plausibility (C/C/P) are reviewer-level medians on 1–5 scales. The pooled row is descriptive because reviewers rated the same items; the majority row takes the modal label per item, resolving ties to *ambiguous*.

Reviewer	Direction agreement	κ vs. key	Premise usable	C/C/P
R1	12/18 (66.7%)	.54	10/18	4/4/4
R3	12/18 (66.7%)	.51	13/18	4/4/3
R4	18/18 (100.0%)	1.00	18/18	5/5/5
R5	10/18 (55.6%)	.38	18/18	4/4/4
R6	13/18 (72.2%)	.61	12/18	4/4/5
R7	15/18 (83.3%)	.76	17/18	5/4/3
R8	12/18 (66.7%)	.55	18/18	4/5/4
R9	16/18 (88.9%)	.84	18/18	5/5/5
Pooled	108/144 (75.0%)	—	124/144	—
Majority label	17/18 (94.4%)	.92	—	—

Table 13: **Item-level Stage 3 judgments for eight quality-eligible reviewers.** Key is the frozen conditional direction; Majority is the modal panel response, with ties assigned to *ambiguous*. Agree and Usable give counts out of eight; C/L/P gives median direction confidence, clarity, and plausibility on 1–5 scales. Item numbers follow the frozen public packet order.

Question	Key	Majority	Agree	C/L/P	Usable
1. Hormuz traffic normal by July 31	More	More	8/8	5/5/4.5	8/8
2. Iran reaches World Cup knockouts	Less	Less	6/8	4.5/4.5/4.5	7/8
3. Israel strikes Yemen by June 30	No effect	No effect	8/8	4.5/4/3.5	6/8
4. Netanyahu exits election by July 31	More	More	6/8	5/5/4.5	8/8
5. Anthropic valuation reaches \$1.1T	Less	Less	7/8	4.5/4.5/5	8/8
6. Oladokun starts Chiefs Week 1	No effect	No effect	6/8	4/4/3.5	7/8
7. S&P 500 reaches 7,700 in June	More	More	4/8	3.5/4/5	7/8
8. Trump unfreezes Iranian assets	Less	Less	5/8	3.5/4/4	7/8
9. May durable-goods orders: 0–2%	No effect	No effect	4/8	3/3/3.5	6/8
10. China Q2 growth: 4.6–4.9%	More	Ambig.	3/8	3/4/3.5	5/8
11. Eisenkot joins Bennett–Lapid alliance	Less	Less	6/8	5/5/5	7/8
12. Russia enters Orikhiv by July 31	No effect	No effect	8/8	5/4/4.5	6/8
13. Romanian PM is independent/technocrat	More	More	6/8	4.5/4.5/4	8/8
14. Russia captures Kostyantynivka	Less	Less	5/8	3.5/5/5	7/8
15. At least four June ChatGPT outages	No effect	No effect	4/8	3/4/4	7/8
16. El-Sayed wins Michigan primary	More	More	8/8	5/5/5	8/8
17. Little is MN-02 Democratic nominee	Less	Less	8/8	5/5/4	7/8
18. OpenAI IPO close: \$1.25–1.5T	No effect	No effect	6/8	4/3.5/3.5	5/8

953 The response-pattern quality rule excludes any reviewer who selects one direction on at least 17 of 18
 954 balanced items. R2 met this rule by selecting *more likely* on 17 of 18 balanced items, including all
 955 six frozen negative items and five of six frozen null items. Raw responses are retained, and no item is
 956 removed or relabeled. Retaining R2 gives 115/162 (71.0%) exact agreement across all nine reviewers.

957 The task requests no identifying or sensitive information. It was conducted as an internal materials
 958 review under the authors’ institutional guidance; no IRB protocol was opened.

959 Figure 12 records the reviewer-facing workflow. The capture script at `exp1_prospective/
 960 stage3_materials_annotation/capture_review_screenshots.py` loads the frozen item
 961 packet against an empty temporary annotation database. Item order was independently randomized
 962 by reviewer.

963 Orthogonal packets are a useful outcome-relevance control, but are not equivalent to receiving no
 964 new text. A literal no-information update condition would separate generic prompt-induced revision

965 from response to topically adjacent content. The very small hosted-system orthogonal revisions
966 (0.5–1.4 pp) do not substitute for that control.

967 **B.5.5 Prompt, retrieval, and runtime confounding**

968 Agents shared a substantive task contract, while execution was not byte-identical. Hosted models used
969 Azure Foundry search, while local models used Tavily results in an Ollama workflow; temperatures
970 and provider behavior also differ. Hosted–local gaps therefore combine model capability with retrieval
971 and runtime differences.

972 **B.5.6 Scale, missingness, and dependence**

973 Realized repeat counts are uneven because failures and top-ups differ by provider. Equal-weight
974 market aggregation prevents high-coverage markets from dominating EHC, HFC, and ICS. Movement
975 and sensitivity intervals use 20,000 market-clustered bootstrap draws that retain every packet and
976 repeated initial-run update from each selected market. The mixed-scale Qwen 2.5-7B row described
977 in Section B.3.1 remains an appendix compliance diagnostic and is excluded from the main magnitude
978 comparison.

979 **B.5.7 Premise acceptance at Stage 3**

980 Search is intentionally unavailable during updating, and agents are told to accept the synthetic
981 headline as true. This design measures conditional premise use rather than source verification or
982 epistemic refusal. The human materials review uses the same instruction and therefore assesses
983 comprehension and premise usability. These results do not establish whether a model should trust or
984 verify the synthetic report.

985 **B.6 Reproducibility and artifact map**

986 The paper treats the June 17 model-update report and the August 23 final materials-review summary
987 as the frozen numerical authorities. The relevant artifacts are:

- 988 • **Sampling:** `exp1_prospective/fetch_markets/select_markets.py` and the June 9 selec-
989 tion/diversity JSON and manifests in `exp1_prospective/data/selected_markets/`.
- 990 • **Initial elicitation:** `exp1_prospective/agent/forecast_agent.py` for hosted runs and
991 `exp1_prospective/agent/run_local_forecast.py` for Ollama runs, with records and man-
992 ifests in `exp1_prospective/data/initial_forecasts/`.
- 993 • **Interventions:** `exp1_prospective/agent/build_counterfactuals.py` and the frozen
994 `exp1_prospective/data/counterfactuals/counterfactuals_2026-06-10.jsonl`.
- 995 • **Updates:** hosted and local update runners under `exp1_prospective/agent/`, with update
996 records in `exp1_prospective/data/updated_forecasts/`.
- 997 • **Evaluation:** `exp1_prospective/agent/evaluate_consistency.py` and
998 `exp1_prospective/data/results/consistency_report_2026-06-17.json`;
999 `exp1_prospective/agent/evaluate_clustered_uncertainty.py` reconstructs
1000 the 20,000-draw market-clustered intervals in `exp1_prospective/data/results/`
1001 `clustered_movement_uncertainty.json`; `exp1_prospective/agent/audit_`
1002 `orthogonal_abstention.py` writes the abstention/magnitude decomposition of Table 9 to
1003 `exp1_prospective/data/results/orthogonal_abstention.json` and its table source,
1004 under the same resampling scheme.
- 1005 • **Human materials review:** the frozen protocol, blinded packet, browser application, and tests in
1006 `exp1_prospective/stage3_materials_annotation/`; `make_final_summary.py` recon-
1007 structs `data/exports/final_summary_current.json`, while `data/exports/agreement_`
1008 `current.json` is the paper-facing agreement export.
- 1009 • **Figures:** `paper/generate_figures.py` writes the direction-selective panel; `paper/`
1010 `generate_paper_figures.py` writes the combined EHC/sensitivity panel.

1011 The model-update tables are defined by the June 17 frozen JSON. The mutable collection directories
1012 contain later records, so evaluating their present contents would define a different dataset. Bit-for-bit

1013 reconstruction begins from the frozen JSON; a full pipeline reconstruction uses only the manifests
1014 and records included in that freeze.

1015 **C Experiment 2: contextual relationship selection and revision in Coin City**

1016 Experiment 2 tests whether context selects which response relationship a model applies and whether
1017 direct target observations revise that selection. The six primary deployments and symbol-control
1018 comparisons reuse the same 250 seeded episodes, so prompt-arm differences are paired rather than
1019 comparisons between independently sampled tests.

1020 **C.1 Design and data-generating process**

1021 **C.1.1 Design summary**

1022 The design identifier is `coin_city_stable_relationship_claude_n250_v4`. It contains 250 in-
1023 dependent episodes generated from seeds 930000–930249, five target-city prefixes $k \in \{0, 1, 2, 3, 4\}$,
1024 and three semantic-selection arms. Cue-inversion and arbitrary-symbol modules reuse those episode
1025 identities and numerical inputs; each has a hashed task manifest (Section C.3). Every model received
1026 the same episode records, queries, and arm-specific prompt text.

1027 **C.1.2 Balanced latent regimes**

1028 Each episode crosses two binary factors: whether City A or City B is the strongly responsive reference
1029 city, and whether target City C is strongly or weakly responsive. The closest-balanced allocation
1030 contains 63 A-strong/C-strong, 62 A-strong/C-weak, 62 B-strong/C-strong, and 63 B-strong/C-weak
1031 episodes. Within an episode, each city receives an independently sampled stable slope

$$g_{ec} \sim \begin{cases} \mathcal{N}(0.90, 0.03^2), & \text{strong regime,} \\ \mathcal{N}(0.25, 0.03^2), & \text{weak regime.} \end{cases}$$

1032 Thus the cue identifies a response regime but not the realized coefficient.

1033 **C.1.3 Observed cases and forecast query**

1034 Every city has four reference cases arranged as two matched pairs. Rows in a pair share a starting
1035 poll and news magnitude but have opposite signs; the first sign is random. For reference row r ,

$$p_{ecr}^{\text{end}} = \text{round}_{0,1} (p_{ecr}^{\text{start}} + g_{ec}x_{ecr} + \epsilon_{ecr}), \quad \epsilon_{ecr} \sim \mathcal{N}(0, 5^2).$$

1036 Starting polls are uniform on $[35, 65]$, and reference-news magnitudes are drawn from $\{6, 7, 8, 9, 10\}$.
1037 The query shock is drawn from $\{-10, \dots, -4, 4, \dots, 10\}$. Its scoring target is the latent conditional
1038 expectation $p_0 + g_C x$, with no additional query-time observation noise.

1039 The prefix k exposes the first k City C rows in their stored order. Consequently, $k = 2$ completes the
1040 first matched pair and $k = 4$ completes both; $k = 1$ and $k = 3$ deliberately expose partial pairs. City
1041 A and B always contribute all four rows in the ABC arms.

Table 14: Coin City design and evaluation dimensions. Final analytic record counts retain one record per task cell; raw transport-level calls are discussed in Section C.9.

Quantity	Value
Independent episodes	250
Factorial cells	63, 62, 62, 63 across the four A/B-by-C regimes
Target-city prefixes	$k \in \{0, 1, 2, 3, 4\}$
Primary prompt arms	target only; ABC/no C context; ABC/correct C context
Intervention modules	ABC/misleading C context; ABC/episode-randomized arbitrary labels
Primary deployment roster	Claude Opus 4.8; GPT-5.6; DeepSeek-V4-Pro; Kimi K3; Gemini 3.6 Flash; Claude Opus 5
Symbol-comparison roster	Qwen3 4B/8B/14B/32B; Qwen2.5 7B/14B/32B/72B; Llama 3.1 8B/70B; DeepSeek-V4-Pro; Kimi K3; Claude Opus 4.8/5; GPT-5.6 Sol
Primary task cells / final records	22,500 / 22,500 (22,497 parseable)
Misleading task cells / final records	7,500 / 7,500 (all parseable)
Symbol-comparison task cells / response records	56,250 / 56,250 (15 models)
Strong / weak slope means	0.90 / 0.25 poll points per news point
Slope standard deviation	0.03 within each regime
Reference-case noise	5 poll points
Reference cases per city	4 (two matched pairs)

C.1.4 Preflight validation and task integrity

The GPT-5.6, DeepSeek-V4-Pro, Kimi K3, Gemini 3.6 Flash, and Claude Opus 5 replications reuse the exact task files and prompt hashes created for the original Claude Opus 4.8 evaluation; no episode was regenerated. Before model calls, the manifest hashed the engine, task files, answer and scoring keys, validator, runner, renderer, and launcher. Automated validation confirmed the intended dimensions and factorial balance, prompt contracts and hashes, correct cue-to-regime mapping, visible matched pairs, and benchmark reconstruction. No model response or performance statistic was used by this preflight validator.

C.2 Prompt arms and information available

C.2.1 Shared task instructions

All prompts state that rows are separate polling cases rather than a time series, positive news favors the candidate, poll change is end minus start, and within-city responsiveness is stable even though individual polls are noisy. Rows sharing a pair label are described as having the same starting poll and equal-magnitude, opposite-sign news. Every displayed row includes the starting poll, news value, change, and ending poll.

The required answer is a JSON object with exactly a one-sentence `rationale` and a numeric `predicted_poll` between 0 and 100. The prompt asks for no step-by-step derivation.

C.2.2 Five information conditions

Target only. The model sees the first k City C cases and is explicitly told that no additional background information is available. This arm measures recovery from target-city evidence alone.

ABC, no City C context. The model also sees four cases from each reference city, each introduced by a background sentence. The strongly responsive reference city is described verbatim as receiving “national news coverage, where campaign stories receive sustained attention and tend to produce larger immediate poll movements”, and the weakly responsive city as receiving “local news coverage, where campaign stories compete with local issues and tend to produce smaller immediate poll movements”. The direction of the coverage-to-response mapping is therefore stated in words, though no numerical value is. City C receives no background description. This arm tests numeric transfer without a cue that aligns C to either reference regime.

1073 **ABC plus correct City C context.** The numeric rows and query are identical to the preceding arm.
1074 The sole added information is that City C receives national or local coverage, correctly
1075 aligned with its latent strong or weak regime. This is the primary cue-transfer condition.

1076 **ABC plus misleading City C context.** This intervention arm is identical to the correct-context
1077 prompt except that its City C sentence names the opposite coverage regime. Every number,
1078 case row, query, and other instruction is unchanged. This arm tests whether the model
1079 follows the semantic cue when it conflicts with the truth.

1080 **ABC plus an arbitrary City C label.** Every national/local description is replaced by one of two
1081 arbitrary labels, KIV or ZOR. The prompt states only that cities sharing a label share a
1082 typical response regime and that the labels have no inherent meaning or ordering. Their
1083 higher/lower-response mapping reverses on alternating episodes; City A and B’s numerical
1084 cases are therefore the only evidence from which the episode-local mapping can be induced
1085 before applying City C’s label.

1086 No prompt contains a fitted coefficient, an estimator, the strong or weak numerical means, the
1087 case-noise scale, a rule for pooling cities, or any analyst benchmark. The correct-versus-no-context
1088 contrast therefore isolates the one City C background sentence while holding all displayed numeric
1089 evidence fixed; the correct-versus-misleading comparison isolates the sentence’s regime assignment.
1090 The symbol-versus-no-context contrast asks whether the same assignment can be induced without
1091 lexical knowledge of national and local coverage.

1092 Because the reference sentences state the direction of the mapping, the semantic arms are not by
1093 themselves a test of whether that direction can be induced. A policy that reads no displayed number
1094 and assigns slope 1.0 to the nationally covered regime and 0.2 to the locally covered one attains
1095 $k = 0$ forecast MAE 0.523 and implied-slope correlation .996, better than any deployment and
1096 better than the analogical-regression benchmark of Table 16; 42 of 108 prior pairs on the grid
1097 $\{0.4, \dots, 1.5\} \times \{0.0, \dots, 0.8\}$ beat every deployment’s $k = 0$ MAE. The semantic arms therefore
1098 measure how a stated regime assignment is converted into a numerical response. Two separate
1099 controls carry the induction claim: the arbitrary-symbol arm, whose labels have no stated direction
1100 and whose mapping reverses across episodes, and the reference-selection analysis of Section C.6,
1101 which shows that the forecast tracks the named reference city’s own sampling error rather than any
1102 fixed prior.

1103 C.3 Model execution and response processing

1104 C.3.1 Deployment configurations

1105 Table 15 reports the effective API settings used for production. Temperature refers to the value sent
1106 to the provider: GPT-5.6 and Claude Opus 5 omit it even though the local GPT run manifest records
1107 a default of zero. No deployment used tools or retrieval.

Table 15: Production deployment configuration and primary-arm parse counts.

Deployment	Interface	Reasoning / sampling	Output ceiling	Parsed
Claude Opus 4.8	Anthropic Messages via Azure	temperature 0	512	3,750
1108 GPT-5.6	OpenAI Responses	low reasoning; temperature omitted	4,096	3,749
DeepSeek-V4-Pro	OpenAI Chat Completions	temperature 0	4,096	3,750
Kimi K3	OpenAI Chat Completions	low reasoning; temperature 0	16,384	3,750
Gemini 3.6 Flash	Gemini OpenAI-compatible Chat	low reasoning; temperature 0	16,384	3,750
Claude Opus 5	Foundry Anthropic Messages	default extended thinking; temperature omitted	4,096	3,748

1109 Claude Opus 5 returns extended thinking before its answer; the runner therefore selects its first text
1110 block. Gemini is the sole fast-tier deployment, so the six systems should be read as a replication set
1111 rather than a tier-matched leaderboard. Authentication material was supplied at runtime and was not
1112 written to source, manifests, logs, or response files.

1113 C.3.2 Deployment attempts, retries, and pilots

1114 Every primary model-prompt pair received one application-level attempt. A failed or answerless
1115 task was not re-asked. HTTP-client retries were disabled for GPT, DeepSeek, Kimi, and Opus 5.
1116 The original Opus 4.8 client allowed up to two transport retries. Gemini’s runner allowed adaptive
1117 pre-model transport retries but recorded none in the completed shards. These transport policies do
1118 not alter the one-answer-per-task analytic rule.

1119 Before production, 119 Kimi, 15 Gemini, and 17 Opus 5 task-shaped calls were used only to select
1120 configurations that reliably returned the required JSON schema. Forecast accuracy was not inspected.
1121 The calls are archived separately and are excluded from every denominator and analysis; each
1122 production namespace began empty. Production tasks were sharded only for throughput.

1123 C.3.3 Response parsing and record selection

1124 The parser scans the raw reply for JSON objects, retains the last object with an exact
1125 `predicted_poll` key whose value converts to a number in $[0, 100]$, truncates the retained rationale
1126 to 500 characters, and stores at most 8,000 raw characters. When duplicate task records exist, the
1127 analysis keeps the latest parseable response, allowing a valid response to supersede an earlier error. A
1128 single offline normalization removed whitespace from JSON keys in one otherwise valid Opus 4.8
1129 reply; this recovered one primary record without another model call. Other record-level corrections
1130 are documented in Section C.9.

1131 C.3.4 Misleading-context design

1132 The cue-inversion module begins from the frozen semantic-selection tasks. Before any inversion
1133 call, its builder verifies that every prompt equals its correct-context counterpart under exactly one
1134 sentence substitution and that all numeric content is unchanged. The cue is inverted for all 1,250
1135 episode-prefix tasks: 625 strong-city tasks are described as weak and 625 weak-city tasks as strong.
1136 Each deployment uses the same client and production configuration as its correct-context arm.

1137 C.3.5 Arbitrary-symbol design

1138 The symbol builder generates 1,250 matched prompts from the same frozen episodes and verifies
1139 that their complete sequence of numeric tokens is identical to the correct-context arm. The higher-
1140 response label is KIV in 125 episodes and ZOR in 125; the target regime-by-label cells contain 62–63
1141 episodes each. A prompt validator rejects national/local and strong/weak semantic descriptions. At
1142 $k = 0$, City C supplies no numerical cases, so its correct regime can be recovered only by decoding
1143 the label from City A and B’s examples.

1144 For DeepSeek-V4-Pro, we collected all 1,250 symbol tasks and contemporaneously re-collected
1145 all 2,500 no-context and semantic-context tasks on the same reachable Azure resource, using one
1146 deployment identifier and configuration. The released 15-model comparison combines that triplet
1147 with Qwen3-4B, Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-32B,
1148 Qwen2.5-72B, Llama 3.1-8B, Llama 3.1-70B, Kimi K3, Claude Opus 4.8, Claude Opus 5, and
1149 GPT-5.6 Sol.

1150 The eight Qwen and two Llama checkpoints received all 3,750 prompts through vLLM in bfloat16,
1151 with temperature-zero decoding, a 3,072-token model context, and a 512-token output ceiling. Qwen
1152 uses the no-thinking chat template; Llama uses its native instruction template. Qwen2.5-14B uses
1153 two-way and Qwen3-32B and Qwen2.5-32B four-way tensor parallelism, while Qwen2.5-72B and
1154 Llama 3.1-70B use eight-way tensor parallelism.

1155 For Kimi K3, Claude Opus 5, and GPT-5.6 Sol, the comparison reuses each deployment’s preserved
1156 no-context and semantic-context responses and adds only the 1,250 symbol prompts under the same
1157 model-specific production protocol. Claude Opus 4.8’s symbol arm instead ran on a second Azure
1158 resource serving the same deployment and used a 1,024-token rather than 512-token ceiling; all 1,250
1159 responses returned valid JSON. We therefore treat its arm contrast as a cross-run robustness result
1160 and do not use it to rank providers.

1161 Every runner verifies the frozen task hash before generation and writes a timestamped manifest.
1162 Each released model has 1,250 unique response records in every arm; this count does not imply that

every prediction parsed. No primary response file or frozen task is overwritten. Original outputs are preserved, and the final tables use matched complete cases after the no-call arithmetic reparse described in Section C.9.

C.4 Estimands, benchmarks, and statistical inference

C.4.1 Forecast and latent-response estimands

Forecast error is measured against the latent conditional expectation rather than a newly sampled noisy ending poll:

$$\text{MAE} = \frac{1}{n} \sum_{e=1}^n |\hat{p}_e - (p_{0e} + g_{Ce}x_e)|.$$

For every nonzero query shock, a forecast also identifies an implied slope $\hat{g}_e = (\hat{p}_e - p_{0e})/x_e$. We report its mean absolute error from g_{Ce} and its Pearson correlation with g_{Ce} across episodes. Correlation is needed because a nearly constant prior-mean slope can attain moderate MAE while failing to distinguish strong from weak cities.

Within each deployment and prefix, the three primary-arm metrics use the common intersection of episodes with parseable responses in all three arms. Arm contrasts are paired by episode and prefix. The misleading-arm table uses its final unique parseable record for every task. The arbitrary-symbol analysis uses each model’s corresponding common intersection across its three matched arms.

C.4.2 Analyst-only benchmarks

The ABC OLS benchmark fits poll change on news through the origin using every displayed A, B, and available C row. The target-only OLS benchmark uses only displayed City C rows and is undefined at $k = 0$. Both are reconstructed from the visible data by the analyst; neither the estimator nor its output appears in a model prompt.

The label-conditioned analogical regression implements the minimal structured inductive policy defined in Methods. It fits separate through-origin slopes to City A and City B, uses City C’s coverage label to select the reference assigned to the same response regime, and predicts with that slope. At $k > 0$, it pools the selected reference rows with the visible City C rows before refitting. In the misleading arm, the inverted label selects the opposite reference. The implementation uses only displayed rows; neither the estimator nor its output appears in a model prompt.

Table 16: Zero-shot behavior of the analogical policy and the six model deployments. At $k = 0$, City C has no numerical cases. Model columns give the range across deployments; forecast MAE is in poll points.

Cue	Policy MAE	Policy ρ	Model MAE range	Model ρ range
Correct	1.842	.656	1.596–2.474	[.52, .70]
Inverted	4.561	–.667	4.204–4.361	[–.68, –.50]

C.4.3 Uncertainty and comparison families

Primary arm contrasts use percentile intervals from 2,000 bootstrap resamples of the paired episode-level absolute-error difference. Deterministic, model-specific seeds make these intervals reproducible. The $k = 0$ pairwise model table uses 4,000 paired bootstrap resamples with seed 20260812 and the common episode intersection for each model pair. Negative differences favor the first-listed condition or model.

The episodes are the resampling units. Intervals are pointwise descriptions of this fixed seeded synthetic evaluation: they are not simultaneous bands across prefixes, and no multiple-comparison correction is applied. Accordingly, the number of pairwise intervals excluding zero is not used as an estimand; each interval supports only its displayed cell.

1200 C.5 Response-inference diagnostics

1201 The tables below retain a common arm and prefix ordering. Within a deployment and prefix, all three
 1202 primary arms use the same complete-case episode set; n is repeated so that every exception is visible.
 1203 Boldface identifies the lowest forecast MAE or slope MAE, and the highest slope correlation, within
 1204 that deployment and prefix.

1205 C.5.1 Per-deployment primary-arm recovery

Table 17: Claude Opus 4.8 forecast and latent-response diagnostics for every prompt arm. Forecast MAE is in poll points; slope MAE and ρ compare the implied response with the true City C slope.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.321	.332	.019
	ABC, no C context	250	2.651	.379	-.063
	ABC + C context	250	1.817	.265	.700
1	Target only	250	3.730	.549	.371
	ABC, no C context	250	2.438	.358	.322
	ABC + C context	250	2.147	.316	.584
2	Target only	250	2.668	.403	.383
	ABC, no C context	250	2.041	.296	.492
	ABC + C context	250	2.079	.308	.491
3	Target only	250	2.485	.358	.449
	ABC, no C context	250	2.128	.305	.513
	ABC + C context	250	1.977	.291	.627
4	Target only	250	2.209	.324	.498
	ABC, no C context	250	1.929	.281	.542
	ABC + C context	250	1.760	.257	.658

Table 18: GPT-5.6 forecast and latent-response diagnostics on the same frozen prompts. Metrics use episodes parseable in all three arms at each prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.596	.511	.054
	ABC, no C context	250	2.518	.352	-.041
	ABC + C context	250	1.596	.224	.678
1	Target only	249	2.993	.439	.394
	ABC, no C context	249	2.768	.399	.302
	ABC + C context	249	1.586	.225	.669
2	Target only	250	2.560	.387	.459
	ABC, no C context	250	2.050	.299	.534
	ABC + C context	250	1.854	.268	.606
3	Target only	250	2.240	.328	.528
	ABC, no C context	250	1.967	.286	.584
	ABC + C context	250	1.773	.256	.600
4	Target only	250	2.153	.318	.577
	ABC, no C context	250	1.746	.253	.653
	ABC + C context	250	1.594	.228	.664

Table 19: DeepSeek-V4-Pro forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.447	.353	-.027
	ABC, no C context	250	2.841	.401	-.027
	ABC + C context	250	2.474	.344	.523
1	Target only	250	3.585	.525	.334
	ABC, no C context	250	3.130	.458	.302
	ABC + C context	250	2.456	.347	.578
2	Target only	250	3.380	.501	.319
	ABC, no C context	250	3.423	.502	.365
	ABC + C context	250	2.687	.391	.492
3	Target only	250	2.908	.424	.454
	ABC, no C context	250	3.224	.463	.403
	ABC + C context	250	2.797	.403	.504
4	Target only	250	2.812	.415	.430
	ABC, no C context	250	3.040	.459	.366
	ABC + C context	250	2.809	.409	.470

Table 20: Kimi K3 forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.352	.476	-.016
	ABC, no C context	250	2.548	.357	-.021
	ABC + C context	250	1.886	.266	.593
1	Target only	250	3.362	.494	.365
	ABC, no C context	250	3.370	.490	.345
	ABC + C context	250	2.382	.341	.579
2	Target only	250	3.194	.491	.387
	ABC, no C context	250	3.114	.487	.352
	ABC + C context	250	2.306	.349	.455
3	Target only	250	2.816	.412	.430
	ABC, no C context	250	2.636	.407	.410
	ABC + C context	250	2.005	.292	.609
4	Target only	250	2.378	.357	.444
	ABC, no C context	250	1.957	.284	.622
	ABC + C context	250	1.840	.263	.601

Table 21: Gemini 3.6 Flash forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.534	.370	.002
	ABC, no C context	250	2.552	.356	.078
	ABC + C context	250	2.174	.302	.559
1	Target only	250	3.733	.550	.403
	ABC, no C context	250	3.717	.548	.361
	ABC + C context	250	3.300	.500	.292
2	Target only	250	3.564	.558	.348
	ABC, no C context	250	3.703	.579	.335
	ABC + C context	250	3.600	.569	.330
3	Target only	250	3.370	.513	.314
	ABC, no C context	250	3.092	.460	.397
	ABC + C context	250	3.047	.443	.471
4	Target only	250	3.308	.498	.389
	ABC, no C context	250	2.911	.447	.433
	ABC + C context	250	2.793	.419	.447

Table 22: Claude Opus 5 forecast and latent-response diagnostics on the same frozen prompts. Two responses exhausted the output budget on extended thinking and returned no answer, one in each ABC arm.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.359	.333	-.018
	ABC, no C context	250	2.619	.371	-.007
	ABC + C context	250	1.764	.247	.617
1	Target only	249	3.549	.520	.391
	ABC, no C context	249	2.303	.338	.370
	ABC + C context	249	1.741	.254	.601
2	Target only	250	3.804	.596	.309
	ABC, no C context	250	2.680	.412	.389
	ABC + C context	250	2.060	.314	.524
3	Target only	250	3.022	.461	.379
	ABC, no C context	250	2.363	.356	.459
	ABC + C context	250	2.075	.309	.550
4	Target only	249	2.700	.418	.443
	ABC, no C context	249	2.369	.366	.459
	ABC + C context	249	2.201	.335	.499

C.5.2 Misleading-cue results

Table 23: Misleading-cue arm: forecast MAE and latent-slope correlation at each City C prefix. The prompt is byte-identical to the correct-context arm except that the City C sentence names the opposite response regime. Negative ρ means the implied response is anti-correlated with the truth, i.e. the deployment has adopted the stated regime. Each of the six deployments contributes all 1,250 prompts.

Model	$k=0$		$k=1$		$k=2$		$k=3$		$k=4$	
	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ
Claude Opus 4.8	4.20	-.66	3.88	-.51	3.04	.09	2.61	.22	2.31	.34
GPT-5.6	4.23	-.68	4.15	-.63	2.78	.17	2.71	.25	2.10	.46
DeepSeek-V4-Pro	4.36	-.62	4.46	-.48	3.61	.07	3.39	.14	3.04	.32
Kimi K3	4.31	-.62	4.12	-.25	3.03	.15	2.72	.14	2.18	.44
Gemini 3.6 Flash	4.29	-.50	4.40	-.06	3.93	.24	3.52	.25	2.93	.36
Claude Opus 5	4.25	-.62	3.15	-.21	2.59	.27	2.41	.41	2.36	.47

1214 **C.5.3 Arbitrary-symbol induction results**

Table 24: **Fifteen-system episode-randomized arbitrary-symbol comparison at $k = 0$.** Panel A reports forecast calibration; Panel B reports response-regime discrimination. All arms share the frozen numeric task content; Section C.3 documents model-specific execution and the Opus 4.8 cross-resource deviation. MAE is in poll points; ρ correlates implied and true latent slopes; Acc. classifies the response regime at slope 0.575. Each Δ is symbol minus no context, with a paired 95% episode-bootstrap interval. With no City C examples, its label can be decoded only from City A/B examples.

Panel A: Forecast calibration						
Model	n	No context	Semantic	Symbol	Δ MAE [95% CI]	
Qwen3-4B	250	2.896	2.365	2.892	−.005	[−.088, .074]
Qwen2.5-7B	250	2.911	2.669	3.248	+.338	[.027, .682]
Qwen3-8B	250	2.985	2.859	2.973	−.011	[−.239, .207]
Qwen2.5-14B	250	2.606	2.548	2.711	+.105	[−.178, .388]
Qwen3-14B	250	2.632	2.241	2.812	+.180	[−.188, .548]
Qwen3-32B	250	2.482	2.069	2.914	+.432	[.119, .752]
Qwen2.5-32B	250	2.679	2.404	3.208	+.529	[.169, .924]
Qwen2.5-72B	250	2.474	2.383	3.398	+.924	[.582, 1.310]
Llama 3.1 8B	242	3.113	3.386	3.605	+.492	[.029, .980]
Llama 3.1 70B	250	3.483	3.761	3.931	+.448	[.030, .886]
DeepSeek V4-Pro	250	2.622	2.588	2.729	+.107	[−.222, .452]
Kimi K3	250	2.548	1.886	2.051	−.497	[−.765, −.211]
Claude Opus 4.8	250	2.651	1.817	2.067	−.584	[−.858, −.314]
Claude Opus 5	250	2.619	1.764	2.017	−.603	[−.869, −.336]
GPT-5.6 Sol	250	2.518	1.596	1.744	−.774	[−1.011, −.533]

Panel B: Response-regime discrimination							
Model	n	$\rho(\hat{g}, g_C)$			Regime accuracy		
		No context	Symbol	$\Delta\rho$ [95% CI]	No context	Symbol	Δ Acc. [95% CI]
Qwen3-4B	250	−.011	.085	+.095 [−.060, .230]	.496	.512	+.016 [−.008, .044]
Qwen2.5-7B	250	−.034	.230	+.264 [.112, .406]	.500	.572	+.072 [.020, .128]
Qwen3-8B	250	−.014	.009	+.023 [−.128, .171]	.492	.516	+.024 [−.012, .060]
Qwen2.5-14B	250	−.029	.208	+.236 [.107, .356]	.492	.580	+.088 [.020, .152]
Qwen3-14B	250	−.057	.342	+.398 [.253, .541]	.496	.660	+.164 [.088, .240]
Qwen3-32B	250	.075	.377	+.302 [.177, .433]	.524	.656	+.132 [.060, .200]
Qwen2.5-32B	250	−.015	.358	+.372 [.226, .502]	.500	.648	+.148 [.088, .208]
Qwen2.5-72B	250	.027	.408	+.381 [.250, .508]	.516	.632	+.116 [.056, .176]
Llama 3.1 8B	242	−.042	.003	+.045 [−.123, .250]	.467	.512	+.045 [−.029, .120]
Llama 3.1 70B	250	−.039	.385	+.424 [.314, .543]	.500	.592	+.092 [.036, .152]
DeepSeek V4-Pro	250	.018	.454	+.436 [.323, .558]	.520	.692	+.172 [.104, .240]
Kimi K3	250	−.021	.612	+.633 [.520, .742]	.468	.804	+.336 [.272, .400]
Claude Opus 4.8	250	−.063	.560	+.623 [.492, .744]	.448	.760	+.312 [.240, .380]
Claude Opus 5	250	−.007	.583	+.591 [.485, .697]	.496	.780	+.284 [.220, .348]
GPT-5.6 Sol	250	−.041	.655	+.696 [.597, .807]	.464	.836	+.372 [.312, .432]

Table 25: **Matched open-weight scaling across target-city observations.** Symbol-minus-no-context effects are reported for Qwen2.5 32B/72B and Llama 3.1 8B/70B at each City C prefix. Negative ΔMAE favors the induced symbol mapping; positive $\Delta\rho$ and $\Delta\text{accuracy}$ indicate better regime discrimination.

Model	k	n	ΔMAE [95% CI]	$\Delta\rho$ [95% CI]	$\Delta\text{Acc.}$ [95% CI]
Qwen2.5-32B	0	250	+.529 [.169, .924]	+.372 [.226, .502]	+.148 [.088, .208]
	1	250	+.124 [−.188, .441]	+.096 [.010, .190]	+.068 [.016, .124]
	2	250	+.184 [−.044, .421]	−.010 [−.059, .037]	−.012 [−.048, .024]
	3	250	+.462 [.222, .710]	−.033 [−.096, .024]	−.004 [−.044, .036]
	4	250	+.255 [−.001, .524]	−.010 [−.059, .033]	−.016 [−.048, .016]
Qwen2.5-72B	0	250	+.924 [.582, 1.310]	+.381 [.250, .508]	+.116 [.056, .176]
	1	250	+.189 [−.148, .520]	+.166 [.075, .265]	+.076 [.024, .128]
	2	250	−.024 [−.286, .239]	+.070 [.017, .127]	+.016 [−.028, .060]
	3	250	+.252 [−.050, .549]	+.019 [−.051, .092]	+.024 [−.024, .068]
	4	250	+.019 [−.280, .322]	+.011 [−.067, .088]	+.020 [−.024, .064]
Llama 3.1 8B	0	242	+.492 [.029, .980]	+.045 [−.123, .250]	+.045 [−.029, .120]
	1	240	−2.318 [−3.256, −1.373]	+.120 [−.022, .263]	−.050 [−.129, .029]
	2	238	−2.454 [−3.433, −1.529]	+.106 [−.038, .260]	−.017 [−.097, .055]
	3	244	−.867 [−1.739, .085]	−.088 [−.238, .061]	−.078 [−.156, .000]
	4	246	−1.396 [−2.321, −.489]	+.100 [−.067, .245]	−.049 [−.122, .024]
Llama 3.1 70B	0	250	+.448 [.030, .886]	+.424 [.314, .543]	+.092 [.036, .152]
	1	250	+.194 [−.245, .658]	+.077 [−.026, .190]	−.016 [−.076, .040]
	2	250	−.200 [−.548, .149]	+.039 [−.026, .106]	−.016 [−.068, .036]
	3	250	−.345 [−.621, −.075]	+.037 [−.015, .087]	+.020 [−.016, .056]
	4	250	−.079 [−.306, .158]	+.034 [−.006, .082]	+.012 [−.020, .048]

At $k = 0$, Qwen3 provides a within-family contrast. Symbol-minus-no-context $\Delta\rho$ is .095 [−.060, .230] at 4B and .023 [−.128, .171] at 8B, but .398 [.253, .541] at 14B and .302 [.177, .433] at 32B; the regime-accuracy contrasts follow the same pattern. The 4B and 8B intervals span zero, whereas the 14B and 32B intervals do not. The 32B point estimate sits below the 14B one, so this is a boundary among the tested Qwen3 checkpoints rather than a monotone or universal parameter threshold. The smaller checkpoints are not wholesale task failures: their semantic-context slopes correlate .370 and .286 with truth, versus −.011 and −.014 without context.

All four tested Qwen2.5 checkpoints discriminate under arbitrary labels ($\Delta\rho = .264/.236/.372/.381$ at 7B/14B/32B/72B), reinforcing the family qualification. Their symbol-minus-no-context MAE changes are +.338/+.105/+.529/+.924: mapping discrimination does not ensure calibrated numerical use. Direct City C examples make the large-Qwen symbol advantage redundant; by $k = 3$ –4, the 32B and 72B $\Delta\rho$ and $\Delta\text{accuracy}$ intervals span zero.

Llama 3.1 provides a second within-family contrast: $\Delta\rho = .045 [−.123, .250]$ at 8B and .424 [.314, .543] at 70B. The 70B regime-accuracy contrast is .092 [.036, .152], although its MAE rises by .448 [.030, .886]. In Qwen3 the mapping-recovery interval excludes zero only from 14B upward, and in Llama only at 70B. The positive Qwen2.5-7B and Qwen2.5-14B estimates, despite Qwen3-8B’s interval spanning zero, indicate that the tested transition depends on model family rather than parameter count. Mapping recovery and numerical calibration remain separate outcomes.

DeepSeek improves discrimination ($\Delta\rho = .436 [.323, .558]$) without a detectable zero-shot MAE change. Kimi, both Opus deployments, and GPT combine mapping recovery with lower forecast error: their $\Delta\rho$ values are .633/.623/.591/.696, while MAE falls by .497/.584/.603/.774 points. Cross-provider magnitudes remain descriptive because runtimes, reasoning settings, and output ceilings differ.

1240 C.5.4 Paired cue effects and interpretation

Table 26: Paired forecast-MAE difference between the correct-context arm and the context-blind ABC OLS benchmark, which the analyst fits through the origin on every displayed A, B and available City C row. Negative values favour the deployment. Boldface marks intervals excluding zero.

Model	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Claude Opus 4.8	−.73	−.18	−.05	−.01	−.08
GPT-5.6	−.95	−.74	−.28	−.21	−.24
DeepSeek-V4-Pro	−.07	.13	.56	.81	.97
Kimi K3	−.66	.06	.18	.02	.01
Gemini 3.6 Flash	−.37	.98	1.47	1.06	.96
Claude Opus 5	−.78	−.57	−.07	.09	.36

At $k = 0$ every deployment matches or beats the context-blind benchmark, and five of the six intervals exclude zero: with no City C cases, the only way to separate City C from the pooled A/B fit is to use its description. The advantage does not persist once City C supplies its own rows. By $k = 2$ DeepSeek-V4-Pro and Gemini 3.6 Flash are significantly worse than the benchmark, Claude Opus 5 becomes so at $k = 4$, and the remaining deployments are indistinguishable from it. The cue effect this experiment measures is therefore a zero-shot phenomenon, and a plain through-origin estimator remains competitive once direct target evidence accumulates.

In the zero-shot matched-reference comparison, the target-city description changes the influence of the A/B cases on forecasts. All six $k = 0$ bootstrap intervals exclude zero. A label-conditioned benchmark that selects a reference, estimates its slope, and transfers that estimate reproduces the aggregate pattern; the comparison does not identify the model’s internal algorithm. More target rows do not monotonically reduce error: DeepSeek and Opus 5 drift by 0.34 and 0.44 points from $k = 0$ to $k = 4$, and Gemini’s context advantage is significant only at $k \in \{0, 1\}$. Short, noisy prefixes can temporarily mislead either arm.

1256 C.6 Does the forecast track the cue-named reference?

Two accounts explain the correct-versus-no-context contrast equally well. Under a lexical-prior account the model maps the City C coverage sentence straight to a remembered response magnitude and never reads City A or City B’s numbers. Under a reference-selection account it uses the cue to choose a reference city, estimates that city’s slope from its four displayed cases, and transfers the estimate. Both predict a regime-level correlation, because the cue and the latent regime coincide.

The design separates them. A through-origin fit to four noisy reference cases has a within-regime standard deviation of .332 against a generator standard deviation of .03, so the matched reference’s fitted slope correlates only .656 with the true City C slope where the regime indicator correlates .996. That sampling error is the discriminating signal: a forecaster that is deterministic given the cue cannot reproduce it, whereas one that transfers a fitted reference slope inherits it. For every deployment, arm and prefix we therefore report

$$r\left(\hat{g}_e, \hat{\beta}_e^{\text{ref}} \mid R_e\right),$$

the partial correlation between the forecast-implied slope and a reference city’s through-origin OLS slope, controlling for the response regime R_e that the City C label names. Arms with no City C label are scored against the true regime and report the regime-matched and regime-mismatched reference instead. Intervals are 2,000-draw episode bootstraps with deterministic per-cell seeds, and the analysis makes no model call.

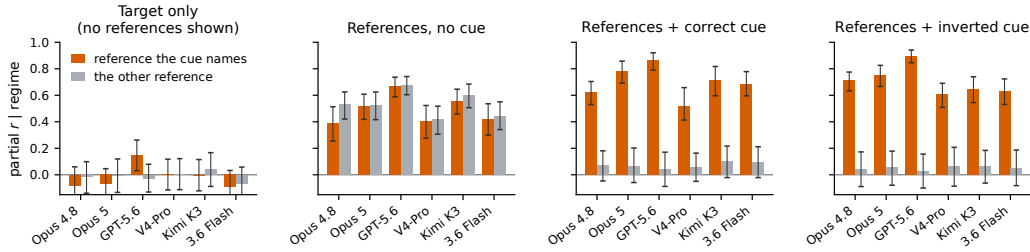


Figure 13: **The cue selects one of the two displayed references.** Partial correlation between each deployment’s forecast-implied City C slope and a reference city’s own fitted slope at $k = 0$, controlling for the response regime the City C sentence names; bars are 95% episode-bootstrap intervals. Orange is the reference that sentence points to, gray the other. With no reference cases displayed neither loads; with both displayed and no cue they load equally, the signature of pooling; the correct cue concentrates loading on the reference it names; and an inverted cue moves that loading onto the reference it falsely names. A prior attached to the words *national* and *local* predicts no loading in any panel. Per-deployment values and the arbitrary-symbol version are in Appendix C.6.

Table 27: Partial correlation between the forecast-implied City C slope and each reference city’s fitted slope at $k = 0$, controlling for the regime the City C label names. “Named” is the reference the label points to, or the regime-matched reference in the two arms without a label. Boldface marks named-reference intervals excluding zero.

Model	Target only		ABC, no label		ABC + correct		ABC + inverted	
	Named	Other	Named	Other	Named	Other	Named	Other
Claude Opus 4.8	-.08	-.02	.39	.53	.62	.07	.71	.04
GPT-5.6	.15	-.03	.67	.68	.86	.04	.90	.03
DeepSeek V4-Pro	.00	.01	.40	.42	.51	.05	.61	.06
Kimi K3	-.01	.04	.55	.60	.71	.10	.64	.06
Gemini 3.6 Flash	-.09	-.07	.42	.44	.69	.09	.63	.05
Claude Opus 5	-.07	-.00	.52	.53	.78	.07	.75	.05

The four arms make four different predictions and all four hold. With no reference cases displayed at all, the target-only arm loads on neither reference: eleven of its twelve intervals span zero. With both references displayed but no City C label, every deployment loads on the two references almost equally, the pooling signature of a model that cannot tell which reference applies. Adding the correct label concentrates the loading on the named reference and removes it from the other, every “other” interval spanning zero. Inverting the label moves the loading wholesale onto the reference that the false cue names, leaving the truth-matched reference near zero.

The forecast therefore tracks the particular reference city the label names, including that city’s sampling error, whether or not the label is correct. No prior attached to the words *national* and *local* reproduces this. It also explains why implied-slope correlations plateau near .66 rather than near .996: faithful transfer of a noisy reference estimate is capped at that reference’s own .656 correlation with the truth. GPT-5.6 and Claude Opus 4.8 slightly exceed the cap at $k = 0$, which is what blending a reference estimate with a regime prior produces.

The estimator only detects an inefficiency. The signal this analysis reads is noise the forecast did not have to import. Each city draws its slope independently, so a reference city’s deviation from its regime mean is uncorrelated with City C’s: simulating the generator gives $\text{corr}(\hat{\beta}^{\text{ref}} - \mu, g_C - \mu) = -.0000$ (Section C.7.4). Within a regime the reference’s fitted slope is therefore pure sampling error about a mean the regime already supplies, and the response-recovery-optimal policy is to read the cue, ignore the reference numbers, and predict the regime mean—which attains $\rho = .996$ against the .656 a faithful transfer is capped at. A forecaster that did this would show no loading on either reference and would be indistinguishable, under this estimator, from the lexical-prior account the section argues against. The estimator thus identifies reference selection only in forecasters that fail to shrink, and has least power on those closest to the shrinking policy: the two deployments above the transfer cap are the two with the highest slope correlation, one of which also has the lowest $k = 0$ forecast error of

the six. We read a positive loading as evidence for reference selection, and a null under this estimator as uninformative rather than as evidence for a prior.

Table 28: The same partial correlation in the arbitrary-symbol arm at $k = 0$, where KIV and ZOR carry no stated direction and their mapping reverses across episodes. The no-label columns repeat each model’s pooling control. Boldface marks symbol named-reference intervals excluding zero.

Model	n	ABC, no label		ABC + symbol	
		Named	Other	Named	Other
Qwen3-4B	250	.05	−.09	.00	−.03
Qwen2.5-7B	250	.11	.15	.28	.10
Qwen3-8B	250	.12	.06	.09	.11
Qwen2.5-14B	250	.31	.20	.32	.03
Qwen3-14B	250	.26	.20	.46	.06
Qwen3-32B	250	.20	.19	.40	.03
Qwen2.5-32B	250	.22	.15	.38	.02
Qwen2.5-72B	250	.25	.18	.53	.04
Llama 3.1 8B	249	−.02	.07	−.00	−.08
Llama 3.1 70B	250	.28	.19	.54	.00
DeepSeek V4-Pro	250	.40	.42	.51	.07
Kimi K3	250	.55	.60	.86	.01
Claude Opus 4.8	250	.39	.53	.71	−.05
Claude Opus 5	250	.52	.53	.83	.02
GPT-5.6 Sol	250	.67	.68	.98	.02

Applying the estimator to the arbitrary-symbol arm removes the lexical account entirely. Twelve of the fifteen released systems have a named-reference interval excluding zero, and the three that do not—Qwen3-4B, Qwen3-8B and Llama 3.1-8B—are exactly the three whose symbol-minus-no-context $\Delta\rho$ intervals span zero in Table 24. The two statistics are computed from different quantities and agree on all fifteen systems.

C.7 Linear-probe analysis of cue-conditioned representations

C.7.1 Behavioral screening criteria

We screen the same held-out prompts with Qwen3-8B, Qwen3-14B, dense Qwen3-32B, and Qwen2.5-72B-Instruct. Each checkpoint received 16 test episodes crossed with $k \in \{0, 4\}$ and correct, absent, or inverted City C context (96 prompts), with 96/96 parseable forecasts.

Table 29: **Behavioral screening results for the linear-probe analysis.** Entries are paired mean differences with 95% episode-bootstrap intervals ($n = 16$). Evidence is no-context $k = 4$ minus $k = 0$ aligned slope; selection is correct minus absent context at $k = 0$; reversal is inverted minus correct context at $k = 0$; forecast penalty is inverted minus correct poll MAE at $k = 0$. Positive evidence and selection, negative reversal, and a positive forecast penalty are the predicted directions.

Model	Evidence	Selection	Reversal	Forecast penalty
Qwen3-8B	+ .119 [−.094, +.374]	+ .078 [−.012, +.219]	− .140 [−.315, +.007]	+ .812 [−.063, +2.023]
Qwen3-14B	+ .394 [+ .152, +.734]	+ .261 [+ .134, +.400]	− .380 [−.597, −.172]	+1.662 [+ .252, +3.155]
Qwen3-32B	+ .315 [+ .017, +.646]	+ .234 [+ .111, +.407]	− .390 [−.554, −.221]	+1.965 [+1.013, +3.083]
Qwen2.5-72B	+ .246 [+ .001, +.529]	+ .096 [−.077, +.259]	− .433 [−.674, −.220]	+2.206 [+ .957, +3.620]

The 14B and 32B intervals exclude zero in all four predicted directions. For 72B, the evidence, reversal, and forecast-penalty intervals exclude zero, while the correct-versus-absent selection interval spans zero. All four 8B intervals span zero. We use 32B for the dense Qwen3 scaling comparison and 72B for a cross-generation comparison; neither is interpreted as another point on a monotonic size curve. Qwen3-4B and Llama-3.1-8B use the identical task file, held-out split, and model, layer, target, and endpoint selection rules.

1318 C.7.2 Activation and probe protocol

1319 We selected 64 episodes without replacement, balanced over which reference city was strongly
 1320 responsive and whether City C’s latent regime was strong or weak. Forty-eight episodes form a
 1321 development set and 16 form a sealed test set. Each episode is rendered at $k = 0$ and $k = 4$ under
 1322 correct, absent, and inverted context, giving 384 activation records. Within an episode-prefix pair
 1323 the three prompts have identical numeric content. The only intervention is the City C relationship
 1324 description.

1325 For the final input token, we save the embedding output and every transformer-block output.
 1326 The resulting tensor dimensions are [384, 37, 2560] for Qwen3-4B; [384, 37, 4096] for Qwen3-8B;
 1327 [384, 33, 4096] for Llama-3.1-8B; [384, 41, 5120] for Qwen3-14B; [384, 65, 5120] for Qwen3-32B;
 1328 and [384, 81, 8192] for Qwen2.5-72B-Instruct. We fit separate dual-ridge probes at each layer and k ,
 1329 using only correct-context development episodes. Four cell-balanced development folds select the
 1330 ridge strength from $\{10^{-4}, 3 \times 10^{-3}, 10^{-1}, 3, 100\}$ and then select one reported layer. The untouched
 1331 test episodes are evaluated in all three context arms ($n = 16$ per arm). A 10,000-repeat held-out label
 1332 permutation test keeps the development-selected layer, ridge strength, and fitted probe fixed. Thus
 1333 neither the test labels nor test-arm scores select the reported layer.

1334 We probe three increasingly strict targets: the binary strong/weak regime; the full realized slope g_C ;
 1335 and the within-regime residual $g_C - .90$ for strong episodes or $g_C - .25$ for weak episodes. The first
 1336 two are recoverable from the displayed evidence and the third is not, by construction: Section C.7.4
 1337 derives what a Bayes-optimal reader of the same prompt could score on each, and the residual is
 1338 included as a specificity control rather than as a target a model could hit. Mean-pooled embedding
 1339 and final-layer states are additional pooling controls.

Table 30: **Six-model correct-context probe summary.** Each entry gives $k = 0/k = 4$ on the 16 sealed episodes. Layers and ridge strengths were selected on development episodes only. Residual p is the one-sided 10,000-repeat held-out label-permutation value.

Model	Regime AUC	Full-slope R^2	Residual-slope R^2	Residual p
1340 Qwen3-4B	1.000/1.000	.960/.885	-.389/-.227	.927/.614
Qwen3-8B	.844/1.000	.713/.653	-.359/-.374	.502/.735
Llama-3.1-8B	1.000/.984	.913/.731	-.126/-.207	.380/.572
Qwen3-14B	1.000/.984	.898/.627	-.531/-.128	.315/.450
Qwen3-32B	1.000/1.000	.967/.811	-.269/.003	.477/.221
Qwen2.5-72B	1.000/1.000	.975/.869	-.451/-.053	.675/.447

Table 31: **Held-out layer-wise open-weight probe.** AUC is reported for the regime target and R^2 for both slope targets. Correct, none, and wrong score the same development-selected probe against the true City C target under the three context arms; wrong/cue instead scores the inverted arm against the relationship named by its cue. p is the one-sided held-out label-permutation value on the correct-context test episodes.

Model	k	Target	Layer	Dev CV	Correct	None	Wrong/true	Wrong/cue	p
Qwen3-4B	0	Regime (AUC)	1	1.000	1.000	.562	.000	1.000	< .001
		Full slope (R^2)	24	.936	.960	-.517	-2.614	.956	< .001
		Residual slope (R^2)	15	.054	-.389	-.296	-.258	–	.927
	4	Regime (AUC)	19	1.000	1.000	.609	.094	.906	< .001
		Full slope (R^2)	20	.805	.885	-.065	-1.147	.492	< .001
		Residual slope (R^2)	14	.004	-.227	.089	-.183	–	.614
	0	Regime (AUC)	19	1.000	.844	.453	.078	.922	.009
		Full slope (R^2)	23	.870	.713	-.697	-2.344	.771	< .001
		Residual slope (R^2)	1	.063	-.359	-.673	-.378	–	.502
Qwen3-8B	4	Regime (AUC)	23	.925	1.000	.578	.000	1.000	< .001
		Full slope (R^2)	23	.462	.653	-.512	-1.550	.693	< .001
		Residual slope (R^2)	33	.030	-.374	-1.003	-.029	–	.735
	0	Regime (AUC)	1	1.000	1.000	.359	.000	1.000	< .001
		Full slope (R^2)	1	.907	.913	-18.953	-2.629	.972	< .001
		Residual slope (R^2)	12	.018	-.126	-.458	-.227	–	.380
	4	Regime (AUC)	1	1.000	.984	.438	.000	1.000	< .001
		Full slope (R^2)	1	.751	.731	-15.450	-2.082	.808	< .001
		Residual slope (R^2)	10	.006	-.207	-.148	-.150	–	.572
Qwen3-14B	0	Regime (AUC)	2	1.000	1.000	.422	.000	1.000	< .001
		Full slope (R^2)	26	.874	.898	-.522	-2.423	.943	< .001
		Residual slope (R^2)	34	.159	-.531	-9.255	-.096	–	.315
	4	Regime (AUC)	26	.880	.984	.688	.109	.891	< .001
		Full slope (R^2)	26	.408	.627	-.411	-1.191	.409	< .001
		Residual slope (R^2)	1	-.013	-.128	-.605	-.159	–	.450
	0	Regime (AUC)	27	1.000	1.000	.359	.000	1.000	< .001
		Full slope (R^2)	47	.930	.967	-.027	-2.570	.958	< .001
		Residual slope (R^2)	1	.062	-.269	-.232	-.259	–	.477
Qwen3-32B	4	Regime (AUC)	44	1.000	1.000	.562	.000	1.000	< .001
		Full slope (R^2)	47	.797	.811	-.160	-1.694	.666	< .001
		Residual slope (R^2)	1	.043	.003	.062	-.000	–	.221
	0	Regime (AUC)	53	1.000	1.000	.469	.000	1.000	< .001
		Full slope (R^2)	58	.955	.975	-1.050	-2.958	.969	< .001
		Residual slope (R^2)	32	.061	-.451	-1.192	-.401	–	.675
	4	Regime (AUC)	58	.995	1.000	.922	.062	.938	< .001
		Full slope (R^2)	58	.678	.869	-.468	-1.072	.493	< .001
		Residual slope (R^2)	1	-.017	-.053	-.063	-.053	–	.447

C.7.3 Regime and full-slope decodability

Across the six checkpoints and two evidence depths, correct-context regime AUC is .844–1.000 and full-slope R^2 is .627–.975 (Table 30). This information is not a context-invariant encoding of the true coefficient: removing the City C description drives selected-layer full-slope R^2 below zero in all 12 model- k cells. Under inversion, true-target regime AUC falls to .000–.109 while cue-target AUC is .891–1.000; the full-slope probe likewise fits the cue-assigned relationship better than the true one. The Llama replication shows that this pattern is not specific to the Qwen checkpoints. Across both

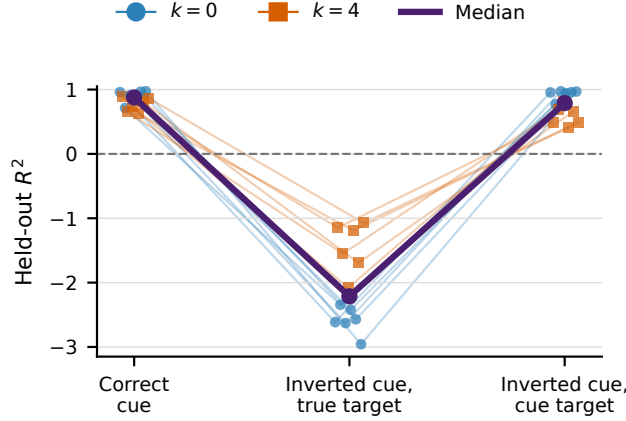


Figure 14: **Semantic-arm slope decoding follows cue assignment.** Each line is a sealed model- k cell across the six open checkpoints of the 16-episode roster; purple is the median. Probes are fit on correct-context development episodes. Under inversion, held-out R^2 collapses against the true slope and recovers against the cue-assigned slope. The input-embedding control below shows this signal is available before any computation, which is why the arbitrary-label replication rather than this figure carries the representational claim.

1349 model families, the linearly available signal therefore tracks the response family that context assigns
1350 to City C.

1351 C.7.4 What each probe target can be scored against

1352 A probe is scored against a latent generator quantity, so a null is only a fact about a model if the
1353 displayed evidence determines that quantity. We therefore derive, for each target and evidence depth,
1354 the score a Bayes-optimal reader of the same prompt would obtain. This is an analytic property of
1355 Equation 3.2 and its row construction, computed in closed form and re-derived by simulating the
1356 generator’s own row logic; the two agree to within .005 and no model output enters; the derivation is
1357 `exp2_v2/biased_news/analysis/compute_bayes_ceilings.py`. Table 32 gives the ceilings.
1358 The same module reports forecast-MAE floors: 2.28 poll points at $k = 0$ with no cue, 0.17 for an
1359 oracle told the regime means, and 1.78 for a reader who must instead estimate the named regime’s
1360 slope from its four displayed rows.

1361 The same simulation records one quantity the reference-selection analysis of Section C.6 depends
1362 on. Cities draw their slopes independently, so a reference city’s deviation from its regime mean
1363 carries no information about City C’s: $\text{corr}(\hat{\beta}^{\text{ref}} - \mu, g_C - \mu) = -.0000$ over 400,000 simulated
1364 episodes, against a reference-fit standard deviation of .318 within regime. That is why a loading on
1365 the reference’s own sampling error identifies reference selection, and equally why a forecaster that
1366 shrank optimally would show none.

Table 32: Maximum score a Bayes-optimal reader of the same prompt can obtain on each probe target, under the frozen data-generating process. The regime and the full slope are recoverable; the within-regime residual is not, because it is generated with standard deviation .03 while a through-origin fit to City C’s displayed rows carries a standard deviation of .318. At $k = 0$ City C displays no rows at all and the residual ceiling is exactly zero.

1367

Probe target		$k = 0$	$k = 4$
Regime, correct semantic cue	AUC	1.000	1.000
Regime, arbitrary label	AUC	0.979	—
Full slope g_C	R^2	0.992	0.992
Residual slope $g_C - \mu$	R^2	0.000	0.009

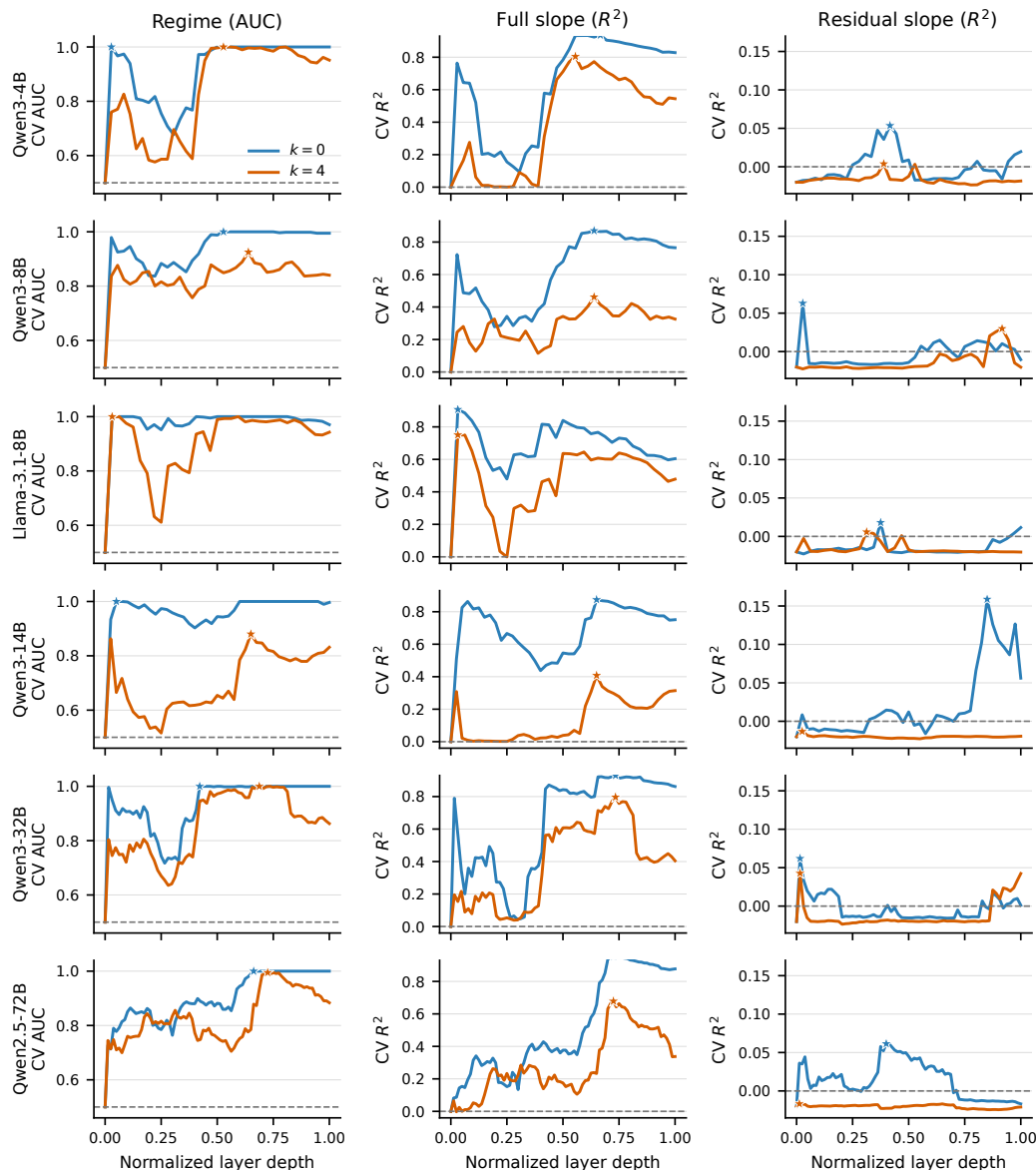


Figure 15: **Development layer trajectories for the six-model probe.** Curves show four-fold development CV rather than test performance; stars mark the development-selected layer. Layer depth is normalized because the checkpoints have 32–80 transformer blocks. The regime and full-slope signals are broad, whereas residual-slope CV is weak and does not survive held-out evaluation.

1368 C.7.5 Specificity: within-regime residuals

1369 The residual target is the specificity control, and it behaves as a target with no available signal should.
 1370 The 12 correct-context residual-slope R^2 values range from $-.531$ to $.003$, none of their permutation
 1371 tests rejects the null ($p = .221-.927$), and 33 of 36 arm-level residual scores are nonpositive. Against
 1372 a ceiling of $.000$ at $k = 0$ and $.009$ at $k = 4$, that is the result the design permits: no reader of
 1373 these prompts, and no representation computed from them, could score materially above zero. We
 1374 therefore read the residual null as evidence that the probe does not manufacture structure where none
 1375 is available, and not as a limit on what these checkpoints encode. A probe that returned a positive
 1376 residual score here would be the finding, and it would be a warning about the probe.

1377 The two recoverable targets do carry information, and the input-embedding control is what bounds
1378 their interpretation. Mean-pooled input embeddings for every Qwen checkpoint already give perfect
1379 regime AUC and full-slope $R^2 = .988-.993$ under correct context—at the .992 ceiling—because the
1380 explicit description names a regime and the regime means account for essentially all slope variance.
1381 Llama input embeddings likewise give perfect regime AUC and full-slope $R^2 = .756/.827$. The
1382 layer-wise semantic result is therefore evidence that the cue-selected response family is linearly
1383 represented, and reaches nothing further about the continuous g_C : the part of g_C beyond the family is
1384 not in the prompt to begin with. The original behavioral inversion screen connects this representation
1385 to output sensitivity in the three larger screened checkpoints: inverted context raises $k = 0$ forecast
1386 MAE by 1.66, 1.96, and 2.21 points for 14B, 32B, and 72B, respectively.

1387 **C.7.6 Power-matched and cross-model arbitrary-symbol replication**

1388 The six-checkpoint roster above seals 16 test episodes per arm. Its exact label-permutation null spans
1389 ROC AUC $[.203, .797]$, so it can separate a score from chance only above about .75. That resolution
1390 suffices for the semantic arm, whose scores sit at ceiling, but it cannot evaluate a weaker signal
1391 at all. We therefore evaluate Qwen3-14B at the design’s balanced maximum—248 episodes, 160
1392 development and 88 sealed test, 1,488 activation records—once with the semantic cue and once with
1393 the episode-randomized KIV/ZOR labels. Anchor, targets, fold structure, ridge grid, layer-selection
1394 rule, pooling controls and the 10,000-repeat held-out permutation test are unchanged; only the number
1395 of episodes differs. The 88-episode null reaches AUC .603. The enlarged development set is disjoint
1396 from every episode used by the 16-episode analysis, providing an independent estimate rather than a
1397 re-slice.

Table 33: **Power-matched Qwen3-14B probe, 88 sealed episodes per arm.** Correct, none and wrong score the development-selected probe against the true City C target under the three cue arms; wrong/cue scores the inverted arm against the relationship its cue names. “Layer” is the development-selected block of 40 and “Embed.” is the mean-pooled input-embedding control under correct context. p is the one-sided held-out label-permutation value.

Run	Target	Correct	None	Wrong	Wrong/cue	Layer	Embed.	p	Behav. $\Delta\rho$
Qwen3-14B, semantic	$k=0$ regime AUC	1.000	.429	.000	1.000	1	1.000	<.001	—
	$k=4$ regime AUC	1.000	.560	.000	1.000	1	1.000	<.001	
	$k=0$ slope R^2	.942	-.300	-2.806	.961	27	.989	<.001	
	$k=4$ slope R^2	.910	-24.039	-2.387	.906	1	.989	<.001	
Qwen3-14B, symbol	$k=0$ regime AUC	.678	.467	.321	.679	26	.555	.0016	.398
	$k=4$ regime AUC	.713	.689	.747	.253	40	.582	<.001	
	$k=0$ slope R^2	.016	-.094	-.130	.065	26	.003	.1162	
	$k=4$ slope R^2	.030	-.237	.126	-.855	40	-.032	<.001	
Qwen3-8B, symbol	$k=0$ regime AUC	.566	.535	.567	.433	10	.534	.1471	.023
	$k=4$ regime AUC	.610	.545	.593	.407	19	.621	.0401	
	$k=0$ slope R^2	-.001	.005	.004	-.007	25	.003	.4883	
	$k=4$ slope R^2	-.151	-.219	-.188	-.504	20	-.018	.0314	
Qwen3-4B, symbol	$k=0$ regime AUC	.520	.419	.329	.671	23	.498	.3756	.095
	$k=4$ regime AUC	.685	.689	.637	.363	22	.585	.0017	
	$k=0$ slope R^2	-.009	-.065	-.079	.023	23	.004	.2899	
	$k=4$ slope R^2	-.021	-.265	-.031	-.687	23	.004	.0014	
Qwen2.5-32B, symbol	$k=0$ regime AUC	.778	.571	.200	.800	50	.550	<.001	.372
	$k=4$ regime AUC	.797	.831	.662	.338	46	.589	<.001	
	$k=0$ slope R^2	.217	-.256	-1.033	.254	50	.004	<.001	
	$k=4$ slope R^2	.263	-.563	-.044	-.774	46	-.029	<.001	
Qwen2.5-72B, symbol	$k=0$ regime AUC	.662	.492	.210	.790	63	.556	.0026	.381
	$k=4$ regime AUC	.793	.745	.651	.349	61	.575	<.001	
	$k=0$ slope R^2	-.005	-.490	-1.034	.187	62	.004	.0072	
	$k=4$ slope R^2	.222	-.624	-.129	-.870	61	-.047	<.001	
Llama 3.1-8B, symbol	$k=0$ regime AUC	.423	.447	.452	.548	11	.539	.8925	.045
	$k=4$ regime AUC	.543	.604	.549	.451	31	.604	.2470	
	$k=0$ slope R^2	-.097	-.089	-.086	-.001	11	.003	.8607	
	$k=4$ slope R^2	-.275	-.091	-.156	-.358	31	.009	.4529	
Llama 3.1-70B, symbol	$k=0$ regime AUC	.728	.468	.292	.708	75	.495	<.001	.424
	$k=4$ regime AUC	.810	.740	.721	.279	42	.606	<.001	
	$k=0$ slope R^2	.076	-8.200	-.961	.082	75	.003	<.001	
	$k=4$ slope R^2	.222	-3.224	.087	-1.007	55	.007	<.001	

The two arms separate sharply, and the input-embedding control is what separates them. Under the semantic cue the development-selected layer is the first transformer block for three of the four conditions, and mean-pooled input embeddings alone already reach regime AUC 1.000 and full-slope $R^2 = .989$: the decodable quantity is the cue token itself. Under arbitrary labels the same control collapses to AUC .555 and $R^2 = .003$, the selected layers move to blocks 26 and 40 of 40, and held-out regime AUC is .678 at $k = 0$ ($p = .0016$) and .713 at $k = 4$ ($p < .001$). The induced label-to-regime mapping is therefore linearly represented, but only after computation, far more weakly than a regime the model was told outright, and below the .979 AUC a Bayes-optimal reader of the same two reference cities could reach. Unlike the residual target, the full slope is recoverable here—its ceiling is .992—so its symbol-arm score of $R^2 = .016$ at $k = 0$, which does not reject its null, is a genuine gap rather than a property of the design.

Representation and behavior agree closely once the cue must be induced. On the same 88 sealed episodes the model’s own forecasts give regime AUC .702, .518 and .331 under correct, absent and inverted labels, against probe AUCs of .678, .467 and .321; inversion raises $k = 0$ poll MAE from 2.49 to 4.28. With four City C cases the inverted-arm representation reverses: true-target regime AUC rises from .321 to .747, and the full-slope probe fits the cue-named relationship at $R^2 = -.855$ while fitting the true one at $+.126$. Direct target evidence overrides the induced label in the representation as it does in the forecast.

Six further checkpoints received the identical symbol protocol at the same 88 sealed episodes: Qwen2.5-32B and Qwen2.5-72B, which change model generation; Qwen3-8B and Qwen3-4B, which hold family, tokenizer and architecture fixed and change only scale; and Llama 3.1-8B and Llama 3.1-

1420 70B, which do the same within a second vendor and architecture. Held-out regime AUC at $k = 0$ is
1421 .778 for Qwen2.5-32B ($p < .001$, selected layer 50 of 64), .728 for Llama 3.1-70B ($p < .001$, layer
1422 75 of 81), .662 for Qwen2.5-72B ($p = .003$, layer 63 of 80), .566 for Qwen3-8B ($p = .15$), .520 for
1423 Qwen3-4B ($p = .38$) and .423 for Llama 3.1-8B ($p = .89$). No null checkpoint rejects in any $k = 0$
1424 condition—regime, slope or residual slope. Mean-pooled input embeddings stay between .495 and
1425 .556 in every symbol run, so no symbol arm carries the mapping before computation.

1426 Decodability tracks behavioral recovery across all seven checkpoints, four positive and three null.
1427 The four whose forecasts recover the mapping are the four in which it decodes (Llama 3.1-70B,
1428 $\Delta\rho = .424$; Qwen3-14B, .398; Qwen2.5-72B, .381; Qwen2.5-32B, .372), and the three whose
1429 forecasts do not are the three in which it does not (Qwen3-4B, .095; Llama 3.1-8B, .045; Qwen3-8B,
1430 .023; all three intervals spanning zero). The Llama 3.1 contrast carries the weight, because it holds
1431 family, tokenizer and architecture fixed, varies only scale, and is the one within-family contrast whose
1432 behavioral classification survives the harder generator population of Section C.8: 8B is null on both
1433 populations and 70B positive on both, while the probe runs .423 to .728 across them. Because the
1434 null and the positive share an architecture and a tokenizer, the split is not an artifact of either.

1435 The Qwen3 sequence—.520, .566, .678 from 4B to 14B—is consistent with this but does not add
1436 independent weight, because the population that would test it is the one on which its behavioral
1437 boundary moves. That movement is smaller than the $\Delta\rho$ column alone suggests. Symbol-arm
1438 regime accuracy is .512, .516, .660, .656 originally against .528, .554, .589, .606 on the harder
1439 population. The larger pair remains above the smaller pair, but 14B and 32B exchange order and
1440 the between-group separation shrinks. The $\Delta\rho$ contrast then subtracts a no-context baseline whose
1441 standard error is about .064, and baselines of $-.036$ for Qwen3-8B and $+.025$ for Qwen3-14B are
1442 enough to put one interval above zero and the other .004 below it, with the two intervals overlapping
1443 by 78%. We therefore read the harder population as preserving the broad size-group contrast without
1444 locating the significance threshold at a particular checkpoint. Section C.8 should be read alongside
1445 the seven-for-seven agreement, which is evidence that the probe reads what the forecasts express
1446 on the episodes both were measured on and not evidence for a scale law. The positives span three
1447 model generations and two vendors, and every positive shows the same shape: the decoded regime
1448 follows the cue at $k = 0$ and the truth at $k = 4$. Qwen3-4B and Qwen3-8B do reject at $k = 4$ (.685,
1449 $p = .002$ and .610, $p = .04$), which is expected rather than contrary: at $k = 4$ the displayed target
1450 cases exhibit the regime whatever the label says, and the mean-pooled input embeddings rise to .585
1451 and .621 accordingly, in Qwen3-8B’s case above the probe itself. The induction claim is a $k = 0$
1452 claim throughout. The sealed test is 88 episodes per model, well above the 16 the roster began with
1453 but not a large sample, and the permutation nulls are computed rather than assumed.

1454 At 16 episodes, the same arbitrary-symbol procedure returns $k = 0$ regime AUC .312 with $p = .90$.
1455 Its [.203, .797] null is too broad to resolve a signal of the magnitude observed at 88 episodes.

1456 C.7.7 Causal test: patching the City C label representation

1457 Decodability is not mediation. To test whether the representation the probe reads is the one the
1458 forecast uses, we intervene on it directly. Under arbitrary labels, an episode’s correct and swapped
1459 prompts are positionally aligned through the City C label and differ at exactly two label-token
1460 positions. We replace the residual stream at those two positions with the states the opposite arm
1461 produces, in both directions, and regenerate the forecast greedily. The endpoint is the episode-level
1462 mean of the regime-aligned correct-into-swapped shift and the sign-reversed swapped-into-correct
1463 shift, tested one-sided against an episode sign-flip null.

1464 The analysis uses three patch sites and two controls. The primary site is a three-layer window centred
1465 on the development-selected relational layer; patching at the embedding output only, and at every
1466 causally effective layer, are the comparison sites; the unpatched arm and a self-patch that writes
1467 each state back over itself are the controls. Layer, window and ridge were selected on development
1468 episodes with the sealed test unread.

Table 34: **Causal effect of patching the two City C symbol subtokens, Qwen3-14B.** Shift is the regime-aligned symmetric patch mean; positive means the forecast moved toward the relationship the donor label names. The 16- and 88-episode analyses select their windows independently on their own development sets. The null interval is the 88-episode sign-flip null.

Patch site	k	16 sealed, window 33–35			88 sealed, window 19–21			Null 95%
		Shift	p	n	Shift	p	n	
Selected window	$k=0$	-.033	.75	14	.333	<.0001	82	[-.177, .178]
	$k=4$	-.007	.75	15	-.035	.8256	87	[-.073, .073]
Embedding only	$k=0$	.236	.15	14	.319	.0002	84	[-.182, .182]
	$k=4$	.115	.23	15	-.017	.6419	87	[-.087, .087]
All layers	$k=0$	.236	.15	14	.319	.0002	84	[-.182, .182]
	$k=4$	.115	.23	15	-.017	.6419	87	[-.087, .087]

At $k = 0$ the intervention moves the forecast: patching the selected window shifts the regime-aligned prediction by .333 against a null interval of $[-.177, .178]$ ($p < .0001$, 82 complete episodes), and patching the embedding alone gives .319. All eight stability checks pass, including 347/347 exact self-patch matches.

Two readings of the table matter more than the significance. First, the embedding-only and all-layer rows are identical, and necessarily so: the two prefixes differ only at the patched label positions, so overwriting their embeddings and letting the model recompute reproduces what overwriting every later state does. They are one intervention reported at two depths, not two converging results. What distinguishes the mid-window row is that it reaches the same magnitude while touching only three layers late in the network.

Second, the effect is confined to $k = 0$. With four City C cases every patch site returns a null ($p = .83$ and $p = .64$), so the induced label no longer controls the forecast once direct target evidence is available. That is the revision claim of Section C.5 in causal form, and it matches the representational result: at $k = 4$ the inverted-arm probe recovers the true regime rather than the cue-assigned one.

At 16 episodes, the primary patch estimate is $-.033$ at $p = .75$ and the sign-flip null spans $[-.387, .387]$. Its 64-episode development set selects layers 33–35, whereas the 160-episode development set selects 19–21 at a higher cross-validation score (.903 against .806).

Permutation implementation. The test enumerates all 2^n sign assignments for 16 episodes; exhaustive enumeration is infeasible for 88 ($2^{88} \approx 3 \times 10^{26}$). Enumeration is retained for $n \leq 20$; above it the same null is sampled with 200,000 draws under a fixed seed, giving a Monte Carlo standard error of .0005 at $p = .05$. The estimand, endpoint, and direction are identical under both implementations.

Scope. This is one checkpoint under arbitrary labels. It shows that the label representation Qwen3-14B computes is causally responsible for its zero-shot regime assignment, and that direct evidence displaces it. It does not establish the same for the hosted deployments, the semantic cue, or other architectures.

C.7.8 Reproducibility and scope

The checkpoints are pinned to commits cdbee75f17c01a7cc42f958dc650907174af0554 (Qwen3-4B), b968826d9c46dd6066d109eabc6255188de91218 (Qwen3-8B), 0e9e39f249a16976918f6564b8830bc894c89659 (Llama-3.1-8B), 40c069824f4251a91eefaf281ebe4c544efd3e18 (Qwen3-14B), 9216db5781bf21249d130ec9da846c4624c16137 (Qwen3-32B), and 495f39366efef23836d0cfaf4f635880d2be31 (Qwen2.5-72B). The activation hashes are e191385963d639aa5b8574a9abbe91b3519aed934ac491f863655631b26dbbc (Qwen3-4B) and 7f65aeec7a91e404cc60068618fedf25c679abde5d48c2b8477f736670fbc54f (Llama-3.1-8B); all six hashes and tensor dimensions are enforced by the fail-closed generator. Frozen tasks, activation tensors, extraction manifests, generated behavior, selected predictions, complete layer curves, and probe results are under exp2_v2/biased_news/data/coin_city_stable_relationship_claude_n250_v4/mechanistic_probe/. The analysis and asset generator are exp2_v2/biased_news/mechanistic_probe/analyze_probe.py and paper/generate_

coin_city_mechanistic_comparison.py. The causal run is mechanistic_probe/qwen3_14b_symbol_relational_v2/, frozen by freeze_symbol_relational_protocol.py and executed by extract_symbol_label_states.py, patch_symbol_label_activations.py and analyze_symbol_relational_probe.py; every count in that pipeline is derived from the frozen task manifest and checked by tests/test_symbol_relational_scaling.py. paper/generate_coin_city_causal_patch_table.py fails closed on the run’s study name, sealed-test size and stability gate. The power-matched runs are mechanistic_probe/qwen3_14b_probe_v2/, mechanistic_probe/qwen3_14b_symbol_probe_v2/, mechanistic_probe/qwen3_8b_symbol_probe_v2/, mechanistic_probe/qwen3_4b_symbol_probe_v2/, mechanistic_probe/llama3_1_70b_symbol_probe_v2/, mechanistic_probe/qwen2_5_72b_symbol_probe_v2/, mechanistic_probe/qwen2_5_32b_symbol_probe_v2/ (checkpoint 5ede1c97bbab6ce5cda5812749b4c0bdf79b18dd) and mechanistic_probe/llama3_1_8b_symbol_probe_v2/ (checkpoint 0e9e39f249a16976918f6564b8830bc894c89659), built by exp2_v2/biased_news/mechanistic_probe/build_probe_tasks.py and build_arbitrary_symbol_probe_tasks.py with 62 and 31 episodes per factorial cell; paper/generate_coin_city_probe_power_table.py regenerates their table and validates the sealed-test size and task hash. Each run’s activation tensor is regenerable from its frozen task file and pinned checkpoint; the extraction manifests record the tensor hashes rather than the tensors. Cross-vendor probe evidence here is the Llama 3.1-8B null rather than a positive. The diagnostic covers six checkpoints spanning Qwen and Llama at 16 sealed episodes per arm, plus the matched 88-episode semantic and arbitrary-symbol Qwen3-14B pair. For the semantic cue it is correlational and does not establish that the decoded direction mediates the forecast; under arbitrary labels Section C.7.7 provides a causal intervention for one checkpoint. Both complement the paired behavioral interventions above.

C.8 Robustness across generator populations and deployments

C.8.1 Generator-population robustness

Fresh seeds 940000–940249 target strong/weak means of .75/.40 with slope SD .10, jointly narrowing the regime gap and widening within-regime dispersion. The realized target slopes are $.752 \pm .100$ and $.404 \pm .105$. Draws were neither truncated nor reordered: two reference pairs reverse their nominal latent ordering and remain in the analysis. Case noise, factorial balance, prompts, prefix set, local checkpoint settings, and episode-randomized symbol mapping are unchanged.

All ten open checkpoints were run on the three matched arms, producing 37,500 response records; 37,355 contain a numeric forecast, and unresolved formats remain missing. Table 35 compares the $k = 0$ symbol-minus-no-context contrast across generator populations. Bold intervals exclude zero; negative ΔMAE would favor symbol context. Because this population changes separation and dispersion together, it tests the joint perturbation rather than identifying either parameter’s effect separately.

Table 35: Arbitrary-symbol recovery on the original and harder generator populations. Intervals are 95% episode-bootstrap intervals with 5,000 draws; n is the harder-population complete-case count.

Model	n	Original $\Delta\rho$ [95% CI]	Harder $\Delta\rho$ [95% CI]	Symbol acc.	Harder ΔMAE [95% CI]
Qwen3-4B	250	+ .095 [−.060, .230]	+ .050 [−.079, .197]	.528	+ .052 [−.056, .161]
Qwen3-8B	249	+ .023 [−.128, .171]	+ .231 [.074, .370]	.554	−.113 [−.269, .036]
Qwen3-14B	246	+ .398 [.253, .541]	+ .155 [−.004, .303]	.589	+ .361 [.076, .644]
Qwen3-32B	249	+ .302 [.177, .433]	+ .193 [.063, .325]	.606	+ .848 [.539, 1.168]
Qwen2.5-7B	250	+ .264 [.112, .406]	+ .149 [.009, .279]	.540	+ .303 [.000, .627]
Qwen2.5-14B	250	+ .236 [.107, .356]	+ .133 [.005, .254]	.584	+ .365 [.115, .609]
Qwen2.5-32B	250	+ .372 [.226, .502]	+ .247 [.109, .383]	.616	+ .882 [.590, 1.186]
Qwen2.5-72B	250	+ .381 [.250, .508]	+ .336 [.204, .463]	.604	+ 1.012 [.717, 1.316]
Llama 3.1-8B	229	+ .045 [−.123, .250]	−.077 [−.216, .079]	.524	+ .597 [.229, .952]
Llama 3.1-70B	249	+ .424 [.314, .543]	+ .242 [.119, .369]	.558	+ .856 [.465, 1.242]

Which Qwen3 checkpoints clear significance changes, while the larger pair retains higher symbol-arm regime accuracy than the smaller pair. Qwen3-8B moves from an original null to $\Delta\rho = .231$ [.074, .370] while Qwen3-14B falls to .155 [−.004, .303]; Qwen3-4B remains null and Qwen3-32B

remains positive. Regime accuracy in the symbol arm, which subtracts no baseline, is .512, .516, .660, .656 originally against .528, .554, .589, .606 here. The 14B and 32B checkpoints exchange order, and the smaller models improve slightly, but the gap between 8B and 14B shrinks from .144 to .035. The $\Delta\rho$ contrast then subtracts a no-context baseline carrying a standard error of about .064, and baselines of $-.036$ for Qwen3-8B against $+.025$ for Qwen3-14B suffice to place one interval above zero and the other .004 below it, with the two overlapping by 78%. We read this as a detection threshold that is no longer locatable to a checkpoint on the compressed population; it does not support a fixed scale threshold.

All four Qwen2.5 intervals remain positive, from .149 [.009, .279] at 7B to .336 [.204, .463] at 72B, and both Llama checkpoints keep their classification, 8B null in each population and 70B positive in each. Seven of ten intervals therefore exclude zero in each population and eight of ten checkpoints keep their classification, the Qwen3 8B/14B pair being the two that do not. Mapping recovery survives the substantially less separable population; the checkpoint at which it becomes detectable does not, and neither does a scale reading across families, since Qwen2.5-14B sits below Qwen2.5-7B in both. Part of the shrinkage is mechanical: the regime explains less of the slope here, capping ρ at .87 against .996. The observed $\Delta\rho$ falls further than that alone predicts. Numerical use stays distinct: Qwen2.5-72B’s discrimination improves while symbol context raises forecast MAE by 1.012 [.717, 1.316].

C.8.2 One independent hosted repeat

GPT-5.6 was replayed once on the 250 $k = 0$ prompts in each no-context, semantic-context, and arbitrary-symbol arm. The deployment, low reasoning effort, provider-native sampling, prompt hashes, parser, and one-attempt policy were unchanged; all 750 repeat responses parsed.

Table 36: Independent GPT-5.6 $k = 0$ repeat. Paired intervals resample episodes; mean $|\Delta\hat{y}|$ is the per-task absolute change in forecast.

Arm	Forecast MAE			$\rho(\hat{g}, g_C)$			Mean $ \Delta\hat{y} $
	Original	Repeat	Repeat – original [95% CI]	Original	Repeat	Repeat – original [95% CI]	
No C context	2.518	2.451	$-.068 [-.130, -.012]$	-.041	-.013	$+.029 [-.008, .066]$	.166
Semantic context	1.596	1.613	$+.016 [-.051, .082]$	.678	.678	$+.000 [-.032, .037]$	.193
Arbitrary symbol	1.744	1.708	$-.036 [-.104, .023]$	.655	.648	$-.006 [-.016, .002]$	.167

Per-task mean absolute forecast changes are .166, .193, and .167 points across the three arms. The arbitrary-symbol-minus-no-context correlation effect is .696 originally and .661 on repeat; their difference is $-.035 [-.075, .003]$. The corresponding forecast-MAE benefit is $-.774$ originally and $-.743$ on repeat, with difference $+.031 [-.046, .104]$. This one repeat supports stability of the paired conclusions for this named snapshot, not a general estimate of provider or model variance.

C.9 Missingness, scope, and artifact map

C.9.1 Missingness and record recovery

Claude Opus 5 has two unparseable primary task cells: one correct-context response at $k = 1$ and one no-context response at $k = 4$. Both exhausted the output budget during extended thinking and returned no answer. GPT-5.6 has one no-context failure at $k = 1$, where the provider returned a model-side HTTP 500 before any response. None was rerun. Complete-case pairing therefore gives $n = 249$ for the corresponding deployment-prefix rows and $n = 250$ elsewhere. These three failures constitute 0.013% of the 22,500 primary task cells.

Claude Opus 4.8, DeepSeek-V4-Pro, Kimi K3, and Gemini 3.6 Flash have 3,750 parseable primary responses each after the one offline Claude key-whitespace normalization described above. That normalization changed neither the forecast value nor the raw model answer and made no additional API call.

All six misleading-context analytic files contain exactly 1,250 unique parseable tasks. Concurrent writes in the Kimi collection produced 11 duplicate records and 14 malformed intermediate lines. Construction of the analytic file retained the earliest response for each task and identified 11 absent task IDs; those 11 tasks were collected under the unchanged Kimi configuration. Thus 7,500 is the

number of final unique analytic records for the misleading arm rather than the number of intermediate records or provider-level attempts.

The expanded arbitrary-symbol evaluation contains fifteen response-record triplets, or 56,250 records. Within every model, each arm contains 1,250 unique task IDs matched to the frozen task-file prompt hash. Original outputs are preserved. When a model placed an explicit arithmetic expression rather than a number in `predicted_poll`, a no-call reparse accepts only numeric constants, parentheses, unary signs, and the four arithmetic operators, and requires the result to lie in the primary parser’s 0–100 range. For Llama 3.1-8B, this reparse leaves 1,234/1,241/1,232 parseable no-context, semantic-context, and symbol records, respectively. Its matched complete-case n is 242 at $k = 0$ and ranges from 238 to 246 across the five prefixes. Qwen3-32B retains one unresolved symbol response, leaving 1,249 parseable symbol records; every other model parses all 1,250 in every arm, and every model except Llama 3.1-8B has $n = 250$ at $k = 0$. The final table reports these complete-case denominators, and unresolved outputs remain missing rather than being re-asked.

C.9.2 Interpretive scope

The aggregate behavior is consistent with context-conditioned response-regime assignment and integration of direct City C evidence in this controlled synthetic setting. A label-conditioned analogical-regression benchmark—map the label to a reference, estimate that reference’s slope, transfer it to City C, and revise with City C’s own cases—is sufficient to reproduce the pattern but does not identify the model’s internal algorithm. In the misleading arm, reversing only the City C label reverses forecast assignment while all numeric content remains fixed.

The primary binary semantic cue is aligned by construction with the higher- or lower-response family and has a fixed meaning across episodes. The arbitrary-symbol control removes that meaning by reversing the KIV/ZOR mapping across episodes; it therefore tests whether the semantic-arm result can arise solely from a pre-existing national-greater-than-local prior. The expanded 15-model comparison shows detectable symbol-based discrimination for Qwen3-14B and Qwen3-32B, all four tested Qwen2.5 checkpoints, Llama 3.1-70B, and all five hosted systems. In this original population, intervals span zero for Qwen3-4B, Qwen3-8B, and Llama 3.1-8B. Within Qwen3 the mapping is recovered only from 14B upward, and within Llama only at 70B, providing within-family contrasts on the original population; Qwen2.5-7B’s positive result alongside Qwen3-8B’s interval spanning zero, and Qwen3-32B’s smaller point estimate than Qwen3-14B’s, show that the transition is neither monotone in nor determined by parameter count alone. Four quantitative analyses bear on whether a model induces the arbitrary label-to-regime mapping, and they are computed from different quantities: the symbol-minus-no-context correlation contrast on forecasts; the partial correlation between the implied slope and the cue-named reference’s own fitted slope (Section C.6); held-out linear decoding of hidden states; and causal patching of those states (Section C.7.7). Where more than one was run on a checkpoint they agree, including Llama 3.1-8B, where every analysis run is null. Agreement among measures that could have disagreed is the reason to prefer the induction account over a lexical or pooling one. It does not extend the claim to the hosted deployments, whose internals are unavailable, and only one checkpoint carries the causal test. A fifth, descriptive audit asks what models themselves say in their one-sentence rationales.

Deterministic lexical rules were developed on a hash-selected 50-episode subset and then applied without revision to the remaining 200 episodes. Across all three comparison arms, 56,057 of 56,250 response records contain a rationale. At $k = 0$ in the arbitrary-symbol arm, 2,992 of 3,000 held-out records contain one; the coder identifies a selected City A/B reference in 1,832, and every codable statement names the reference that actually shares City C’s episode-local symbol. In the inverted semantic arm, all 1,200 held-out records contain a rationale: all 1,089 codable reference statements name the reference selected by the false cue, and all 1,186 codable national/local statements name the false regime. These outputs are behavioral self-reports, not privileged evidence of computation; in particular, the coder abstains whenever a statement does not make an unambiguous reference or regime claim. The conditional pattern also appears in the three checkpoints with null behavioral symbol contrasts, so it establishes declared cue interpretation rather than successful numerical use.

Because discrimination and forecast MAE can move in opposite directions, mapping induction is not equivalent to calibrated numerical use. Ambiguous, continuous, and adversarial cues remain untested. All prompts also tell models that within-city response is stable and explain the matched-pair structure.

1648 The query target is a latent conditional mean with no newly sampled observation noise. This isolates
1649 response recovery but is not equivalent to calibration on a future noisy poll. The six primary
1650 deployments differ in provider, reasoning mode, output budget, and service tier. The symbol module
1651 mixes local vLLM checkpoints with hosted calls and includes the documented cross-resource Opus 4.8
1652 arm. Cross-model comparisons are therefore descriptive rather than configuration-matched capability
1653 rankings. The cue-inversion module is a matched stress test over the same task identities and numeric
1654 inputs.

1655 The primary and harder designs are two reproducible seeded populations in the same synthetic world.
1656 The second jointly narrows the regime gap and widens slope dispersion, but two settings do not
1657 characterize generator-parameter uncertainty or isolate those changes. Bootstrap intervals resample
1658 episodes within a population and are pointwise; they do not incorporate provider drift or correct for
1659 the many model, arm, and prefix comparisons. The evaluated populations also do not characterize
1660 factorial variation in generator parameters, ambiguous cues, or calibration against noisy held-out
1661 outcomes.

1662 C.9.3 Reproducibility and artifact map

1663 The frozen data root is `exp2_v2/biased_news/data/coin_city_stable_relationship_`
1664 `claude_n250_v4/`, and the experiment-code root is `exp2_v2/biased_news/`. Paths below are
1665 relative to those roots unless a repository-level path is shown.

1666 **Design and preflight.** Under the data root, `design/manifest.json` and `design/validation.`
1667 `json` freeze and validate the primary design; `design/tasks_abc_wrong_context.`
1668 `manifest.json` records the misleading arm, and `design/tasks_abc_symbol_context.`
1669 `manifest.json` records the balanced, episode-randomized symbol control.

1670 **Generator and task builders.** Under the code root, the generator is `engine/`
1671 `coin_city_stable_relationship_claude_n250.py`; the builders are
1672 `eval/build_coin_city_stable_relationship_claude_n250.py` and
1673 `eval/build_coin_city_wrong_context_tasks.py`. The symbol imple-
1674 mentation is `engine/coin_city_symbol_context_arm.py`, with task builder
1675 `eval/build_coin_city_symbol_context_tasks.py`.

1676 **Execution and parsing.** The manifest-locked Opus 4.8 wrapper is `eval/run_coin_city_`
1677 `stable_relationship_claude_n250.py`; shared append-only execution and parsing are
1678 in `eval/run_frozen_task_shard.py`. Model-specific runners and `eval/run_coin_`
1679 `city_wrong_context.py` preserve configurations and response records under the data root.
1680 The symbol runner is `eval/run_coin_city_symbol_context.py`; its contemporaneous
1681 comparison arms are under `responses/symbol_control_matched_20260813/`. The
1682 open-model runner is `eval/run_coin_city_symbol_context_open_model.py`; the
1683 module-specific Qwen and Llama responses, hosted symbol arms, launchers, and run mani-
1684 fests are under `local_results/symbol_context_model_comparison_20260824/` rel-
1685 ative to the code root.

1686 **Primary analysis.** `analysis/analyze_coin_city_stable_relationship_multimodel.py`
1687 under the code root produces `analysis/multimodel_results.json` under the data root,
1688 the authority for the primary-arm metrics and paired contrasts.

1689 **Arbitrary-symbol analysis.** `analysis/analyze_coin_city_symbol_context.py`
1690 supplies the common estimator; `local_results/symbol_context_model_`
1691 `comparison_20260824/build_final_comparison.py` writes the paper au-
1692 thority `analysis/symbol_context_model_comparison_20260824.json`. Fo-
1693 cused design checks are in `tests/test_coin_city_symbol_context.py`;
1694 `paper/generate_coin_city_symbol_context_tables.py` generates both appendix
1695 tables from the final comparison.

1696 **Reference-selection analysis.** `analysis/analyze_coin_city_reference_selection.py` un-
1697 der the code root recomputes the partial correlations of Section C.6 from the frozen re-
1698 sponses and writes `analysis/reference_selection_results.json` under the data
1699 root; it makes no model call. Checks are in `tests/test_coin_city_reference_`
1700 `selection.py`, including one that requires the partial correlation and the symbol-arm
1701 $\Delta\rho$ contrast to name the same systems. `paper/generate_coin_city_reference_`
1702 `selection_tables.py` writes both appendix tables.

1703 **Probe-target ceilings.** `analysis/compute_bayes_ceilings.py` under the code root derives the
1704 Bayes-optimal ceiling for each probe target in Table 32 and the forecast-MAE floors drawn
1705 in Figure 5, importing the generator constants from the engine module so they cannot drift,
1706 and writes `analysis/bayes_ceilings.json` under the data root. It reads no response
1707 file and makes no model call, and fails closed if its closed form and its simulation of the
1708 generator’s own row logic disagree. Checks are in `tests/test_bayes_ceilings.py`.

1709 **Misleading arm and paper outputs.** The repository-level script `paper/generate_coin_city_`
1710 `split_figures.py` recomputes misleading-arm metrics from the final response files and
1711 scoring key. `paper/generate_coin_city_appendix_tables.py` generates the primary
1712 tables and pairwise contrasts.

1713 **Generator and deployment robustness.** `engine/coin_city_robustness_population.py`
1714 and `eval/build_coin_city_robustness_population.py` build the fresh-seed
1715 population; its design, responses, scheduler ledgers, and analysis are under
1716 `data/coin_city_population_075_040_sd010_v1/`. `analysis/analyze_`
1717 `coin_city_rationales.py` writes `analysis/rationale_audit_v1.json` under
1718 the primary data root. `eval/run_coin_city_gpt56_k0_repeat.py` and
1719 `analysis/analyze_coin_city_gpt56_k0_repeat.py` isolate and score the hosted
1720 repeat. `paper/generate_coin_city_robustness_tables.py` renders both appendix
1721 tables from the saved analysis artifacts.

1722 D Experiment 3: controlled relationship transfer

1723 As the controlled transfer step, Experiment 3 asks whether outcome-based training learns to select and
1724 apply an episode-matched relationship beyond its training setting. The primary Qwen3-8B endpoint
1725 jointly changes the domain and response mechanism; Qwen3-4B and Llama-3.1-8B provide scale
1726 and architecture comparisons. The original three-seed design and results are retained below as an
1727 execution record. Section D.1.6 reports the later eight-seed Qwen extension used for the main-text
1728 conclusions; the Llama and structureless comparators retain their original seed counts.

1729 D.1 Design and evaluation

1730 D.1.1 Data-generating processes

1731 Training uses only Structure A, the direct relationship from Experiment 2. On the displayed scale of
1732 domain d ,

$$y_{t+1} = b_d + g_j x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_d^2). \quad (6)$$

1733 A pulse acts in the next period and has no residual effect under zero subsequent input. The unseen
1734 evaluation mechanism, Structure B, is

$$m_{t+1} = \rho m_t + g_j x_t, \quad (7)$$

$$y_{t+1} - b_d = \phi(y_t - b_d) + \lambda m_{t+1} + \epsilon_t. \quad (8)$$

1735 This is a linear-Gaussian latent-state process in the sense of Kalman [1960]: it introduces a persistent
1736 mediator and outcome carry-over, so a single pulse has both immediate and delayed effects. Model
1737 prompts disclose neither equation, parameter values, internal variable, nor structure label.

1738 Coin City uses $b = 50$, range $[0, 100]$, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$, and $\sigma = 1.6$.
1739 Held-out Coin Harbor uses $b = 180$, range $[40, 320]$, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$,
1740 and $\sigma = 4.5$, together with new terminology, units, and qualitative descriptions. Evaluation
1741 gains follow $g \sim \text{Uniform}(.22, .92)$. Structure B independently samples $\rho \sim \text{Uniform}(.48, .74)$,
1742 $\phi \sim \text{Uniform}(.18, .42)$, and $\lambda \sim \text{Uniform}(.68, .94)$. A reference and its matched target share these
1743 parameters but have independent inputs and noise.

Table 37: **Exact Coin City–Coin Harbor domain contrast.** Both surfaces instantiate the same Structure A and B simulators and task template; Coin Harbor changes the semantic setting, time unit, numeric scale, interventions, noise, and pathway descriptions.

Field	Coin City (train and test)	Coin Harbor (test only)
Driver → outcome	Net campaign news → candidate support	Net shipping bulletin → cargo-flow index
Time unit	Week	Tide cycle
Resting level / support	50 / [0, 100]	180 / [40, 320]
Observed-input pool	$\{-12, -8, -4, 0, 4, 8, 12\}$	$\{-24, -16, -8, 0, 8, 16, 24\}$
Evaluation shocks	$\{-12, -6, 0, 6, 12\}$	$\{-24, -12, 0, 12, 24\}$
Observation-noise SD	1.6 support points	4.5 cargo-index points
Structure A description	Synchronized citywide alerts; reactions occur mostly in the same week	Synchronized fleetwide radio; responses occur mostly in the same tide cycle
Structure B description	Neighborhood conversations and local organizations; reactions build gradually and persist	Dockside crews and harbor groups; responses build gradually and persist

D.1.2 Training episodes and matched control

Each training prompt contains two 14-case Coin City reference systems, both governed by Equation 6. One draws $g \in [.18, .46]$ and the other $g \in [.68, .98]$; their order is randomized. The target shares exactly one reference gain, selected in balanced fashion, and shows $k \in \{0, 2, 4, 8\}$ independently generated cases. Its national/local background description identifies the matching weak or strong response as in Experiment 2. There are 1,200 rows at each k , or 4,800 per training run.

The episode-matched and population-prior datasets have byte-identical prompts. Episode-matched reward uses the target’s noise-free forecasts. For the control, targets are deranged within each k stratum with a fixed permutation: no row keeps its own target, while the complete multiset of target vectors is exactly preserved. Consequently the control can learn the output format and population distribution but cannot improve reward by using that row’s references or target history. The Qwen3-4B structureless diagnostic additionally scrambles displayed drivers and rewards a flat intervention response. It isolates task-format learning and is excluded from the episode-matched-versus-prior estimand.

Table 38: **One training episode, and what each policy is rewarded against.** Panel A is the prompt for episode `train:1:k2`, abridged from the rendered text. Panel B is the prompt–key pairing used during training, drawn for four episodes at one evidence depth: P_i is episode i ’s prompt and y_i^* its ten-answer key. Episode-matched and population-prior training use byte-identical prompts and the same multiset of keys; only the pairing differs. The control shifts cyclically across all 1,200 episodes at a depth, so its key is a real key drawn from the same distribution but never this prompt’s, and no reward can be earned by reading this row’s references. Base is untrained and receives no supervision.

A. Shared prompt. Coin City; resting level 50.0, bounded to [0, 100]; no equation or coefficient is shown.

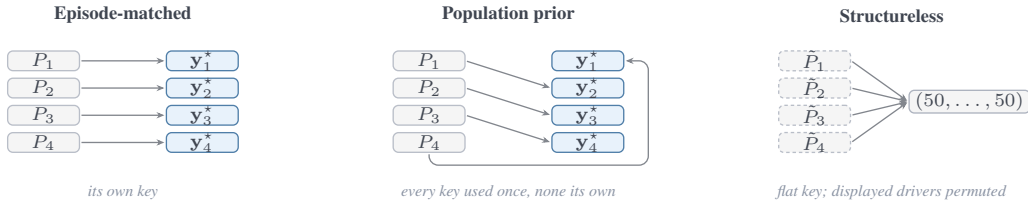
REFERENCE 1 "...immediate national alerts ...react **strongly**" 14 wk: $-8.0 \rightarrow 43.8, -12.0 \rightarrow 40.9, \dots, 0.0 \rightarrow 51.6$

REFERENCE 2 "...local civic briefings ...react **weakly**" 14 wk: $+8.0 \rightarrow 52.6, +4.0 \rightarrow 51.3, \dots, -12.0 \rightarrow 46.9$

TARGET ($k=2$) "...immediate national alerts ...react **strongly**" $-8.0 \rightarrow 42.3, +4.0 \rightarrow 53.9$

QUERY ten counterfactuals A–J: shock $\in \{-12, -6, 0, 6, 12\} \times \text{horizon} \in \{1, 3\}$; return {"forecasts": [A, ..., J]}

B. How prompts are paired with answer keys during training.



D.1.3 Evaluation interface and qualitative hint

Every evaluation prompt contains a complete 14-period Structure A reference, a complete 14-period Structure B reference, and a target governed by one of them. For Coin City the two reference descriptions are:

1764 Residents receive campaign developments through synchronized citywide alerts. Reactions
 1765 occur mostly during the same week as the news.
 1766 Campaign developments usually spread through neighborhood conversations and local
 1767 organizations. Reactions may build gradually and persist after the original news has faded.

1768 Coin Harbor uses the matched fleetwide-radio versus dockside-network descriptions. The target
 1769 receives the correct description, no description, or the other pathway’s description. The three prompt
 1770 variants reuse the exact reference and target trajectories, scenario grid, and gold forecasts; the manifest
 1771 records a common numeric hash for every cue triplet.

1772 Each task requests ten counterfactuals: shocks $\{-2u_d, -u_d, 0, u_d, 2u_d\}$ at horizons one and three,
 1773 with $u_d = 6$ in Coin City and $u_d = 12$ in Coin Harbor. The shock occurs in the next period and
 1774 all later inputs are zero. The response vector consists of the eight nonzero-shock minus zero-shock
 1775 contrasts. The full endpoint contains

$$2 \text{ domains} \times 2 \text{ structures} \times 3 \text{ hints} \times 4 \text{ } k \text{ values} \times 30 \text{ episodes} = 1,440$$

1776 paired prompts. Training and evaluation seed bands begin at 81,000,000 and 91,000,000, respectively.

1777 **Exact task template and prompt variants.** The following is the task string emitted by the prompt
 1778 generator before the model-specific chat wrapper is applied. Bracketed uppercase fields below denote
 1779 row-specific substitutions; they are not shown to the model. Line wrapping is typographic, while the
 1780 blank-line-separated sections and all fixed wording are part of the prompt.

```

1781     Forecasting task in [DOMAIN].
1782     The resting level of [OUTCOME] is [BASELINE]. Values are bounded to [LOWER, UPPER].
1783     The reference systems are complete examples. The target system may share a stable
1784     response pattern with one reference. Infer from the information shown; no response
1785     equation or coefficient is provided.
1786     REFERENCE SYSTEM 1
1787     Background: [REFERENCE 1 CONTEXT]
1788     [REFERENCE 1 TABLE: period, driver, and 14 observed outcomes]
1789     REFERENCE SYSTEM 2
1790     Background: [REFERENCE 2 CONTEXT]
1791     [REFERENCE 2 TABLE: period, driver, and 14 observed outcomes]
1792     TARGET SYSTEM
1793     [TARGET BACKGROUND, WHEN PRESENT]
1794     [TARGET TABLE WITH  $k$  OBSERVATIONS, OR THE EXACT NO-OBSERVATION SENTENCE]
1795     COUNTERFACTUAL FORECASTS
1796     For each scenario, apply the stated [DRIVER] in the next [PERIOD] and then set it to 0
1797     in later periods. Forecast [OUTCOME] at the stated horizon.
1798     A: shock=[VALUE]; horizon=1 ... E: shock=[VALUE]; horizon=1
1799     F: shock=[VALUE]; horizon=3 ... J: shock=[VALUE]; horizon=3
1800     Return only strict JSON with exactly ten finite numbers in scenario order:
1801     {"forecasts": [A,B,C,D,E,F,G,H,I,J]}
```

1802 The renderer realizes the variants as follows. Training always includes the correct strong/weak
 1803 background and uses the same input string for episode-matched and population-prior training; only
 1804 the hidden reward target differs. At evaluation, correct inserts the target pathway’s description,
 1805 none omits the background line entirely, and misleading inserts the other pathway’s description.
 1806 Every cue triplet otherwise has identical tables, interventions, and targets. If $k = 0$, the target block
 1807 contains the exact sentence *No target-system observations are available yet.*; otherwise it contains
 1808 the same table format as the references with the first k rows. The Qwen3-4B structureless
 1809 diagnostic retains this shell but swaps every displayed $-12.0 \leftrightarrow 12.0$ and $-8.0 \leftrightarrow 8.0$ token through
 1810 a collision-safe substitution. Untrained bases receive the same held-out prompts as trained models.
 1811 Thus cue, reward-arm, structureless, and base comparisons do not introduce an unreported instruction
 1812 change.

1813 D.1.4 Estimands, roster, and validation criteria

1814 The original factorial crosses Qwen3-4B, Qwen3-8B, and Llama-3.1-8B, two reward arms, and three
 1815 training seeds, for 18 runs. Three Qwen3-4B structureless runs and three untrained endpoints provide

1816 diagnostic comparisons. The primary estimand is population-prior minus episode-matched response
1817 MAE on Structure B in Coin Harbor with the correct hint, averaging the four k values equally.
1818 Positive values favor episode-matched training. The same estimator defines factorial contrasts for
1819 Structure A/Coin Harbor (domain transfer), Structure B/Coin City (structural transfer), and the
1820 in-distribution cell. Paired absent-minus-correct and misleading-minus-correct contrasts measure
1821 hint use. Their paired $k = 8$ minus $k = 0$ difference tests whether observations override a misleading
1822 prior; negative values indicate a shrinking misleading-hint penalty.

1823 Every reported round-300 endpoint uses five draws per prompt at temperature .7; temperature-zero
1824 checkpoints serve only as online training monitors.

1825 Each episode’s score is the mean of its five per-draw response MAEs, so “averaging draws” averages
1826 errors and not forecasts. The hierarchical bootstrap resamples the three training seeds at the top
1827 level and paired episodes within each selected seed. Parse failures receive the full observable-range
1828 error penalty rather than being discarded. That penalty is the clip width—280 points in Coin Harbor,
1829 roughly fifty times a typical error—so it carries large leverage: a single unparsed draw of 150 raises
1830 the Qwen3-8B base joint-cell MAE from 5.26 to 5.72, which is most of that arm’s reported gap
1831 to matched training. Both trained arms parse completely in that cell, so the primary estimand is
1832 unaffected; the base comparison is not. The launch ledger contains all 24 endpoints and 172,800
1833 stochastic rows; 172,732 parse (99.96%), and every endpoint has at least 99.82% coverage.

1834 **Original three-seed uncertainty analysis.** The original hierarchical interval resamples three top-
1835 level clusters, and a nonparametric cluster bootstrap at $G = 3$ is known to understate between-cluster
1836 uncertainty [Cameron et al., 2008]. We therefore report a deliberately conservative alternative
1837 alongside it: a Student- t interval on the three seed-level means with two degrees of freedom, which
1838 uses the same paired episode differences but charges the full small-sample penalty for estimating a
1839 variance from three numbers. Under that alternative none of the twelve transfer-cell intervals excludes
1840 zero, the primary Qwen3-8B contrast becoming .251 [−.259, .762] and the Qwen3-4B comparator
1841 .319 [−.120, .758]. Nine of the twelve intervals that exclude zero under the hierarchical bootstrap
1842 therefore do not survive the t alternative, and we do not treat any original three-seed interval as robust
1843 to the choice of small-cluster procedure.

1844 In that original roster, the sign pattern was consistent without estimating a variance. Across the four
1845 transfer cells and three training seeds there are twelve seed-level contrasts per model. All twelve
1846 favor episode-matched training for Qwen3-8B and all twelve for Qwen3-4B, so all 24 Qwen seed-
1847 by-cell contrasts point the same way; these reuse training seeds across cells and are not independent
1848 replications. The corresponding count for Llama-3.1-8B is three of twelve. This sign pattern
1849 motivated the later precision extension; its interpretation is superseded by the eight-seed results in
1850 Section D.1.6.

1851 Before training, the dataset passed 17 fail-closed checks: exact row counts and factorial balance;
1852 byte-identical episode-matched/population-prior prompts and held-out datasets; exact within- k target
1853 marginals with no fixed reward links; absence of Structure B from training prompts and metadata;
1854 cue-triplet numeric identity; train/evaluation identifier disjointness; hidden-equation leak scans;
1855 strict gold round trips; and oracle separation from the blind pathway mixture. Mean A–B response
1856 separation is 2.65 points in Coin City and 4.80 in Coin Harbor. Across both tokenizers, the longest
1857 prompt is 1,071 tokens against the configured 2,304-token limit.

1858 We estimate population-prior-minus-episode-matched intervals in all four transfer cells, paired
1859 intervals for both cue penalties at every k , and the paired $k = 8$ minus $k = 0$ change in misleading-
1860 minus-correct error to evidence override. The protocol archive records versioned input, analysis, and
1861 output hashes. Full optimization and compute parameters are in Section F.

1862 D.1.5 Original three-seed scale and architecture comparators

1863 This subsection preserves the original estimates. Its Qwen contrasts and cross-size synthesis precede
1864 the eight-seed extension and do not establish the current transfer claim; see Section D.1.6.

1865 **Qwen3-4B scale replication.** The Qwen3-4B comparison includes three seeds in each of the episode-
1866 matched, population-prior, and structureless arms. Each is evaluated on the same 1,440 tasks at round
1867 300 with five stochastic draws per task; the untrained base uses the same endpoint. Parse rates are
1868 100% for every trained arm and 99.97% for the stochastic base.

Joint-cell MAE is 6.15, 6.76, 4.81, and 4.49 for base, structureless, population-prior, and episode-matched training, respectively. The prior-minus-episode-matched difference is .319 [.134, .489]; episode-matched training has the lowest point estimate in all four cells, and all four original intervals exclude zero. The original Qwen3-8B estimate is likewise positive, .251 [.058, .470], giving the same interval-excluding-zero result at both tested Qwen sizes.

Qwen3-family summary. In the hardest joint-transfer cell, Qwen3-8B response MAE is 5.72 for the untrained base, 4.71 after population-prior training, and 4.46 after episode-matched training. The prior-minus-matched contrasts are .319 [.134, .489], .251 [.058, .470], and .120 [−.027, .261] at 4B, 8B, and 14B. An equal-weight synthesis across the Qwen3 family is .230 [.128, .333], a 4.9% MAE reduction. All three size-level estimates, eight of nine size–seed estimates, and 11 of 12 size–cell estimates agree in direction. The original estimates suggested an advantage that attenuated with scale; the eight-seed extension does not sustain that cross-size transfer conclusion. At three seeds and 8B, about one fifth of the improvement over the untrained base comes from preserving which evidence belongs to which outcome rather than learning the format or common answers.

Averaged over evidence depths, omitting the hint raises episode-matched response MAE by .151 [.079, .235], and a misleading hint raises it by .313 [.228, .415]. The misleading-hint penalty is smaller at $k = 8$ than at $k = 0$, but the evidence-override contrast −.116 [−.309, .100] includes zero.

Llama-3.1-8B architecture comparator. In the joint-transfer cell the prior-minus-episode-matched training contrast is positive, .420 [−.115, .958], but its interval spans zero. The three seed-level effects are +.762, +.622, and −.125, so this result does not establish a robust cross-architecture training advantage.

The joint cell is also the most favorable of the four Llama cells, and reporting it alone would misrepresent the factorial, so we give all four. The contrasts are −.351 [−.680, −.130] in the in-distribution cell, −.167 [−.862, .306] under structure-only shift, −.581 [−.868, −.243] under domain-only shift, and +.420 [−.115, .958] under joint shift. In the two Structure A cells the population-prior arm is therefore better than the episode-matched arm by an interval that excludes zero, and only three of the twelve Llama seed $\times$ cell contrasts favor episode-matched training. On this architecture the factorial runs against the hypothesis in the cells whose response mechanism was seen during training, and recovers a positive point estimate only where both shifts apply. We report this as a failure of the episode-matching effect to replicate on Llama-3.1-8B rather than as a directionally-consistent-but-underpowered result. The later Qwen extension also weakens the original cross-size transfer interpretation.

Llama nevertheless uses the qualitative pathway cue at the stochastic endpoint: absent-minus-correct MAE is .219 [.123, .313] and misleading-minus-correct is .210 [.092, .320]. Its misleading-hint penalty shrinks from $k = 0$ to $k = 8$ by −.621 [−.957, −.323]. All three cue-related intervals exclude zero. These contrasts indicate sensitivity to the supplied mechanism cue and reduced misleading-cue effects as observations accumulate.

Additional Llama-3.1-70B scale check. To distinguish the Llama-8B result from a general failure of the architecture, we launched a Llama-3.1-70B extension using the same training and held-out data. We report a fixed exploratory snapshot: seed 42 at round 100 of 300, temperature zero, one draw on each of the same 1,440 held-out tasks. The intervals below bootstrap paired tasks conditional on this single training seed; they do not measure between-seed uncertainty and are excluded from the primary three-seed factorial estimates.

The capacity result is already pronounced on the response mechanism seen in training. In the correct-hint Coin City/Structure A cell, response MAE falls from 4.114 for the untrained 70B checkpoint to .410 under population-prior training and .450 under episode-matched training at round 100. For comparison, the completed round-300 Llama-8B temperature-zero endpoints average 4.159 and 4.328 in the corresponding arms. Thus, after only one third of the planned schedule, 70B has roughly one tenth the direct-response error of 8B; both 70B arms parse every output and classify Structure A correctly on every task in this cell.

This snapshot does not yet establish the treatment-specific transfer effect. Across the in-distribution, domain-only, structure-only, and joint-transfer cells, the round-100 prior-minus-episode-matched contrasts are respectively −.040 [−.098, .020], +.132 [−6.166, 6.820], −.068 [−.156, .018], and +1.778 [−.143, 5.406]. Positive values favor episode-matched training. The large joint-cell point estimate is driven by one additional unparsed population-prior response: among tasks parsed by both

arms, the contrast is only $+.039 [-.209, .276]$, and Structure B classification is near zero in both arms. The in-distribution contrast also changes from $+.291 [.209, .371]$ at round 50 to $-.040$ at round 100. We therefore draw the narrow conclusion from this snapshot that scale lets Llama execute the seen numerical relationship far more accurately, while episode-conditioned structural transfer remains unresolved. This additional snapshot is distinct from the primary three-seed, five-draw round-300 comparison.

Table 39: **Stochastic Coin City appendix comparators in the joint-transfer cell, original three-seed roster.** The updated eight-seed Qwen results appear in Table 40. Entries are response MAE under the correct hint after averaging five draws per task; lower is better. Positive prior-minus-episode-matched contrasts favor episode-matched training. Intervals use the paired seed-first bootstrap with a fixed cell-specific bootstrap seed.

Model	Base	Prior	Episode-matched	Δ [95% CI]
Qwen3-4B	6.15	4.81	4.49	0.32 [0.13,0.49]
Llama-3.1-8B	9.43	4.69	4.27	0.42 [-0.12,0.96]

D.1.6 Later eight-seed precision extension

Qwen3-8B and Qwen3-4B were extended to seeds 42–49 in both reward arms, holding fixed the 300-round endpoint, five-draw decoder, data, and estimands. The added seed rosters and four estimators were recorded before their added endpoints existed. The seed count was chosen after seeing the original three-seed results; this is a later precision extension, not an eight-seed preregistration of the original study. The Qwen3-4B extension followed the disclosed Qwen3-8B interim look at seven seeds, whose joint-cell estimate was $.108 [-.035, .265]$ while a crashed seed-48 run was repeated. Eight seeds remained the fixed endpoint. The amendment and interim records are in `exp3_training_transfer/five_seed_extension/`.

Table 40: **Eight-seed prior-minus-episode-matched response MAE.** Positive values favor episode matching; brackets give hierarchical 95% intervals. All cells use the correct cue and average evidence depths equally.

Cell	Qwen3-8B	Qwen3-4B
In distribution	.060 [.018, .112]	.067 [-.056, .194]
Domain shift	.110 [-.026, .269]	.087 [-.091, .255]
Structure shift	-.003 [-.056, .040]	.052 [-.075, .179]
Joint shift	.126 [-.002, .265]	.136 [-.032, .298]

The Qwen3-8B joint estimate falls from $.251 [.058, .470]$ to $.126 [-.002, .265]$; all four Qwen3-4B intervals now span zero. The sole cell supported by all four procedures is Qwen3-8B in distribution: $.060 [.018, .112]$, Student- t interval $[-.005, .115]$, wild-cluster $p = .0089$, and 8/8 positive seeds (one-sided sign-test $p = .0039$). The hierarchical bootstrap uses 5,000 repetitions; the Student- t interval has seven degrees of freedom; the wild cluster bootstrap uses 9,999 Webb-weight repetitions; the sign test is exact. Seed-level contrasts and all four procedures for every cell are retained in `qwen3_8b_step300_stochastic_eight_seed.json` and `qwen3_4b_step300_stochastic_eight_seed.json` under the Coin City reports root. These artifacts supply `tables/exp3_rebuilt_data.tex`.

The wrapper deviation record separately documents changed checkpoint-saving flags in the Qwen3-8B added-seed block and the crossover reruns of seeds 42–44. Their mean joint-cell contrast is $.255$, close to the original $.251$, although individual seed contrasts change. This supports robustness of the mean at three seeds, not exact equivalence of the wrappers. Those diagnostic runs are excluded from the endpoint roster. Qwen3-4B’s added runs restored the parent flags before producing endpoints. The original Llama-8B and Qwen3-4B structureless comparisons remain three-seed analyses.

1955 D.1.7 Structure-selection experiment

1956 Protocol `coin_city_structure_selection_v1`. The parent Coin City design cannot test whether
1957 cue-driven selection transfers on functional form, because its training set contains only the direct
1958 structure. This dataset makes structure selectable. Each episode displays one direct and one mediated
1959 reference whose horizon-1 responses are matched exactly—the direct gain is scaled by the mediated
1960 system’s λ , since a mediated horizon-1 response is $\lambda g s$ against a direct $g s$ —so persistence at
1961 horizon 3 is the only discriminating feature and no model can select on response magnitude instead.
1962 Mediated parameters are resampled every episode, so a memorised template does not solve the task.

1963 Solvability was verified before any data was generated, the check the parent design lacked: a one-
1964 feature classifier on the coefficient of the lagged outcome, applied to the fourteen reference rows,
1965 separates the structures with $d = 4.47$ in Coin City and 3.10 in Coin Harbor and classifies at 99.0%
1966 and 97.2%. Choosing correctly is worth 2.18 response-MAE points.

1967 Training is Coin City only, semantic labels only, correct cue only, balanced 2,400/2,400 across
1968 target structures. Held out entirely: Coin Harbor, and an arbitrary-label variant in which each
1969 structure is denoted by a symbol whose mapping is redrawn every episode, so it can be recovered
1970 only from the reference trajectories. Twenty-eight fail-closed checks gate the dataset, including that
1971 the dimension under test varies in training and that the arbitrary symbols never appear in a training
1972 prompt. Evaluation is fully crossed over two domains, two label kinds, two target structures, three
1973 cue conditions and four evidence depths, 120 tasks per cell.

1974 The primary measure is the implied persistence ratio, mean $|\text{predicted horizon-3 response}|$ over
1975 mean $|\text{predicted horizon-1 response}|$, computed as a ratio of means so it stays finite when individual
1976 horizon-1 predictions are near zero. Its true value is 0 for direct targets and approximately .65 for
1977 mediated. The cue-following index is the difference in that ratio between episodes whose cue names
1978 a mediated structure and those whose cue names a direct one.

1979 The untrained reference was evaluated on the same 2,880-task set at 99.91% parse. Its index is .020
1980 and -.005 on Coin City under semantic and arbitrary labels and .068 and .002 on Coin Harbor, with
1981 implied persistence between 1.05 and 1.21 in every condition: the untrained model predicts full
1982 persistence irrespective of cue or truth.

1983 Results are a three-seed pilot of the episode-matched arm, reported as a pilot. No claim about the
1984 episode-matched minus population-prior contrast is made at three seeds; that contrast requires the
1985 eight-seed treatment applied to the parent roster. A mid-training interim look at the greedy monitors
1986 was taken at the author’s direction and is recorded in `INTERIM_LOOK_DISCLOSURE.md`; it agreed
1987 with the endpoint.

1988 D.1.8 Component diagnosis with paired cue flips

1989 The preceding RL pilot motivates a fixed, exploratory decomposition, frozen before these inference
1990 outputs. We sample 96 fresh latent episodes, 24 in each of the two domain $\times$ two label cells.
1991 Each episode is presented with both cue assignments under four interfaces, giving 768 prompts per
1992 checkpoint: (i) identify the named reference without forecasting; (ii) forecast with exact unit-shock
1993 response coefficients at horizons 1 and 3 supplied for both references; (iii) forecast with the selected
1994 reference and its coefficients supplied; and (iv) the original forecasting task. The coefficients are
1995 latent simulator summaries, not the requested ten forecasts; this assistance tests whether a supplied
1996 relationship is usable, not whether exact coefficients are recoverable from fourteen noisy observations.
1997 We use $k = 0$ target observations so numerical evidence is identical across the original-task cue flips.
1998 This diagnostic does not test integration with target observations at $k > 0$.

1999 The fixed roster comprises Qwen3-8B base, its matched and fixed-prior RL checkpoints at seeds
2000 42–44, and an untrained Qwen3-32B feasibility comparison. All 6,144 greedy generations parse.
2001 Selection-only accuracy is 100% for every checkpoint in every domain/label cell. Table 41 reports
2002 the frozen scores, including every assistance condition. At 8B, supplying patterns does not reduce
2003 average response error in any domain/label cell for the base or either trained arm. This locates a
2004 failure beyond merely identifying the reference named by the cue; it does not uniquely separate
2005 numerical application from sensitivity to the assisted prompt format.

2006 Forecast structure accuracy assigns each predicted eight-coordinate response vector to the closer of
2007 the episode’s direct and mediated simulator templates in MAE; ties count as incorrect. The paired

Domain	Labels	Model	Name (%)	Original	Both	Selected
City	semantic	8B base	100	2.105	2.393	2.413
City	semantic	8B matched RL	100	1.053	2.213	1.440
City	semantic	8B prior RL	100	0.990	3.657	1.224
City	semantic	32B base	100	4.110	1.710	1.423
City	arbitrary	8B base	100	2.117	2.710	2.961
City	arbitrary	8B matched RL	100	1.130	3.793	1.567
City	arbitrary	8B prior RL	100	1.084	4.156	1.342
City	arbitrary	32B base	100	2.671	1.993	1.604
Harbor	semantic	8B base	100	4.967	7.841	7.872
Harbor	semantic	8B matched RL	100	3.318	9.657	8.633
Harbor	semantic	8B prior RL	100	2.903	7.938	7.197
Harbor	semantic	32B base	100	7.543	5.288	4.730
Harbor	arbitrary	8B base	100	5.353	12.155	8.826
Harbor	arbitrary	8B matched RL	100	3.418	13.837	9.303
Harbor	arbitrary	8B prior RL	100	3.015	10.971	8.224
Harbor	arbitrary	32B base	100	7.012	5.242	5.200

Table 41: **Reference naming succeeds while forecasting remains difficult.** Name is selection-only accuracy; Original, Both, and Selected are response MAE for the three forecast interfaces. Trained rows average all three seeds, not selected endpoints. Each checkpoint contributes 24 paired episodes per cell. Domain scales differ, so compare interfaces within a domain.

2008 cue-change calibration is

$$\alpha = \frac{1}{N} \sum_{i=1}^N \frac{(\hat{r}_{i,M} - \hat{r}_{i,D})^\top (r_{i,M} - r_{i,D})}{\|r_{i,M} - r_{i,D}\|_2^2},$$

2009 where r subtracts the zero-shock forecast separately within each horizon, and M, D index the
2010 mediated and direct cue assignments. Zero denotes invariance to the cue; one denotes the simulator’s
2011 change along this direction, but need not imply a correct full response vector. This differs from the
2012 preceding persistence-ratio index, whose selecting reference is approximately .65. Matched RL’s
2013 mean α is .000, .017, .136, and .008 for City semantic, City arbitrary, Harbor semantic, and Harbor
2014 arbitrary, respectively. The corresponding fixed-prior means are $-.008$, .027, .085, and .035; these
2015 three-seed descriptions are not a new confirmatory comparison between training arms.

2016 **Stronger-model feasibility and output semantics.** The frozen scorer clips forecasts to the valid
2017 domain range before computing responses. Inspection revealed that Qwen3-32B often returns
2018 deviations from rest instead of absolute levels; clipping those values distorts the response vector,
2019 especially in Harbor, whose lower bound is 40. We therefore report a separately labelled *post hoc*
2020 raw-output audit (Table 42), without replacing the frozen scores or counting offset-corrected answers
2021 as correct forecasts. With the selected pattern supplied in City/semantic, raw response MAE is .384
2022 versus 4.794 on the original interface. Complete relative vectors are accurate to .01 in 52.1% of cases,
2023 but 95.8% omit the resting level. This shows partial conditional numerical ability, not a solution of
2024 the original task. The 8B assistance failures remain predominantly response-pattern errors after a
2025 common-offset correction.

Domain	Labels	Original	Both	Selected	Exact (%)	Omit rest (%)
City	semantic	4.794	0.883	0.384	52.1	95.8
City	arbitrary	2.721	1.225	0.525	41.7	91.7
Harbor	semantic	10.754	2.910	3.187	18.8	85.4
Harbor	arbitrary	7.652	1.500	2.494	35.4	85.4

Table 42: **Post hoc Qwen3-32B audit of incorrectly offset answers.** The first three numeric columns are unclipped response MAE. Exact means all eight relative response coordinates are within .01 of truth; Omit rest means both zero-shock predictions are near zero. These last two columns concern the selected-pattern interface only. They do not measure correct absolute forecasts.

2026 D.1.9 Supervised learnability and its transfer boundary

2027 We train two fixed forecast-SFT recipes from the same Qwen3-8B base on the original 4,800 balanced
 2028 Coin City/semantic prompts, with simulator-correct absolute forecasts as targets. Both use seed
 2029 42, one epoch, effective batch 16, 300 optimizer updates, completion-only token cross entropy,
 2030 LoRA rank 32 and alpha 64 on all seven projection modules, AdamW (.9, .95), no weight decay,
 2031 constant learning rate, and gradient norm cap 1.0. Rates are 10^{-6} (the RL rate) and 2×10^{-5} (an
 2032 optimization-feasibility condition). Neither Harbor nor arbitrary labels appear in training. Final
 2033 endpoints alone are evaluated, on the unchanged 768-prompt component diagnostic.

2034 This matches distinct prompts and prompt presentations, not computation or optimizer updates: RL
 2035 samples eight completions per prompt and uses about 2,400 updates. The higher-rate SFT recipe also
 2036 changes learning rate. Both supervised outcomes are reported in Table 43; the seed-42 RL checkpoint
 2037 is the primary diagnostic comparison. The other two matched RL seeds yield City/semantic structure
 2038 accuracies 52.1% and 50.0%, with response MAE 1.029 and 1.053.

Domain	Labels	Recipe	Structure (%)	Resp. MAE	α	95% interval
City	semantic	Base	45.8	2.105	-0.211	[-0.467, 0.014]
City	semantic	RL	50.0	1.077	0.005	[-0.074, 0.078]
City	semantic	SFT 10^{-6}	58.3	1.253	0.166	[-0.235, 0.549]
City	semantic	SFT 2×10^{-5}	97.9	0.658	1.413	[1.205, 1.620]
City	arbitrary	Base	54.2	2.117	0.344	[0.162, 0.529]
City	arbitrary	RL	50.0	1.222	0.012	[-0.068, 0.102]
City	arbitrary	SFT 10^{-6}	43.8	1.727	0.001	[-0.424, 0.489]
City	arbitrary	SFT 2×10^{-5}	39.6	1.274	-0.211	[-0.516, 0.107]
Harbor	semantic	Base	43.8	4.967	0.170	[-0.157, 0.499]
Harbor	semantic	RL	50.0	3.445	0.180	[0.050, 0.325]
Harbor	semantic	SFT 10^{-6}	54.2	5.317	0.058	[-0.447, 0.538]
Harbor	semantic	SFT 2×10^{-5}	45.8	12.192	-0.932	[-1.528, -0.375]
Harbor	arbitrary	Base	50.0	5.353	-0.094	[-0.301, 0.109]
Harbor	arbitrary	RL	52.1	3.619	0.026	[-0.115, 0.190]
Harbor	arbitrary	SFT 10^{-6}	47.9	5.965	0.120	[-0.132, 0.381]
Harbor	arbitrary	SFT 2×10^{-5}	50.0	10.655	0.004	[-0.282, 0.255]

Table 43: **A supervised recipe learns in-domain structure selection but fails to transfer.** All trained recipes here use seed 42. Structure accuracy is derived from forecasts, not reference naming. All outputs parse. Intervals resample the 24 paired episodes within each cell (2,000 draws); they do not capture training-seed uncertainty. Both fixed SFT learning rates are retained.

2039 At the higher rate, City/semantic structure accuracy is 47/48 (97.9%), response MAE is .658, and
 2040 forecast-level MAE is .527. Calibration $\alpha = 1.413$ indicates overshoot despite accurate choice of
 2041 response form. The lower-rate recipe reaches only 58.3% structure accuracy. Higher-rate SFT does
 2042 not recover structure selection under arbitrary labels or in Harbor; its Harbor response errors increase
 2043 substantially. Original-task SFT response scores are unchanged when clipping is removed, and an
 2044 additive-offset audit does not remove the transfer failure. Reference naming remains perfect. Thus
 2045 the successful recipe establishes in-domain learnability, while a semantic cue-to-pattern association
 2046 remains a possible explanation. It does not establish transferable induction from reference trajectories
 2047 or an isolated effect of SFT versus RL.

2048 **Adaptive decision disclosure.** The initially specified numerical launch gate was not met. Before
 2049 any SFT training, a documented outcome-informed amendment used the stronger model’s conditional
 2050 numerical ability and absolute-level errors to justify running the two already prepared, unchanged
 2051 recipes. This is an exploratory learnability pilot, not a passed preregistered gate or a confirma-
 2052 tory method comparison. There was no SFT endpoint selection. Separate reference-identification
 2053 supervision was unnecessary given the perfect selection-only results.

2054 D.1.10 Mechanism-family evidence use under coherent gain changes

2055 The mechanism-family accuracy comparison uses a fixed population-prior training control. It is
 2056 distinct from the parent Coin City shuffled-pair control and does not preserve the matched arm’s

exact target-vector distribution. To test whether the accuracy gain depends on displayed episode-specific relationships, we freeze a fresh gain-intervention set before inference: 16 pairs in each of 12 mechanisms, comprising 128 pairs from eight seen mechanisms and 64 from four held-out compositions or parameter-extrapolation mechanisms. Both mechanism-disclosure conditions use the same episodes, with $k = 9$ target observations.

Within each pair the target gain changes from the 20th to the 80th percentile of that world’s native uniform support. Inputs, exact noise draws, calibration trajectories, baseline, and mechanism are fixed; affected target observations, terminal states, and future outcomes are regenerated coherently. The disclosed description provides the same gain range on both sides, never the actual gain. This tests a gain change, not a change of mechanism identity. Native supports are retained, including the narrow, noisy extrapolation world.

The primary metric is response-change MAE: subtract the zero-shock forecast within each horizon, take the high-minus-low change of the resulting eight responses, and compare with the simulator’s change. This removes purely additive level tracking. Greedy inference evaluates Qwen3-4B-Instruct-2507 base and all retained final matched/fixed-prior adapters, seeds 45 and 46; weights for seeds 42–44 are unavailable. All pairs parse. The two-seed mean contrasts and bootstrap intervals in Table 44 resample paired training seeds and world-stratified episodes (2,000 draws). With only two training seeds these are exploratory intervals, not seed-generalized confirmation. Individual seed results are retained in the released analysis.

Disclosure	Mechanisms	Base	Matched	Prior	No change	Prior—matched	95% interval
Disclosed	Seen	3.405	1.308	1.503	1.591	0.195	[0.147, 0.244]
Disclosed	Held out	2.629	0.889	0.871	0.826	-0.019	[-0.059, 0.024]
Undisclosed	Seen	3.691	1.163	1.549	1.591	0.386	[0.338, 0.430]
Undisclosed	Held out	3.658	0.840	0.847	0.826	0.007	[-0.039, 0.051]

Table 44: **Better gain tracking on seen mechanisms does not establish transferred evidence use.** Entries are response-change MAE (lower is better); trained entries average seeds 45 and 46. No change is an invariant predictor. Positive prior-minus-matched contrasts favor matched training. Both held-out intervals include zero, and matched held-out errors exceed the no-change baseline.

Matched training improves on the prior for seen mechanisms by .195 (disclosed) and .386 (undisclosed), but held-out contrasts are $-.019$ and $.007$. Held-out matched tracking slopes are .102 and .233 (ideal 1), and its errors exceed the .826 no-change baseline in both conditions. The previously reported ordinary forecast-accuracy advantage therefore remains a narrower result than transferred evidence use. These interventions neither show a complete absence of sensitivity nor establish robust tracking of held-out episode-specific gains.

D.1.11 Distribution-matched shuffled-target control

The accuracy comparison above uses a fixed population-prior control, which does not preserve the matched arm’s exact target-vector distribution. We therefore ran a separately frozen shuffled-target control. It holds the prompts, the training budget and the exact joint target-vector multiset fixed within compatible world/evidence-depth groups, changing only which compatible episode receives which target vector, so the two arms differ in the episode-to-target pairing and in nothing else. Qwen3-4B-Instruct-2507 trains for 300 rounds in each of 5 seeds per disclosure condition. The matched arm is rerun on the same A5000 hardware and actor-memory allocation, so both arms share hardware; endpoints use a frozen fresh held-out set of 720 tasks per seed under greedy and five-draw stochastic decoding. The pre-registered primary estimand is the stochastic shuffled-minus-matched response MAE. All 40 endpoint/decode cells completed and every response parsed.

Matched training retains an advantage on the primary endpoint in both disclosure conditions. With the mechanism inferred it is .094 (5/5 seeds positive, Student- t [.041,.146], hierarchical [.032,.188], wild cluster $p = .0138$, exact sign test $p = .0312$), the only cell that clears every estimator. With the mechanism described the point estimate is larger, .155 [.046,.260], but rests on 4/5 seeds, one of which reverses sign, and does not clear the cluster-robust test ($p = .0794$). Greedy decoding is directionally consistent in both conditions—.120 described and .053 inferred—but neither cell clears the cluster-robust test.

Disclosure	Decode	Shuffled—matched	95% Student- <i>t</i>	95% hierarchical	Webb <i>p</i>	Seeds > 0
Disclosed	Greedy	+0.120	[-0.027, +0.267]	[+0.027, +0.210]	0.094	4/5
Disclosed	Stochastic	+0.155	[+0.005, +0.305]	[+0.046, +0.260]	0.079	4/5
Undisclosed	Greedy	+0.053	[-0.022, +0.128]	[+0.000, +0.099]	0.149	4/5
Undisclosed	Stochastic	+0.094	[+0.041, +0.146]	[+0.032, +0.188]	0.014	5/5

Table 45: **Matched training retains a smaller advantage over a distribution-matched shuffled-target control.** Entries are shuffled-minus-matched response MAE at 5 training seeds; positive favors matched training. The hierarchical interval resamples seeds and trajectories; Webb *p* is a two-sided wild cluster bootstrap. The stochastic inferred-mechanism cell is the pre-registered primary endpoint and the only cell clearing every estimator.

Disclosure	Decode	Shuffled—matched	95% Student- <i>t</i>	95% hierarchical	Webb <i>p</i>	Seeds > 0
Disclosed	Greedy	+0.048	[-0.014, +0.110]	[+0.011, +0.092]	0.088	4/5
Disclosed	Stochastic	+0.076	[-0.018, +0.170]	[+0.017, +0.173]	0.070	4/5
Undisclosed	Greedy	-0.019	[-0.057, +0.018]	[-0.046, +0.007]	0.206	1/5
Undisclosed	Stochastic	+0.018	[-0.093, +0.129]	[-0.040, +0.142]	0.881	1/5

Table 46: **The matched advantage does not appear in level error.** Shuffled-minus-matched level MAE, same runs and estimators as Table 45. The inferred-mechanism cells show no matched benefit. This contrast was not pre-registered and is descriptive.

2100 The benefit is concentrated in response error, which is what the manipulation predicts. Because
2101 donors are drawn from the same compatibility group, the shuffle changes an episode’s target *level*
2102 almost not at all while changing its response *shape* almost entirely: averaged over all 4,800 episodes,
2103 the recipient-donor level difference is .0086–.0093 against a shape difference of .892–.908, a ratio
2104 of 96–104 across both disclosure conditions and all five seeds, with a within-group level standard
2105 deviation of .0076. Both arms therefore receive essentially correct level information and only the
2106 matched arm receives the correct episode-specific relationship, so a level-error advantage is not
2107 expected. Its absence when the mechanism is inferred (.018 [-.040,.142], *p* = .8812 stochastic; -.019
2108 [-.046,.007] greedy) is a manipulation check rather than a negative result, and it argues against the
2109 benefit being generic improvement, which would raise both metrics.

2110 We report this as supporting rather than decisive, for three reasons. The level contrast was not
2111 pre-registered. Level MAE is computed on the raw forecast vector, so it partly contains the response
2112 component and is roughly twice its magnitude, diluting any response-driven gain. And the pattern is
2113 not uniform: the described condition does show positive level contrasts (.048 greedy, .076 stochastic).
2114 The asymmetry above is a property of the frozen builder, not of any prediction; `measure_level_`
2115 `shape_asymmetry.py` recomputes it from the training targets alone and records it in `level_`
2116 `shape_asymmetry.json`. None of this speaks to held-out mechanisms, where gain tracking remains
2117 unestablished (App. D.1.10). These estimates use the same-hardware comparator; a supplementary
2118 historical-hardware comparison reusing the earlier A6000 matched seeds 45–46 is retained in the
2119 released results and supports no claim here.

2120 **Control status and reproducibility.** Only completed diagnostics enter the tables and
2121 claims here. Their artifacts, manifests, raw outputs, and the outcome-informed SFT
2122 amendment reside under `exp3_training_transfer/overnight_diagnostics_20260921/`, in
2123 `structure/`, `learnability/`, `evidence_use/`, and `shuffled_control/`. `paper/generate_`
2124 `exp3_diagnostic_tables.py` renders the four diagnostic tables from completed summaries
2125 and records source hashes in `paper/tables/exp3_diagnostics_sources.json`; `paper/`
2126 `generate_exp3_shuffled_control_tables.py` renders the two control tables and records
2127 hashes, the design freeze and the analysis-code hash in `paper/tables/exp3_shuffled_control_`
2128 `sources.json`. Both refuse to emit a partial roster.

2129 E Experiment 4: historical-market training and held-out evaluation

2130 Experiment 4 asks whether base-relative gains extend from the controlled setting to events and
2131 normalized question families unseen during training. It uses proper-score training, a point-in-time
2132 cutoff, and family-disjoint held-out data.

2133 E.1 Sample construction and leakage controls

2134 **Source archive and sample.** The source export² has 1,788,703 rows and SHA-256
2135 5697f767...f638e7d (the full digest is stored in the manifest). The builder scans the entire
2136 file and fails on checksum drift. The calendar splits contain 1,736 train, 512 development, and 1,024
2137 test markets, with YES rates .311, .275, and .301. Train contains 991 distinct events and 1,456
2138 normalized question templates; development contains 292 and 453; test contains 493 and 866. The
2139 selected archive rows have no official series identifiers, so event and template components supply the
2140 operative family groups.

2141 **Eligibility and point-in-time censoring.** We admit literal Yes/No markets resolved by an extreme
2142 final token price. A boundary-aware domain rule selects political, governmental, macroeconomic, and
2143 geopolitical questions while explicitly rejecting culture, awards, sports, esports, and crypto categories.
2144 This rule is fixed before model training. One task is built 30 days before the earliest archived terminal
2145 timestamp. At least four distinct UTC dates must have genuine pre-cutoff YES candles, the latest
2146 candle must be no more than seven days stale, and its price must remain in [.02, .98]. We keep at
2147 most the latest 14 dates. Empty, carry-forward, and post-cutoff candles are rejected. The prompt
2148 omits the outcome, final prices and volume, market status, realized terminal time, and the tags used
2149 for domain selection.

2150 The archive records timestamped price history but not revisions to static question or resolution-rules
2151 text. We therefore treat those strings as time-invariant. This is a residual point-in-time limitation: the
2152 audit verifies that no timestamped or outcome field crosses the cutoff, but cannot determine whether
2153 an archived rule changed between the forecasting cutoff and resolution.

2154 **Family-disjoint temporal split.** Before applying sample caps, every development row sharing an
2155 event, available series, or normalized template with train is removed; the same rule removes test rows
2156 matching train or development. Numeric values and month names are normalized in template keys.
2157 The final intersection is zero for each pair of splits. Sampling within a split is a stable hash of task
2158 identifier and protocol version, so row order in the export cannot change membership.

2159 E.2 Training and held-out evaluation

2160 **Matched update budget.** Training comprises 1,736 unique markets arranged in a deterministic,
2161 balanced schedule of 4,800 presentations, matching Experiment 3’s 300-round update budget. Every
2162 market appears twice and 1,328 appear a third time. Schedule order and task files are checksum-
2163 verified. With batch size 16, this gives exactly 300 rollout-and-update rounds (2,400 AdamW steps).
2164 The manifests and tables distinguish unique markets from presentations. The longest Qwen3-4B
2165 wrapped prompt contains 1,012 tokens against the 1,792-token training limit. Data, schedule, reward,
2166 optimizer, update count, seeds, and scoring rule are held constant across models; each uses its native
2167 chat template. Shared optimizer, batching, and runtime details appear in Section F.

²  [Full Polymarket archive](#). Link anonymized for review.

Table 47: Experiment 4 training configuration. Held-out data are absent from training and evaluated once per model after all three training seeds.

Quantity	Value
Model roster / adapter	Qwen3-4B-Instruct-2507, Qwen3-8B, and Llama-3.1-8B-Instruct; LoRA rank 32, $\alpha = 64$
Algorithm	Dr. GRPO; $\beta = 0$; AdamW, lr 10^{-6}
Unique train / presentations	1,736 / 4,800 balanced repetitions
Rounds / seeds	300; 42, 43, 44
Rollout	16 prompts, 8 samples each; temperature 1.3
Output	one JSON probability; 128 new-token limit
Reward	$1 - (\hat{p} - y)^2$ only
Chat formatting	Qwen3-4B uses its frozen ChatML wrapper; Qwen3-8B uses its non-thinking template; Llama uses its native instruct template.
Development	512 tasks; temperature-zero evaluation every 25 rounds
Held-out evaluation	1,024 tasks; temperature zero; one evaluation per model

E.3 Held-out results

Decision rule. The primary learned contrast is the mean loss over three trained seeds minus base loss. Invalid JSON receives Brier loss one. Intervals use 5,000 bootstrap resamples of connected event/template components.

All Qwen3-8B test forecasts parse. Its three-seed trained mean obtains Brier .11642 versus .12867 for the base, a difference of $-.01225$ with 95% interval $[-.01948, -.00602]$. Every training seed improves on the base. The interval excludes zero, supporting transfer to held-out events for Qwen3-8B.

Table 48: **Qwen3-8B on the 1,024-market held-out set.** The mean-row interval resamples connected event/template families; lower is better.

Forecaster	Brier ↓	Δ base	Log loss ↓
Untrained Qwen3-8B	.1287	—	.5003
Trained, seed 42	.1164	-.0123	.4058
Trained, seed 43	.1168	-.0119	.4069
Trained, seed 44	.1161	-.0126	.4146
Trained mean	.1164	-.0123 $[-.01948, -.00602]$	.4091

Table 49: **Qwen3-4B and Llama held-out comparisons.** The Qwen3-8B result is immediately above. Δ columns are trained mean minus comparator, so negative values favor training. Parse is the minimum coverage across base and trained endpoints; Llama’s 12 unparsed base outputs receive Brier loss one.

Model	Base	Seed 42	Seed 43	Seed 44	Trained mean	Δ base [95% CI]	Δ market [95% CI]	Δ Platt [95% CI]	Parse
Qwen3-4B	.1316	.1142	.1140	.1194	.1159	-.0157 $[-.02528, -.00667]$	.0022 $[-.00264, .00721]$	.0036 $[-.00125, .00874]$	1.000
Llama-3.1-8B	.1520	.1151	.1149	.1151	.1150	-.0370 $[-.05376, -.02258]$	.0013 $[-.00004, .00371]$	.0027 $[-.00067, .00539]$	.988

Qwen3-4B replicates the base-relative improvement: its trained mean is .11589 versus .13160 for base, a difference of $-.01572$ $[-.02528, -.00667]$. Llama-8B improves from .15203 to .11502, a difference of $-.03701$ $[-.05376, -.02258]$; all three trained Llama endpoints parse.

The latest pre-cutoff market price obtains Brier .11369, and logistic recalibration of its logit fit only on the training split obtains .11232. Neither comparator informs language-model training. The trained-minus-market interval spans zero for every model. Qwen3-4B is also statistically indistinguishable from the calibrated price, but Qwen3-8B and Llama-8B have higher error by $+.00409$ $[+.00098, +.00785]$ and $+.00269$ $[+.00067, +.00539]$, respectively. Thus the complete roster shows base-relative improvement rather than superiority to market-based forecasts.

2188 E.4 Scale and architecture comparison

2189 The scale comparison uses all 318 eligible 2026-window tasks whose identifiers and event, series, and
 2190 normalized question-template families are absent from the training, development, and 1,024-market
 2191 evaluation sets. The holdout contains 227 event families and 288 normalized templates; selection
 2192 does not use outcomes or market probabilities.

2193 Qwen3 checkpoints at 4B, 8B, 14B, and 32B use three seeds, a common reward, a 4,800-presentation
 2194 schedule, and the same development rule. The prospective 32B extension was registered before
 2195 the shared holdout was opened; it retained the 14B scientific settings and changed only memory
 2196 distribution across four A6000 GPUs. Each untrained base and trained adapter receives five non-
 2197 thinking draws per task at temperature .7, top- p .8, and top- k 20. All 25,440 Qwen draws satisfy the
 2198 strict JSON contract.

Table 50: **Exact values behind the main-text scale figure.** Lower Brier is better. Trained means average seeds 42–44. Missing five-draw task groups receive Brier loss one.

Checkpoint	Base	Trained mean	Trained — base	Trained parse
Qwen3-1.7B	0.2253	0.1248	-0.1005	100.0%
2199 Qwen3-4B-Instruct-2507	0.1522	0.1366	-0.0156	100.0%
Qwen3-8B	0.1369	0.1286	-0.0083	100.0%
Qwen3-14B	0.1281	0.1270	-0.0012	100.0%
Qwen3-32B	0.1266	0.1265	-0.0001	100.0%
Llama-3.2-3B	0.2189	0.1367	-0.0822	100.0%
Llama-3.1-8B	0.1642	0.1264	-0.0377	100.0%

2200 The trained means are .13664, .12859, .12695, and .12652 at 4B, 8B, 14B, and 32B; the corresponding
 2201 bases are .15224, .13686, .12812, and .12659. The 32B seed Briers are .12653, .12651, and .12651.
 2202 Trained-minus-base contracts from $-.01559$ at 4B to $-.00827$ at 8B, $-.00117$ at 14B, and $-.00007$
 2203 $[-.00045, +.00029]$ at 32B. Thus the larger untrained checkpoints approach the same absolute
 2204 performance reached by the trained endpoints under the shared recipe. The registered 14B-minus-8B
 2205 trained contrast remains $-.00164 [-.00332, -.00024]$.

2206 On this holdout, contemporaneous Polymarket prices have Brier .12655. Trained 14B minus market
 2207 is $+.00040 [-.00028, +.00139]$; trained 32B minus market is $-.00003 [-.00009, +.00002]$. Both
 2208 intervals cross zero. The scale profile therefore shows convergence to the contemporaneous market
 2209 frontier, not a resolved advantage over it. Because data volume, update count, and optimizer settings
 2210 are held fixed, the comparison does not identify separately tuned optima for each model size.

2211 Llama-3.1-8B uses the same holdout and five-draw endpoint. All 4,770 trained draws parse; nine of
 2212 the 1,590 base draws do not, causing seven base task groups to receive Brier loss one. Trained seed
 2213 Briers are .12638, .12629, and .12662, for a mean of .12643 versus .16417 for base. Trained minus
 2214 base is $-.03774 [-.05860, -.01946]$; trained minus Polymarket is $-.00012 [-.00046, +.00012]$.
 2215 The latter interval crosses zero.

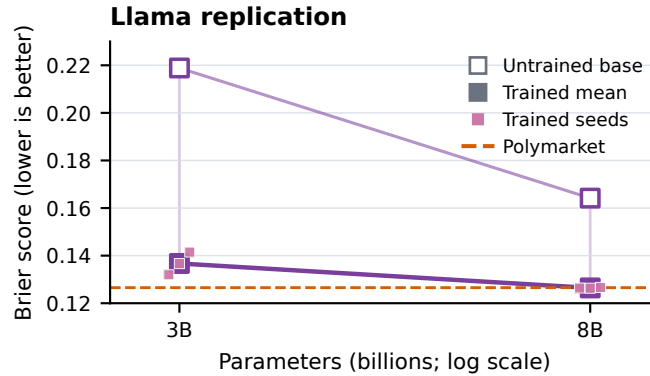


Figure 16: **Llama scale replication.** At 3B and 8B, open/filled squares show base/trained means, small squares show seeds, and the dashed line shows Polymarket. The parameter axis is logarithmic. The 3B base is descriptive because its strict-JSON recovery was specified after observing syntax failures; all other endpoints follow the common protocol.

Qwen3-1.7B and Llama-3.2-3B preserve the same training/development data, three seeds, 4,800-presentation schedule, five stochastic draws, and holdout. Qwen3-1.7B seed Briers are .12413, .12610, and .12415 (mean .12479) versus .22531 for base; trained minus Polymarket is $-.00175$ $[-.00644, +.00265]$. Llama-3.2-3B seed Briers are .13197, .13666, and .14149 (mean .13671); trained minus Polymarket is $+.01016$ $[+.00121, +.02060]$. The pinned public BF16 Llama mirror is recorded because the gated upstream repository was unavailable to the workspace account. Its unconstrained base initially produced only 4/318 complete five-draw groups at the 128-token cap; an output-cap-only rerun reached 21/318. A disclosed, holdout-informed syntax recovery therefore repeated the full base endpoint with identical prompts and sampling settings while constraining output to a JSON object containing one numeric probability. All 1,590 recovered draws parse. Base Brier is .21887, and trained minus base is $-.08216$ $[-.11062, -.05367]$. Because the syntax constraint was introduced after observing the failures and differs from the trained endpoints, this base-relative contrast is descriptive. Both trained models also parse every draw.

Hosted frontier references. The API panel evaluates the same 318-task holdout. It is categorical rather than a parameter-scale curve: provider parameter counts are unavailable or not comparable, and the APIs do not expose identical decoding controls. DeepSeek-V4-Pro uses the common temperature .7/top- p .8 stochastic request. GPT-5.6-Sol uses provider-native low reasoning without user-set temperature or top- p ; Claude Opus 4.8 likewise uses provider-native sampling. Each endpoint receives five independent calls per task and uses the same strict parser, five-draw mean, Brier scoring, and family-clustered 5,000-repetition interval procedure as the local panel.

DeepSeek parses all 1,590 draws and has Brier .12660, differing from the crowd by $+.00005$ $[-.00105, +.00102]$. Claude parses all draws and has Brier .12927, a difference of $+.00272$ $[-.00049, +.00614]$. GPT receives all 1,590 responses but six are malformed; two task groups therefore receive Brier loss one, and task-level coverage is 99.37%. Its Brier is .13335, a difference of $+.00680$ $[+.00016, +.01635]$.

Kimi uses a 1,024-token output budget. Thirteen transport failures are retried without replacing the 1,577 successful responses; all 1,590 final draws parse. Its Brier is .12682, differing from the crowd by $+.00027$ $[-.00046, +.00109]$. Hosted results provide performance context, not evidence for a common scale law or an identical-decoder comparison.

E.5 Qwen3-4B evidence-updating evaluation

This evaluation applies the Experiment 1 update prompt and all nine fixed packets for each of its 100 markets (three pro- H_1 , three anti- H_1 , and three orthogonal). The packet and initial-context SHA-256 digests are 6741be0a...8796d36 and c550cf05...0c098. Every condition receives the same canonical first structured Qwen 2.5-7B forecast for each market, so this paired analysis isolates conditional revision rather than web retrieval or initial event-model construction. Search and thinking

are disabled; decoding is greedy with a 1,500-token output cap. The longest prompt is 1,480 tokens against a 6,144-token context limit. These contrasts characterize revision behavior and are separate from the held-out forecasting evaluation.

The paired evaluation produces all 900 packet-conditioned revisions for the base and each final adapter, with 100% parse coverage and ICS equal to one. Base EHC/HFC is .909 on 591 eligible directional updates. Seeds 42, 43, and 44 obtain .917, .922, and .918 (592, 588, and 591 eligible updates), for a three-seed mean of .919. The paired market-bootstrap trained-minus-base EHC difference is +.0101 with 95% interval $[+.0004, +.0207]$; HFC has the same estimate and interval $[+.0009, +.0210]$. Training reduces mean absolute directional revision by .52 percentage points $[-.70, -.36]$ and orthogonal revision by .22 points $[-.38, -.06]$; the directional-minus-orthogonal gap narrows by .30 points $[-.52, -.09]$. The sensitivity ratio changes from $4.64\times$ to $4.83\times$ $(+.19 [-.07, +.48])$. These components show improved directional consistency alongside smaller revisions overall; they do not indicate an increase in absolute packet selectivity.

Table 51: **Paired evidence-updating contrasts.** Movement rows are mean absolute probability revisions in percentage points; intervals resample the 100 markets.

Measure	Trained — base	95% interval
Directional consistency (EHC)	+ .0101	$[+.0004, +.0207]$
Headline-forecast consistency (HFC)	+ .0101	$[+.0009, +.0210]$
Directional movement	− .52 pp	$[-.70, -.36]$
Orthogonal movement	− .22 pp	$[-.38, -.06]$
Directional — orthogonal movement	− .30 pp	$[-.52, -.09]$
Sensitivity ratio	+ .19 $\times$	$[-.07, +.48]$

F Experiments 3–4: shared training configuration and runtime

Tables 52–54 record the effective OAT/Dr. GRPO [Liu et al., 2024] arguments for the reported Coin City and Polymarket studies, including framework defaults omitted from the command line. Execution manifests hash the data, trainer, launcher, evaluator, prompt renderer, and each cell’s environment.

Step accounting. A “step” in the figures and earlier tables is one rollout-and-update round rather than one AdamW call. Each full run consumes 16 prompts per round and samples eight responses per prompt (128 trajectories). With one learner GPU, per-device microbatch one, one PPO epoch, and gradient accumulation 16, OAT performs eight AdamW parameter updates per round. Thus 4,800 prompt presentations give 300 reported rounds and 2,400 AdamW updates. The command line supplied a .03 warmup ratio, but OAT selected Transformers’ constant scheduler, which does not use the warmup argument. The effective learning rate was therefore 10^{-6} from the first optimizer update. We use “round” below to avoid counting the eight within-round parameter steps as independent data exposures.

Table 52: **Shared Dr. GRPO implementation parameters.** These values apply to the Coin City and Polymarket studies; the study-specific tables below record exceptions.

Quantity	Effective value
Trainable parameters	Frozen base model; LoRA rank 32, $\alpha = 64$, dropout 0, no bias; targets <code>q_proj</code> , <code>k_proj</code> , <code>v_proj</code> , <code>o_proj</code> , <code>down_proj</code> , <code>up_proj</code> , and <code>gate_proj</code> ; no quantization.
Algorithm	OAT Dr. GRPO (<code>critic_type=drgrpo</code>); eight-response group advantage $r_i - \bar{r}$ with no group-standard-deviation division; policy loss is token-summed with the generation limit as a fixed normalizer (no response-length normalization).
Policy objective	PPO ratio clipping 0.2; truncated importance-sampling cap 2.0; one PPO epoch; KL coefficient $\beta = 0$; no critic/value model; reward whitening, advantage whitening, and REINFORCE mode all off.
Optimizer	PyTorch AdamW [Loshchilov and Hutter, 2019] (substituted for DeepSpeed fused/CPU Adam); learning rate 10^{-6} ; $\beta_1 = .9$, $\beta_2 = .95$; weight decay 0; maximum gradient norm 1.0.
Schedule	Constant 10^{-6} learning rate from the first optimizer update; the supplied .03 warmup ratio is ignored by Transformers’ <code>constant</code> scheduler; one prompt epoch; 4,800 prompt presentations = 300 rollout-and-update rounds = 2,400 AdamW steps.
Batching	16 prompts per round; 8 sampled responses per prompt; 128 trajectories; training microbatch 1; gradient accumulation 16; one learner GPU.
Training sampling	Temperature 1.3; top- $p = 1$; top- $k = -1$; one independently sampled completion at a time, repeated eight times per prompt.
Truncation and stopping	Model EOS only (no external stop string); a completion ending at the length limit receives reward zero. Missing or invalid required JSON also receives reward zero.
Precision and memory	bfloat16; FlashAttention 2; gradient checkpointing; DeepSpeed ZeRO-2 [Rajbhandari et al., 2020]; reference-model offload; vLLM prefix caching [Kwon et al., 2023]; dropout disabled.
Seeds and shuffling	Training seeds 42, 43, and 44. Each dataset is deterministically shuffled once, then OAT shuffles prompt batches independently under the training seed.
Checkpointing and logging	Evaluation includes step zero and the final round; intermediate evaluation at the setting-specific frequency; one final LoRA adapter retained; metrics logged each round; Weights & Biases disabled.

E.1 Coin City A-to-B factorial

Table 53 records the original three-seed configuration for Experiment 3. The current Qwen3-8B and Qwen3-4B analyses extend both reward arms to eight seeds (Section D.1.6), retaining the five-draw stochastic round-300 endpoint. The structureless diagnostic and Llama-8B architecture comparator retain their original three seeds. The table supplements the shared optimizer details in Table 52; where the two tables differ, this setting-specific table governs. The execution manifest stores the environment; the protocol archive hashes the preflight manifest, data files, generator, prompt renderer, scorer, and trainer entry point.

Table 53: **Original three-seed configuration for Coin City hidden-structure transfer.** The later Qwen extensions are documented in Section D.1.6.

Quantity	Effective value
Model roster	Qwen/Qwen3-4B-Instruct-2507, Qwen/Qwen3-8B, and meta-llama/Llama-3.1-8B-Instruct.
Chat formatting	Qwen3-4B uses the explicit ChatML wrapper; Qwen3-8B and Llama-3.1-8B use their tokenizer templates, with Qwen thinking disabled. All include an assistant-generation prefix.
Original grid	3 models $\times$ 2 reward arms (episode-matched reward, stored under the legacy artifact key <code>causal</code> , and <code>population_prior</code>) $\times$ 3 seeds (42, 43, 44) = 18 full training runs.
Diagnostics	Qwen3-4B <code>structureless</code> at the same three seeds, plus one untrained endpoint for each model. These six diagnostics are excluded from the primary episode-matched-versus-prior estimand.
Training data	Coin City and Structure A only; 4,800 prompts, balanced across $k \in \{0, 2, 4, 8\}$ and weak/strong target match. Two reference trajectories of length 14 and one target trajectory of maximum length 8.
Comparator target construction	Episode-matched reward uses the episode truth. Population-prior reward targets are a fixed no-self-match derangement within each k ; its target-vector multiset is exactly identical to episode-matched training. Prompt strings are byte-identical across these two arms.
Held-out data	2 domains $\times$ 2 target structures $\times$ 3 hint arms $\times$ 4 evidence depths $\times$ 30 episode seeds = 1,440 rows. The complete held-out dataset is byte-identical across all training arms.
Mechanism ranges	Evaluation $g \in [.22, .92]$; Structure B $\rho \in [.48, .74]$, $\phi \in [.18, .42]$, and $\lambda \in [.68, .94]$. Training reference gains are $[\cdot18, \cdot46]$ and $[\cdot68, \cdot98]$.
Domain ranges	Coin City: baseline 50, support $[0, 100]$, noise 1.6, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$. Coin Harbor: baseline 180, support $[40, 320]$, noise 4.5, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$.
Interventions	Five shocks at horizons one and three; Coin City $\{-12, -6, 0, 6, 12\}$, Coin Harbor $\{-24, -12, 0, 12, 24\}$. Ten joint forecasts produce eight nonzero-minus-zero response contrasts.
Reward	$\cdot4 r_{\text{level}} + \cdot6 r_{\text{response}}$. Each component averages $\max(0, 1 - e /S)$; level tolerance is 10% of the domain’s observable range (10 Coin City points; 28 Coin Harbor units), and response tolerance is half one shock unit (3 and 6, respectively). Invalid or length-truncated output receives zero online reward.
2289 Output contract	Exactly one JSON object with sole key <code>forecasts</code> and exactly ten finite JSON numbers. The same strict parser is used online and offline; no prose, extra/duplicate key, Boolean, non-finite value, or wrong-length list is accepted.
Structured decoding	Syntax-only XGrammar GBNF in training, online evaluation, endpoint evaluation, and base evaluation. It fixes object shape and arity only; numeric values and their relationship to targets remain unconstrained.
Optimization	Frozen base plus LoRA rank 32, $\alpha = 64$; OAT Dr. GRPO; PyTorch AdamW learning rate 10^{-6} with an effective constant schedule and no warmup; $\beta = 0$; one PPO epoch; gradient norm 1; ZeRO-2 and reference offload.
Batch and update accounting	16 prompts per round; eight rollouts per prompt; 128 trajectories; microbatch 1; gradient accumulation 16; 4,800 presentations = 300 rollout-and-update rounds = 2,400 AdamW steps.
Training generation	Temperature 1.3, top- $p = 1$, top- $k = -1$, model EOS only; 192 new-token limit; prompt limit 2,304; total model context 3,072; vLLM prefix caching enabled.
Precision / memory	BF16, FlashAttention 2, gradient checkpointing, dropout off, no quantization; vLLM GPU fraction .38; expandable-segments CUDA allocator.
Online evaluation	All 1,440 rows at temperature 0, one draw, top- $p = 1$, 192-token limit, batch 120, at step zero and every 50 rounds including round 300.
Reported endpoint evaluation	Five draws per prompt at temperature .7 and top- $p = .95$; trained decode seed is training seed +20,260,818 and the base seed is 20,260,819. Temperature-zero checkpoints are retained only as online monitoring artifacts and are not reported as endpoints.
Offline scoring	Level MAE, response MAE, response error normalized by the blind 50/50 pathway mixture, response correlation, strict parse coverage, and nearest behavioral pathway. A parse failure receives the full domain range as both level and response error.
Variance and estimands	Five-draw outputs first average within episode. Original hierarchical intervals resample three training seeds, then paired episodes; the precision extension resamples eight seeds and adds the small-cluster checks in Section D.1.6. Primary: prior-minus-episode-matched response MAE for correct-hint Structure B in Coin Harbor, equal over k . Domain, structure, joint, hint, and hint-by- k contrasts form the factorial analysis.
Preflight validation	A 10-round run for every model and evaluated arm must save an adapter, produce the full temperature-zero monitoring and five-draw stochastic evaluation sets, achieve at least .95 strict parse rate under each decoder, and exhibit nonconstant finite reward. The 17-check data preflight must also pass before estimating the factorial.
Compute	Training and trained-checkpoint evaluation use two NVIDIA A6000 48-GB GPUs (one actor, one learner), 8 CPU cores, and 56–100 GB allocated host memory. Untrained base evaluation uses one A6000, 8 CPU cores, and 24 GB host memory. All runs load models from the same pinned local cache.

Table 54: **Runtime environment for the reported training runs.** Package versions come from the job execution environment.

	Component	Version / configuration
2290	Python / CUDA build	Python 3.10.20; PyTorch 2.6.0 built for CUDA 12.4.
	Training stack	OAT 0.2.4 at Git revision 869706620a35ec5304fa80a0c74f0c566cc2bc57; DeepSpeed 0.16.8; PEFT 0.15.2.
	Generation stack	Transformers 4.51.3; vLLM 0.8.4; FlashAttention 2.7.4.post1.
	Data stack	Datasets 2.16.1; Accelerate 1.14.0.
	Allocator / logging	PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True; Weights & Biases disabled; stdout/stderr and per-round metrics retained with each run.

2291 G Experiments 3–4: reproducibility and artifact map

2292 The artifact roots `exp3_training_transfer/coin_city_structural/` and `exp3_training_`
2293 `transfer/polymarket/scripts/` contain the generators, prompt builders, trainers, evaluators,
2294 and analysis code for the Coin City and historical-market studies. Execution ledgers record effective
2295 hyperparameters, Git revisions, adapter identities, and SHA-256 hashes of data and executable
2296 protocol files; result renderers verify these fields before aggregation. For Coin City, `make_qwen3_`
2297 `8b_endpoint_outputs.py` validates the seven original five-draw files (50,400 scored draws). The
2298 later eight-seed reports in Section D.1.6 supply Figure 9; `render_exp3_paper_macros.py` renders
2299 the current prose values from those reports and the recorded structure-selection and mechanism-family
2300 endpoints. `make_paper_outputs.py` validates the original 24-endpoint roster and all 48 recorded
2301 endpoint files before writing the machine-readable result and the stochastic appendix comparator
2302 table. Temperature-zero files are checked as execution provenance but are excluded from manuscript
2303 estimates. For the historical study, `analyze_exp1_reapplication.py` validates four matched
2304 900-record files and computes the paired-market direction, sensitivity, and movement-component
2305 contrasts. `generate_exp4_scale_figure.py` verifies the frozen 32B protocol, summary, and
2306 raw-output hashes before adding the point to Figure 10 and Table 50; it also writes the separate Llama
2307 replication in Figure 16. The Polymarket manifests preserve sample membership, prompt and adapter
2308 hashes, parse coverage, and bootstrap settings.